

# NL5

## User's Reference

Components  
Operators  
Functions  
Script commands  
HTTP functions

Ver. 4.3.10



## VERSION

NL5 User's Reference version 4.3.10, 04/04/2026

The latest versions of NL5 documents can be found at [sidelinesoft.com/nl5](https://sidelinesoft.com/nl5).

## LIMITED LIABILITY

NL5, together with all accompanying materials, is provided on a “as is” basis, without warranty of any kind. The author makes no warranty, either expressed, implied, or stationery, including but not limited to any implied warranties of merchantability or fitness for any purpose. In no event will the author be liable to anyone for direct, incidental or consequential damages or losses arising from use or inability to use NL5.

## COPYRIGHTS

© 2026, A.Smirnov. The program and User's Manual are copyrighted. No portion of this Manual can be translated or reproduced for commercial purposes without the express written permission from the copyright holder. On publication of results obtained from use of NL5 citation is appreciated.

---

“**Smith**” is a registered trademark of Analog Instruments Company, New Providence, NJ. **Microsoft**, **Windows**, and **Microsoft Visual C++** are registered trademarks of Microsoft Corporation. **MATLAB** is a registered trademark of The MathWorks, Inc. **PYTHON** is a registered trademark of the Python Software Foundation. **Verilog** is a registered trademark of Cadence Design Systems. **Xilinx** and **Vivado** are registered trademarks of Xilinx.

# Table of Contents

|                                                                  |          |
|------------------------------------------------------------------|----------|
| <b>Components.....</b>                                           | <b>7</b> |
| SubCir model .....                                               | 8        |
| Label .....                                                      | 9        |
| A – Amperemeter.....                                             | 16       |
| C – Capacitor .....                                              | 17       |
| C – Voltage controlled capacitor .....                           | 18       |
| C – Current controlled capacitor.....                            | 19       |
| D – Diode.....                                                   | 20       |
| D – Zener .....                                                  | 23       |
| D – Bidirectional zener .....                                    | 24       |
| D – Bridge rectifier .....                                       | 25       |
| D – Diode ring .....                                             | 26       |
| D – Logic controlled thyristor.....                              | 27       |
| D – Voltage controlled thyristor .....                           | 28       |
| D – Current controlled thyristor.....                            | 29       |
| F – Function .....                                               | 30       |
| F – Function with clock .....                                    | 33       |
| F – Function-2.....                                              | 36       |
| F – Function-2 with clock.....                                   | 40       |
| F – Custom function.....                                         | 44       |
| F – Integral.....                                                | 46       |
| F – Integral with reset .....                                    | 47       |
| F – s-function .....                                             | 48       |
| F – z-function.....                                              | 53       |
| I – Current source.....                                          | 55       |
| I – Logic controlled current source .....                        | 61       |
| I – Voltage controlled current source.....                       | 65       |
| I – Current controlled current source .....                      | 68       |
| L – Inductor.....                                                | 71       |
| L – Voltage controlled inductor.....                             | 72       |
| L – Current controlled inductor .....                            | 73       |
| L – Coupled inductors.....                                       | 74       |
| L – Custom coupled inductors .....                               | 75       |
| O – Amplifier.....                                               | 77       |
| O – Differential amplifier .....                                 | 80       |
| O – Differential amplifier with reference .....                  | 83       |
| O – Fully differential amplifier.....                            | 84       |
| O – Differential amplifier with current output.....              | 85       |
| O – Differential amplifier with differential current output..... | 88       |
| O – Current amplifier.....                                       | 91       |
| O – Current amplifier with current output .....                  | 94       |
| O – Summing amplifier .....                                      | 98       |
| O – Voltage controlled amplifier .....                           | 101      |
| O – Current controlled amplifier.....                            | 102      |
| R – Resistor.....                                                | 103      |

|                                              |     |
|----------------------------------------------|-----|
| R – Potentiometer .....                      | 104 |
| R – Voltage controlled resistor .....        | 105 |
| R – Current controlled resistor.....         | 106 |
| S – Switch .....                             | 107 |
| S – Logic controlled switch .....            | 112 |
| S – Voltage controlled switch .....          | 116 |
| S – Current controlled switch .....          | 118 |
| S – SPDT switch .....                        | 120 |
| S – SPDT logic controlled switch.....        | 120 |
| S – SPDT voltage controlled switch .....     | 121 |
| S – SPDT current controlled switch.....      | 121 |
| T – NPN transistor .....                     | 122 |
| T – PNP transistor .....                     | 124 |
| T – N-FET .....                              | 126 |
| T – P-FET .....                              | 130 |
| V – Voltage source.....                      | 134 |
| V – Logic controlled voltage source.....     | 140 |
| V – Voltage controlled voltage source.....   | 144 |
| V – Current controlled voltage source .....  | 147 |
| V – Voltmeter .....                          | 150 |
| W – Winding.....                             | 151 |
| W – Transformer .....                        | 152 |
| W – Differential transformer .....           | 153 |
| W – Custom transformer.....                  | 154 |
| W – Wattmeter.....                           | 155 |
| X – Delay .....                              | 156 |
| X – Transmission line .....                  | 157 |
| X – Sample/hold .....                        | 159 |
| X – Directional coupler.....                 | 160 |
| X – Block-2...Block-8.....                   | 161 |
| X – Custom block .....                       | 162 |
| X – NL5 circuit .....                        | 163 |
| X – C-code .....                             | 164 |
| X – DLL .....                                | 166 |
| X – S-matrix.....                            | 168 |
| Y – Gates.....                               | 169 |
| Y – Logical function .....                   | 171 |
| Y – D flip-flop .....                        | 173 |
| Y – SR trigger .....                         | 174 |
| Y – JK trigger.....                          | 175 |
| Y – Schmitt trigger.....                     | 176 |
| Y – Logic generator .....                    | 177 |
| Y – Logic controlled logic generator .....   | 182 |
| Y – Voltage controlled logic generator.....  | 185 |
| Y – Current controlled logic generator ..... | 187 |
| Y – Bus .....                                | 189 |
| Z – Impedance.....                           | 190 |
| Z – Z-meter .....                            | 194 |

|                                      |            |
|--------------------------------------|------------|
| <b>Operators .....</b>               | <b>195</b> |
| <b>Functions.....</b>                | <b>197</b> |
| bool C-keyword.....                  | 198        |
| (bool) type-casting operator.....    | 198        |
| bool.....                            | 198        |
| int C-keyword.....                   | 199        |
| (int) type-casting operator.....     | 199        |
| int.....                             | 199        |
| int64 C-keyword.....                 | 200        |
| (int64) type-casting operator.....   | 200        |
| int64.....                           | 200        |
| double C-keyword.....                | 201        |
| (double) type-casting operator.....  | 201        |
| double.....                          | 201        |
| complex C-keyword.....               | 202        |
| (complex) type-casting operator..... | 202        |
| complex.....                         | 202        |
| fabs.....                            | 203        |
| fmod.....                            | 203        |
| fmin.....                            | 203        |
| fmax.....                            | 203        |
| abs, mag.....                        | 204        |
| round.....                           | 204        |
| floor.....                           | 204        |
| ceil.....                            | 204        |
| sqrt.....                            | 205        |
| sqr.....                             | 205        |
| sq.....                              | 205        |
| pow.....                             | 205        |
| pwr.....                             | 206        |
| exp.....                             | 206        |
| exp2.....                            | 206        |
| log, ln.....                         | 206        |
| log2, lb.....                        | 207        |
| log10, lg.....                       | 207        |
| log(x,y).....                        | 207        |
| db.....                              | 207        |
| sin, cos, tan, tg.....               | 208        |
| asin, acos, atan.....                | 208        |
| atan2.....                           | 208        |
| phase.....                           | 208        |
| random, rand.....                    | 209        |
| gauss.....                           | 209        |
| islow, ishigh.....                   | 209        |
| lim, limit.....                      | 209        |
| sign.....                            | 210        |
| par.....                             | 210        |
| min.....                             | 210        |

|                                            |            |
|--------------------------------------------|------------|
| max.....                                   | 211        |
| mean, average.....                         | 211        |
| sum.....                                   | 212        |
| re, im.....                                | 212        |
| <b>Script commands.....</b>                | <b>213</b> |
| ac.....                                    | 214        |
| clear.....                                 | 214        |
| close.....                                 | 214        |
| cmd.....                                   | 214        |
| cont.....                                  | 215        |
| cursors.....                               | 215        |
| display.....                               | 215        |
| exit.....                                  | 215        |
| export ( <i>transient</i> ).....           | 216        |
| export (AC).....                           | 216        |
| import ( <i>transient</i> ).....           | 217        |
| logdata.....                               | 218        |
| open.....                                  | 218        |
| pause.....                                 | 218        |
| ready.....                                 | 218        |
| return.....                                | 219        |
| rununtil.....                              | 219        |
| save.....                                  | 219        |
| saveic.....                                | 219        |
| show.....                                  | 220        |
| silent.....                                | 220        |
| sleep.....                                 | 220        |
| stop.....                                  | 220        |
| store.....                                 | 221        |
| storetext.....                             | 221        |
| traces.....                                | 221        |
| <i>tracename</i> ( <i>transient</i> )..... | 222        |
| <i>tracename</i> (AC).....                 | 223        |
| tran.....                                  | 224        |
| <b>Script examples .....</b>               | <b>225</b> |
| <b>HTTP functions.....</b>                 | <b>228</b> |
| NL5_GetInfo.....                           | 229        |
| NL5_New.....                               | 229        |
| NL5_Open.....                              | 229        |
| NL5_Save.....                              | 229        |
| NL5_Close.....                             | 229        |
| NL5_Show.....                              | 229        |
| NL5_GetActive.....                         | 230        |
| NL5_GetParam.....                          | 230        |
| NL5_SetParam.....                          | 230        |
| NL5_DisableCmp.....                        | 231        |
| NL5_EnableCmp.....                         | 231        |

|                            |     |
|----------------------------|-----|
| NL5_SetCmpState .....      | 231 |
| NL5_GetTracesSize .....    | 231 |
| NL5_AddTrace .....         | 232 |
| NL5_DeleteTrace .....      | 232 |
| NL5_GetTrace .....         | 232 |
| NL5_GetTraceValue .....    | 233 |
| NL5_SetTrace .....         | 233 |
| NL5_ShowTran .....         | 234 |
| NL5_SetScreen .....        | 234 |
| NL5_Cursors .....          | 234 |
| NL5_Start .....            | 235 |
| NL5_Pause .....            | 235 |
| NL5_Continue .....         | 235 |
| NL5_Stop .....             | 235 |
| NL5_SaveIC .....           | 235 |
| NL5_SaveAsWithIC .....     | 235 |
| NL5_IsRunning .....        | 236 |
| NL5_GetDataSize .....      | 236 |
| NL5_GetDataAt .....        | 236 |
| NL5_GetData .....          | 237 |
| NL5_AddData .....          | 237 |
| NL5_Export .....           | 238 |
| NL5_GetStorageSize .....   | 238 |
| NL5_Store .....            | 238 |
| NL5_DeleteStorage .....    | 238 |
| NL5_GetACTracesSize .....  | 239 |
| NL5_AddACTrace .....       | 239 |
| NL5_DeleteACTrace .....    | 239 |
| NL5_GetACTrace .....       | 239 |
| NL5_GetACTraceValue .....  | 240 |
| NL5_SetACTrace .....       | 240 |
| NL5_ShowAC .....           | 241 |
| NL5_SetACScreen .....      | 241 |
| NL5_ACCursors .....        | 241 |
| NL5_StartAC .....          | 242 |
| NL5_StopAC .....           | 242 |
| NL5_GetACDataSize .....    | 242 |
| NL5_GetACDataAt .....      | 243 |
| NL5_GetACStorageSize ..... | 243 |
| NL5_StoreAC .....          | 243 |
| NL5_DeleteStorage .....    | 243 |

# Components

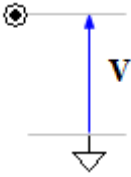
Logical levels and threshold for all components are defined in *Schematic/Settings and tools* window, *Logical levels* tab.



## SubCir model

| Model                                                          | Parameter   | Units | Description                                 |
|----------------------------------------------------------------|-------------|-------|---------------------------------------------|
| SubCir                                                         | <b>File</b> |       | File name of subcircuit schematic.          |
|                                                                | <b>Pin1</b> |       | Name of subcircuit label connected to pin 1 |
|                                                                | <b>...</b>  |       | <b>...</b>                                  |
|                                                                | <b>PinN</b> |       | Name of subcircuit label connected to pin N |
|                                                                | <b>Cmd</b>  |       | Subcircuit start-up command string          |
|                                                                | <b>IC</b>   |       | Subcircuit Initial conditions string        |
| See <i>Subcircuit model</i> chapter of NL5 Manual for details. |             |       |                                             |

## Label

| Symbol                                                                            | Models                                |                                           |                               | Signals                                                                             |
|-----------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------|
|  | Label<br>V<br>Step<br>Single<br>Pulse | Clock<br>Sin<br>Sweep<br>Function<br>List | File<br>Trace<br>IC<br>SubCir |  |

All models (except SubCir) can be used:

- As a voltage trace probe point.
- For connecting schematic points without wires, including points at different sheets.

| Model        | Parameter      | Units | Description |
|--------------|----------------|-------|-------------|
| <b>Label</b> | No parameters. |       |             |

Label can be used:

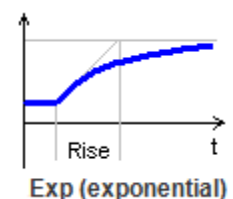
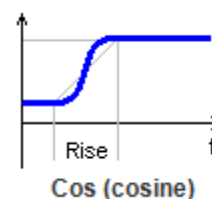
- As a voltage trace probe point.
- For connecting schematic points without wires, including points at different sheets.

| Model    | Parameter | Units | Description |
|----------|-----------|-------|-------------|
| <b>V</b> | <b>V</b>  | V     | Voltage.    |

Constant voltage = **V**.

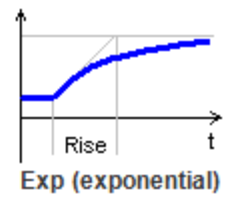
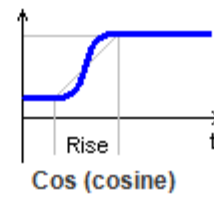
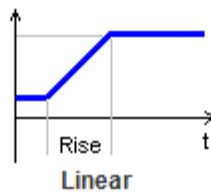
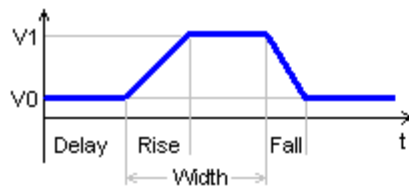
| Model       | Parameter    | Units | Description                |
|-------------|--------------|-------|----------------------------|
| <b>Step</b> | <b>V1</b>    | V     | Step On voltage.           |
|             | <b>V0</b>    | V     | Step Off voltage.          |
|             | <b>Slope</b> |       | Slope type: Linear/Cos/Exp |
|             | <b>Rise</b>  | s     | Step rise length.          |
|             | <b>Delay</b> | s     | Delay before step starts.  |

Step starts after **Delay** time. If **Rise** is non-zero, 3 **Slope** types are available.



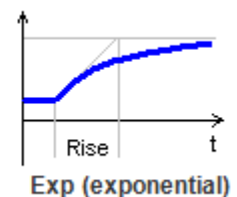
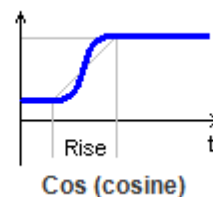
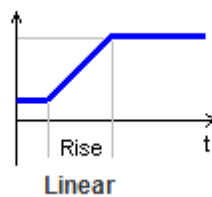
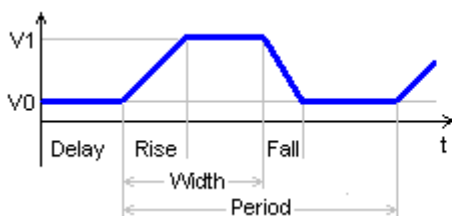
| Model         | Parameter    | Units | Description                |
|---------------|--------------|-------|----------------------------|
| <b>Single</b> | <b>V1</b>    | V     | Pulse On voltage.          |
|               | <b>V0</b>    | V     | Pulse Off voltage.         |
|               | <b>Width</b> | s     | Pulse width.               |
|               | <b>Slope</b> |       | Slope type: Linear/Cos/Exp |
|               | <b>Rise</b>  | s     | Pulse rise length.         |
|               | <b>Fall</b>  | s     | Pulse fall length.         |
|               | <b>Delay</b> | s     | Delay before pulse starts. |

Single pulse starts after **Delay** time. **Rise** time is included into **Width**, **Fall** time is **not** included into **Width**. **Slope** type applies both to pulse rise and fall (if non-zero).



| Model        | Parameter     | Units | Description                      |
|--------------|---------------|-------|----------------------------------|
| <b>Pulse</b> | <b>V1</b>     | V     | Pulse On voltage.                |
|              | <b>V0</b>     | V     | Pulse Off voltage.               |
|              | <b>Period</b> | s     | Period.                          |
|              | <b>Width</b>  | s     | Pulse width.                     |
|              | <b>Slope</b>  |       | Slope type: Linear/Cos/Exp       |
|              | <b>Rise</b>   | s     | Pulse rise length.               |
|              | <b>Fall</b>   | s     | Pulse fall length.               |
|              | <b>Delay</b>  | s     | Delay before first pulse starts. |

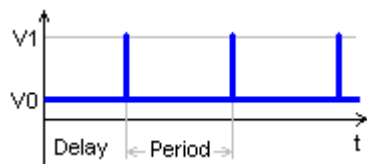
Pulses start after **Delay** time. **Rise** time is included into **Width**, **Fall** time is **not** included into **Width**. **Slope** type applies both to pulse rise and fall (if non-zero).



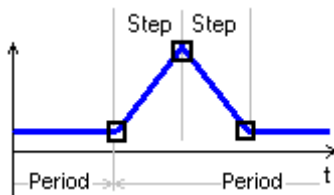
| Model        | Parameter     | Units | Description                       |
|--------------|---------------|-------|-----------------------------------|
| <b>Clock</b> | <b>V1</b>     | V     | Pulse On voltage.                 |
|              | <b>V0</b>     | V     | Pulse Off voltage.                |
|              | <b>Period</b> | s     | Period.                           |
|              | <b>Step</b>   | s     | Simulation step of rise and fall. |
|              | <b>Delay</b>  | s     | Delay before first pulse starts.  |

Periodic pulses with width of one simulation step. Pulses start after **Delay** time.

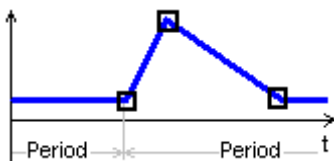
**Clock** model is recommended to produce a constant frequency clock signal for C-code, DLL, logical components, etc. Unlike **Pulse** model, it won't force unnecessary step reduction at the end of the pulse, which may help to accelerate simulation.



If **Step** parameter is not zero **and** is less than current schematic simulation step, the step is adjusted to provide rise and fall of clock pulse to be equal to **Step**:

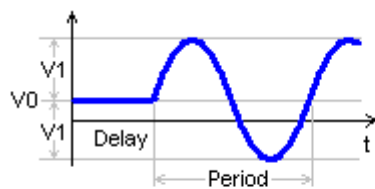


Otherwise, the clock pulse is created using current schematic simulation step, which depends on many factors, and cannot be easily predicted:

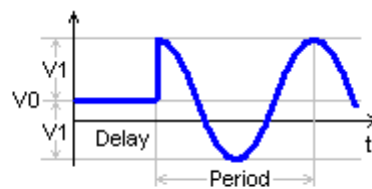


| Model      | Parameter     | Units | Description                      |
|------------|---------------|-------|----------------------------------|
| <b>Sin</b> | <b>V1</b>     | V     | Voltage amplitude.               |
|            | <b>V0</b>     | V     | Voltage baseline.                |
|            | <b>Period</b> | s     | Period.                          |
|            | <b>Phase</b>  | deg   | Phase.                           |
|            | <b>Decay</b>  | 1/s   | Decay constant                   |
|            | <b>Delay</b>  | s     | Delay before sine signal starts. |

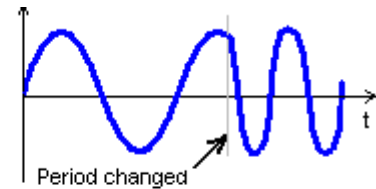
Sinusoidal signal starts after **Delay** time. **Phase** is sine phase in degrees at the moment when signal starts. If transient is paused, sine period changed, and then transient is continued, the phase of the signal remains continuous, providing smooth sine signal of variable frequency. If **Decay** is not zero, the sine signal is exponentially dumped with time constant =  $1/\text{Decay}$ .



Phase = 0



Phase = 90



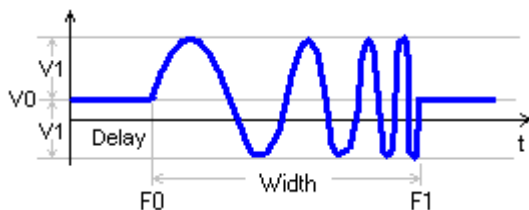
| Model        | Parameter    | Units | Description                 |
|--------------|--------------|-------|-----------------------------|
| <b>Sweep</b> | <b>V1</b>    | V     | Voltage amplitude.          |
|              | <b>V0</b>    | V     | Voltage baseline.           |
|              | <b>Width</b> | s     | Width of the signal.        |
|              | <b>F0</b>    | Hz    | Start frequency.            |
|              | <b>F1</b>    | Hz    | End frequency.              |
|              | <b>Type</b>  |       | Signal type: Linear/Exp.    |
|              | <b>Delay</b> | s     | Delay before signal starts. |

Sinusoidal signal with variable frequency starts after **Delay** time. Signal frequency changes during **Width** interval from **F0** to **F1** linearly or exponentially, depending on specified **Type**.

If **F0** = **F1**, then one period of frequency  $1/\text{Width}$  will be generated.

If lowest frequency is set to zero and **Type** = Exp, then lowest frequency  $0.01/\text{Width}$  will be used.

If needed, the highest frequency will be increased to provide integer number of signal periods, so that signal phase at the beginning and at the end of **Width** interval is exactly zero.



| Model           | Parameter | Units | Description |
|-----------------|-----------|-------|-------------|
| <b>Function</b> | <b>f</b>  | V     | Function    |

Arbitrary function **f** defines voltage as a function of the following variables:

- t** - current time
- V(name)** - voltage on the component **name**
- I(name)** - current through the component **name**
- P(name)** – power on the component **name**
- S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, voltage is zero.

Example:

```
f = sin(t) * (1+cos(t*.01))
f = V(R1) * I(R1)
```

Please note that **V**, **I**, **P**, and **S** variables are taken at previous simulation step. This may affect stability of the schematic with closed loop.

| Model       | Parameter    | Units | Description               |
|-------------|--------------|-------|---------------------------|
| <b>List</b> | <b>List</b>  |       | Comma-separated string.   |
|             | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|             | <b>Delay</b> | s     | Delay.                    |

Piecewise linear signal is defined by **List** parameter in the **csv** (comma-separated values) format, as follows:

```
t0,V0,t1,V1,...,tn,Vn
```

where all **t** and **V** can be numerical values or expressions.

If  $t < t_0$ , signal is  $V_0$ .

If  $t_0 < t < t_1$ , signal value is linearly interpolated between  $V_0$  and  $V_1$ , etc.

If  $t > t_n$ , and **Cycle** parameter is set to **No**, the signal value is  $V_n$ . Otherwise the signal defined in  $t_0 \dots t_n$  interval is repeated continuously.

Signal start is delayed by **Delay** time.

Example:

```
List = 0,0,1,2,4,3,5,0,8,0
```

If **Cycle = Yes**, **Delay = 0**, the following voltage will be generated:

See *List model* chapter of NL5 Manual for more details.

| Model       | Parameter    | Units | Description               |
|-------------|--------------|-------|---------------------------|
| <b>File</b> | <b>File</b>  |       | File name.                |
|             | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|             | <b>Delay</b> | s     | Delay.                    |

Piecewise linear signal is defined in the text file. If **File** parameter does not have a full path, NL5 will search for the file in the directory where schematic file is located, then in the Library directories, and then in the Library directories relative to schematic file directory (see NL5 Manual, *Schematic Properties*).

Signal is defined in the **csv** (comma-separated values) format, as follows:

```
<if first line does not start with a number, it is ignored >
t0,V0
t1,V1
.....
tn,Vn
```

where all t and V can be numerical values or expressions.

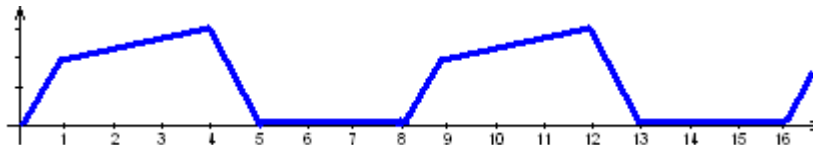
If  $t < t_0$ , signal is V0. If  $t_0 < t < t_1$ , signal value is linearly interpolated between V0 and V1, etc. If  $t > t_n$ , and **Cycle** parameter is set to **No**, the signal value is Vn. otherwise the signal defined in  $t_0 \dots t_n$  interval is repeated continuously.

Signal start is delayed by **Delay** time.

Example. File content:

```
0,0
1,2
4,3
5,0
8,0
```

If **Cycle = Yes**, **Delay = 0**, the following voltage will be generated:



| Model        | Parameter    | Units | Description               |
|--------------|--------------|-------|---------------------------|
| <b>Trace</b> | <b>Trace</b> |       | Trace name.               |
|              | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|              | <b>Delay</b> | s     | Delay.                    |

Signal is defined by an existing trace of **Data** type with “static” data: entered manually, duplicated, imported, pasted from the clipboard, etc. Signal value at required simulation point is obtained by linear interpolation between existing data points of the trace.

**Trace** is a name of the existing trace shown in the Trace list.

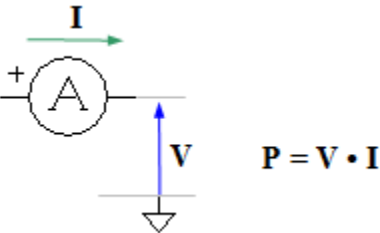
If **Cycle** parameter is set to **Yes**, the signal is repeated continuously.

Signal start is delayed by **Delay** time.

| Model                                                                                                                                                                                                                                                                                                                                                                                | Parameter | Units | Description         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|---------------------|
| <b>IC</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>V</b>  | V     | Initial voltage.    |
|                                                                                                                                                                                                                                                                                                                                                                                      | <b>R</b>  | Ohm   | Initial resistance. |
| <p>Initial condition. The model is used to apply initial voltage during DC operating point calculation. When calculating DC operating point, the temporary voltage source <b>V</b> is connected to the label through initial resistor <b>R</b>. When DC operating point is found, the voltage source is removed, and the <b>IC</b> model operates similar to <b>Label</b> model.</p> |           |       |                     |

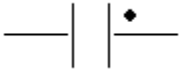
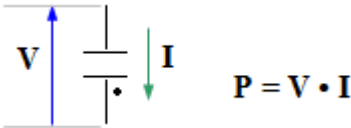


## A – Amperemeter

| Symbol                                                                            | Models      | Signals                                                                             |
|-----------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|
|  | Amperemeter |  |

| Model                                                                                                                            | Parameter      | Units | Description |
|----------------------------------------------------------------------------------------------------------------------------------|----------------|-------|-------------|
| Amperemeter                                                                                                                      | No parameters. |       |             |
| Short circuit. In addition to current, amperemeter can measure voltage relative to ground, and power delivered to grounded load. |                |       |             |

## C – Capacitor

| Symbol                                                                            | Models             | Signals                                                                             |
|-----------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|
|  | C<br>PWL<br>SubCir |  |

| Model    | Parameter | Units | Description                                                |
|----------|-----------|-------|------------------------------------------------------------|
| <b>C</b> | <b>C</b>  | F     | Capacitance                                                |
|          | <b>IC</b> | V     | Initial condition: voltage. Leave blank if IC not defined. |

Linear capacitor:  $I = C \cdot dV/dt$ .

When calculating DC operating point, if **IC** is not blank, capacitor is replaced with voltage source equal to **IC**. Otherwise, capacitor is temporarily removed (open circuit), DC operating point is calculated, and then the voltage found across the capacitor is assigned to the capacitor as its initial voltage.

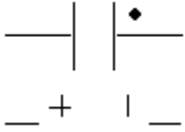
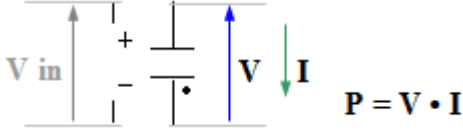
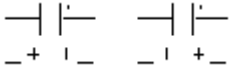
| Model      | Parameter  | Units | Description                                                |
|------------|------------|-------|------------------------------------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, C(V)                               |
|            | <b>IC</b>  | V     | Initial condition: voltage. Leave blank if IC not defined. |

Piecewise constant capacitor: **pwl** string defines capacitance as a function of voltage across the capacitor C(V). Capacitor charge Q(V) is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that C(V) is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

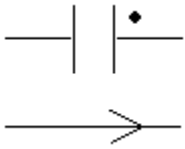
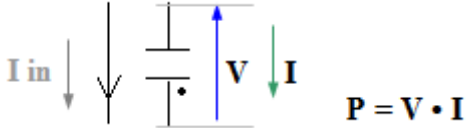
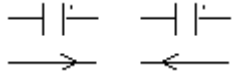
When calculating DC operating point, if **IC** is not blank, capacitor is replaced with voltage source equal to **IC**. Otherwise, capacitor is temporarily removed (open circuit), DC operating point is calculated, and then the voltage found across the capacitor is assigned to the capacitor as its initial voltage.

C – Voltage controlled capacitor

| Symbol                                                                                     | Models | Signals                                                                            |
|--------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------|
|           | PWL    |  |
| Views<br> |        |                                                                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Parameter | Units | Description                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------------------------------|
| PWL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | pwl       |       | Comma-separated string, C(Vin)                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | IC        | V     | Initial condition: voltage. Leave blank if IC not defined. |
| <p>Piecewise constant voltage controlled capacitor. <b>pwl</b> string defines capacitance as a function of control voltage C(Vin). At any moment:</p> $I = C(Vin) * dV/dt.$ <p>See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that C(Vin) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> <p>When calculating DC operating point, if <b>IC</b> is not blank, capacitor is replaced with voltage source equal to <b>IC</b>. Otherwise, the capacitor is temporarily removed (open circuit), DC operating point calculated, and then the voltage found across the capacitor is assigned to the capacitor as its initial voltage.</p> |           |       |                                                            |

## C – Current controlled capacitor

| Symbol                                                                            | Models                                                                            | Signals                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | PWL                                                                               |  |
| Views                                                                             |  |                                                                                    |

| Model | Parameter | Units | Description                                                |
|-------|-----------|-------|------------------------------------------------------------|
| PWL   | pwl       |       | Comma-separated string, C(lin)                             |
|       | IC        | V     | Initial condition: voltage. Leave blank if IC not defined. |

Piecewise constant current controlled capacitor. **pwl** string defines capacitance as a function of control current C(lin). At any moment:

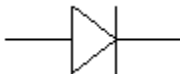
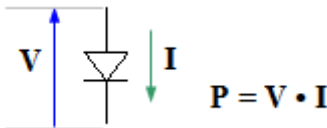
$$I = C(lin) * dV/dt.$$

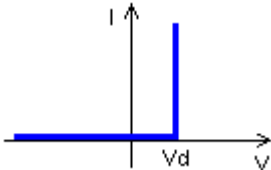
See *PWL model* chapter of NL5 Manual for details.

Please note that C(lin) is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

When calculating DC operating point, if **IC** is not blank, capacitor is replaced with voltage source equal to **IC**. Otherwise, the capacitor is temporarily removed (open circuit), DC operating point calculated, and then the voltage found across the capacitor is assigned to the capacitor as its initial voltage

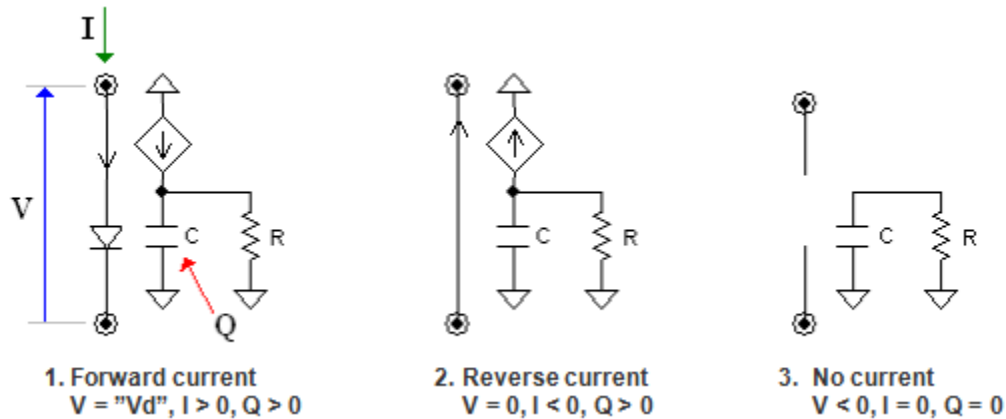
D – Diode

| Symbol                                                                            | Models                                    | Signals                                                                             |
|-----------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------|
|  | Diode<br>Storage<br>Soft<br>PWL<br>SubCir |  |

| Model                                                                                                                                                                                                                                                                                                                | Parameter | Units | Description                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|----------------------------|
| Diode                                                                                                                                                                                                                                                                                                                | Vd        | V     | Forward voltage drop.      |
|                                                                                                                                                                                                                                                                                                                      | IC        |       | Initial condition: On/Off. |
| <p>Ideal diode. If <math>V \geq Vd</math>, diode is On (short circuit). Otherwise diode is Off (open circuit, <math>I=0</math>).</p> <p>When calculating DC operating point diode is set to the state specified in <b>IC</b>.</p>  |           |       |                            |

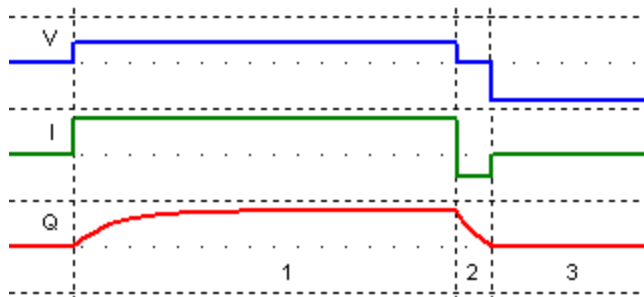
| Model          | Parameter  | Units   | Description                  |
|----------------|------------|---------|------------------------------|
| <b>Storage</b> | <b>Vd</b>  | V       | Forward voltage drop.        |
|                | <b>t</b>   | s       | Recombination time constant. |
|                | <b>IC</b>  |         | Initial condition: Off/On.   |
|                | <b>ICQ</b> | C (A*s) | Initial condition: charge.   |

Charge storage diode. Simplified equivalent schematic of the model is the following:



The diode has internal capacitor  $C$  and resistor  $R$ , with the time constant  $RC = t$ ,  $Q$  is the charge on the capacitor.

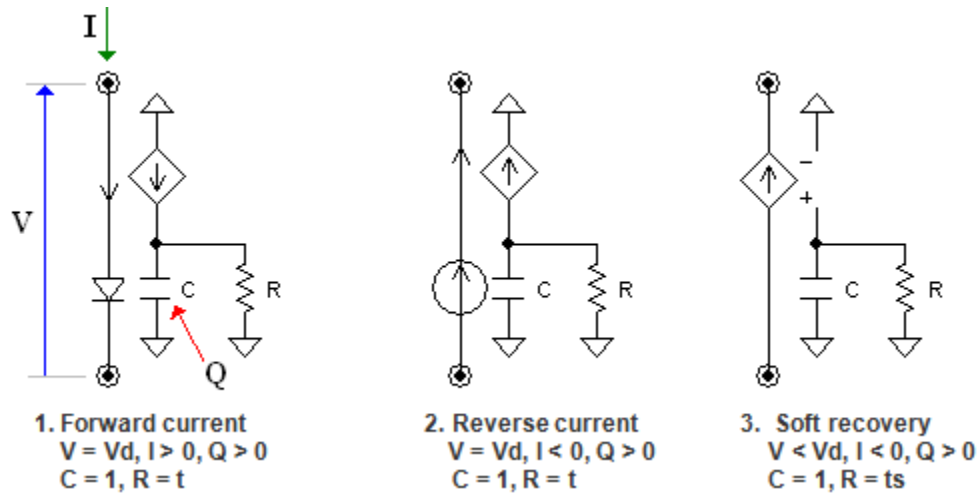
In **mode 1**, forward current flows through the diode and forward voltage drop is **Vd**. At the same time, the current equal to forward current is charging capacitor  $C$ . In **mode 2**, reverse current is applied to the diode, and capacitor  $C$  is being discharged by the current equal to reverse current. As long as charge  $Q$  on the capacitor is positive, the diode is a short circuit with zero voltage drop. Finally, when charge drops to zero, the diode switches to **mode 3**, with zero current and negative voltage drop (open circuit).



When calculating DC operating point the diode is set to the state specified in **IC**, and internal charge  $Q$  is set to specified **ICQ** value.

| Model       | Parameter  | Units   | Description                  |
|-------------|------------|---------|------------------------------|
| <b>Soft</b> | <b>Vd</b>  | V       | Forward voltage drop.        |
|             | <b>t</b>   | s       | Recombination time constant. |
|             | <b>ts</b>  | s       | Soft recovery time constant. |
|             | <b>IC</b>  |         | Initial condition: Off/On.   |
|             | <b>ICQ</b> | C (A*s) | Initial condition: charge.   |

Soft recovery charge storage diode. Simplified equivalent schematic of the model is the following:



The diode has internal capacitor  $C=1$  and resistor  $R$ . Time constant  $RC$  is equal either recombination time constant  $t$ , or soft recovery time constant  $ts$ .  $Q$  is the charge on the capacitor. In **mode 1**, forward current flows through the diode and forward voltage drop is  $V_d$ . At the same time, the current equal to forward current is charging capacitor  $C$ . In **mode 2**, reverse current is applied to the diode, and capacitor  $C$  is being discharged by the current equal to reverse current. Voltage drop on the diode is still  $V_d$ . At the moment when reverse current is equal or less than charge divided by soft recovery time constant  $ts$ , a **mode 3** is turned on. In **mode 3**, capacitor  $C$  is being exponentially discharged by the current through resistor  $R$  with time constant  $ts$  (plus small constant current to ensure full discharge - not shown on the picture). Reverse diode current is proportional to the charge. As soon as charge drops to zero, the diode switches to **mode 4** (not shown), with zero current and negative voltage drop (open circuit).

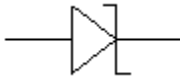
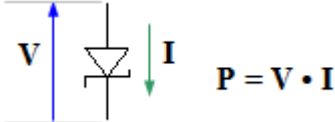
When calculating DC operating point the diode is set to the state specified in **IC**, and internal charge  $Q$  is set to specified **ICQ** value.

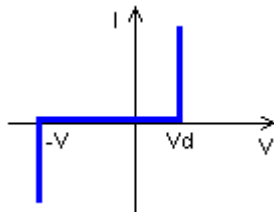
| Model      | Parameter  | Units | Description                    |
|------------|------------|-------|--------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, $R(V)$ |

Piecewise linear diode. **pwl** string defines resistance as a function of voltage across the diode  $R(V)$ . Volt-ampere characteristic of the diode is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that  $R(V)$  is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

## D – Zener

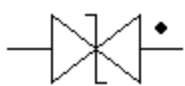
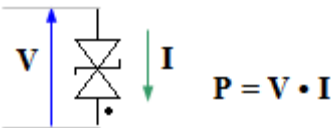
| Symbol                                                                            | Models                 | Signals                                                                             |
|-----------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------|
|  | Zener<br>PWL<br>SubCir |  |

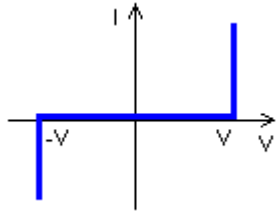
| Model                                                                                                                                                                                                                                                                                                                                                                      | Parameter | Units | Description                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Zener</b>                                                                                                                                                                                                                                                                                                                                                               | <b>V</b>  | V     | Breakdown voltage drop.            |
|                                                                                                                                                                                                                                                                                                                                                                            | <b>Vd</b> | V     | Forward voltage drop.              |
|                                                                                                                                                                                                                                                                                                                                                                            | <b>IC</b> |       | Initial condition: Minus/Off/Plus. |
| <p>Ideal zener. If <math>V</math> (across zener) <math>\leq -V</math> or <math>V \geq Vd</math>, zener is On (short circuit). Otherwise zener is Off (open circuit, <math>I=0</math>).</p> <p>When calculating DC operating point zener is set to the state specified in <b>IC</b>.</p>  |           |       |                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Parameter  | Units | Description                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|--------------------------------|
| <b>PWL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>pwl</b> |       | Comma-separated string, $R(V)$ |
| <p>Piecewise linear zener. <b>pwl</b> string defines resistance as a function of voltage across zener <math>R(V)</math>. Volt-ampere characteristic of zener is piecewise linear function, and it always goes through the origin (0,0). See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that <math>R(V)</math> is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |            |       |                                |



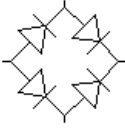
D – Bidirectional zener

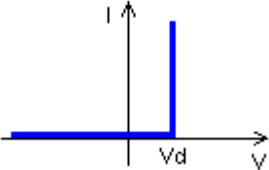
| Symbol                                                                            | Models                 | Signals                                                                             |
|-----------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------|
|  | Zener<br>PWL<br>SubCir |  |

| Model                                                                                                                                                                                                                                                                                                                                                               | Parameter | Units | Description                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| Zener                                                                                                                                                                                                                                                                                                                                                               | V         | V     | Breakdown voltage drop.            |
|                                                                                                                                                                                                                                                                                                                                                                     | IC        |       | Initial condition: Minus/Off/Plus. |
| <p>Ideal bidirectional zener. If V (across zener) <math>\leq -V</math> or <math>V \geq V</math>, zener is On (short circuit). Otherwise zener is Off (open circuit, <math>I=0</math>).</p> <p>When calculating DC operating point zener is set to the state specified in IC.</p>  |           |       |                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                          | Parameter | Units | Description                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------|
| PWL                                                                                                                                                                                                                                                                                                                                                                                                                                            | pwl       |       | Comma-separated string, R(V) |
| <p>Piecewise linear zener. <b>pwl</b> string defines resistance as a function of voltage across zener R(V). Volt-ampere characteristic of zener is piecewise linear function, and it always goes through the origin (0,0). See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that R(V) is <b>piecewise constant</b> function, although he model and parameter are still called <b>pwl</b> for historical reasons.</p> |           |       |                              |

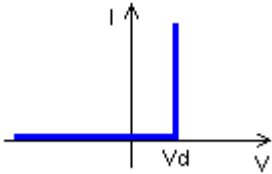
D – Bridge rectifier

| Symbol                                                                            | Models | Signals                                                             |
|-----------------------------------------------------------------------------------|--------|---------------------------------------------------------------------|
|  | Diode  | $P = V_1 \cdot I_1 + V_2 \cdot I_2 + V_3 \cdot I_3 + V_4 \cdot I_4$ |

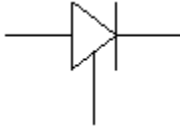
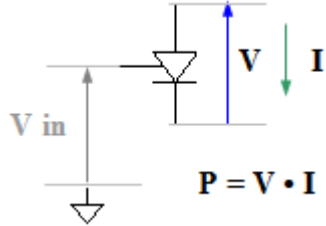
| Model                                                                                                                                                                                                                                                                                                                        | Parameter | Units | Description           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-----------------------|
| Diode                                                                                                                                                                                                                                                                                                                        | Vd        | V     | Forward voltage drop. |
| <p>Bridge rectifier with ideal diodes. For each diode, if <math>V \geq Vd</math>, diode is On (short circuit). Otherwise diode is Off (open circuit, <math>I=0</math>).</p> <p>When calculating DC operating point all diodes are Off.</p>  |           |       |                       |

D – Diode ring

| Symbol                                                                            | Models | Signals                                                             |
|-----------------------------------------------------------------------------------|--------|---------------------------------------------------------------------|
|  | Diode  | $P = V_1 \cdot I_1 + V_2 \cdot I_2 + V_3 \cdot I_3 + V_4 \cdot I_4$ |

| Model                                                                                                                                                                                                                                                                                                                  | Parameter | Units | Description           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-----------------------|
| Diode                                                                                                                                                                                                                                                                                                                  | Vd        | V     | Forward voltage drop. |
| <p>Diode ring with ideal diodes. For each diode, if <math>V \geq Vd</math>, diode is On (short circuit). Otherwise diode is Off (open circuit, <math>I=0</math>).</p> <p>When calculating DC operating point all diodes are Off.</p>  |           |       |                       |

## D – Logic controlled thyristor

| Symbol                                                                            | Models              | Signals                                                                             |
|-----------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|
|  | Thyristor<br>SubCir |  |

| Model            | Parameter    | Units | Description                |
|------------------|--------------|-------|----------------------------|
| <b>Thyristor</b> | <b>Vd</b>    | V     | Forward voltage drop.      |
|                  | <b>Ihold</b> | A     | Holding current.           |
|                  | <b>IC</b>    |       | Initial condition: Off/On. |

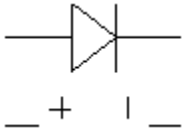
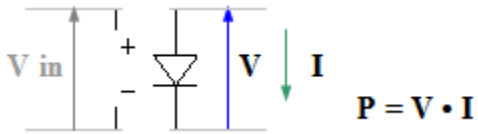
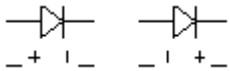
Thyristor has two states:

- Off state (non-conducting): open circuit.
- On state (conducting): ideal diode with **Vd** forward voltage drop.

If control voltage  $V_{in}$  is greater than logical threshold, thyristor is in On state (ideal diode). When control voltage drops below logical threshold, thyristor stays in On state as long as current  $I$  exceeds holding current **Ihold**, and voltage  $V$  is not negative.

When calculating DC operating point thyristor is set to the state specified in **IC**.

D – Voltage controlled thyristor

| Symbol                                                                            | Models                                                                            | Signals                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | Thyristor<br>SubCir                                                               |  |
| Views                                                                             |  |                                                                                    |

| Model     | Parameter | Units | Description                |
|-----------|-----------|-------|----------------------------|
| Thyristor | Vd        | V     | Forward voltage drop.      |
|           | Ihold     | A     | Holding current.           |
|           | Threshold | V     | Voltage threshold.         |
|           | IC        |       | Initial condition: Off/On. |

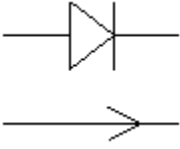
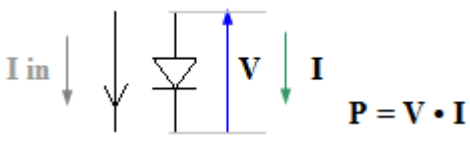
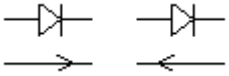
Thyristor has two states:

- Off state (non-conducting): open circuit.
- On state (conducting): ideal diode with **Vd** forward voltage drop.

If control voltage **Vin** is greater than **Threshold**, thyristor is in On state (ideal diode). When control voltage drops below **Threshold**, thyristor stays in On state as long as current **I** exceeds holding current **Ihold**, and voltage **V** is not negative.

When calculating DC operating point thyristor is set to the state specified in **IC**.

## D – Current controlled thyristor

| Symbol                                                                            | Models                                                                            | Signals                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | Thyristor<br>SubCir                                                               |  |
| Views                                                                             |  |                                                                                    |

| Model            | Parameter        | Units | Description                |
|------------------|------------------|-------|----------------------------|
| <b>Thyristor</b> | <b>Vd</b>        | V     | Forward voltage drop.      |
|                  | <b>Ihold</b>     | A     | Holding current.           |
|                  | <b>Threshold</b> | A     | Current threshold.         |
|                  | <b>IC</b>        |       | Initial condition: Off/On. |

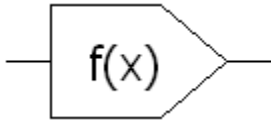
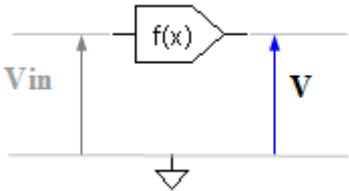
Thyristor has two states:

- Off state (non-conducting): open circuit.
- On state (conducting): ideal diode with **Vd** forward voltage drop.

If control current **Iin** is greater than **Threshold**, thyristor is in On state (ideal diode). When control current drops below **Threshold**, thyristor stays in On state as long as current **I** exceeds holding current **Ihold**, and voltage **V** is not negative.

When calculating DC operating point thyristor is set to the state specified in **IC**.

## F – Function

| Symbol                                                                              | Models                        | Signals                |
|-------------------------------------------------------------------------------------|-------------------------------|------------------------|
|    | Function<br>Pwr<br>Abs<br>Int | Lim<br>Table<br>SubCir |
|  |                               |                        |

The function is calculated and applied to the output on every calculation step.

For all models, when calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Please note that output voltage is always delayed by one calculation step. This may affect stability of the schematic with closed loop.

| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | V     | Output as function of the input.   |
|                 | <b>IC</b> | V     | Initial condition: output voltage. |

Arbitrary function **f** defines output voltage as a function of the following variables:

**x** – input voltage *Vin*

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Examples:

$$f = x * x$$

$$f = x * \sin(t)$$

$$f = P(r1) + P(r2)$$

| Model      | Parameter    | Units | Description                        |
|------------|--------------|-------|------------------------------------|
| <b>Pwr</b> | <b>power</b> |       | Power.                             |
|            | <b>K</b>     | V/V   | Gain.                              |
|            | <b>IC</b>    | V     | Initial condition: output voltage. |

“Signed” power function:  $V = K * \text{pwr}(Vin, \text{power})$ .

The function is calculated as follows:

if **power** = 0:

- if  $Vin < 0$  . . . :  $V = -K$
- if  $Vin = 0$  . . . :  $V = 0$
- if  $Vin > 0$  . . . :  $V = K$

if **power**  $\neq 0$ :

- if  $Vin < 0$  . . . :  $V = -K * (-Vin)^{\text{power}}$
- if  $Vin = 0$  . . . :  $V = 0$
- if  $Vin > 0$  . . . :  $V = K * Vin^{\text{power}}$

| Model      | Parameter | Units | Description                        |
|------------|-----------|-------|------------------------------------|
| <b>Abs</b> | <b>K</b>  | V/V   | Gain.                              |
|            | <b>IC</b> | V     | Initial condition: output voltage. |

Absolute value:  $V = K * \text{abs}(Vin)$ .

| Model      | Parameter         | Units | Description                        |
|------------|-------------------|-------|------------------------------------|
| <b>Int</b> | <b>resolution</b> | V     | Resolution.                        |
|            | <b>K</b>          | V/V   | Gain.                              |
|            | <b>IC</b>         | V     | Initial condition: output voltage. |

Rounding function:  $V = K * \text{round}(Vin, \text{resolution})$ .

Round to the nearest multiple of **resolution**. If **resolution** = 1, round to the nearest integer.

| Model      | Parameter  | Units | Description                        |
|------------|------------|-------|------------------------------------|
| <b>Lim</b> | <b>Max</b> | V     | Maximum.                           |
|            | <b>Min</b> | V     | Minimum.                           |
|            | <b>IC</b>  | V     | Initial condition: output voltage. |

Limiting function is calculated as follows:

- if  $Vin < \text{Min}$  . . . :  $V = \text{Min}$
- if  $Vin > \text{Max}$  . . . :  $V = \text{Max}$
- Otherwise . . . :  $V = Vin$



| Model        | Parameter    | Units | Description                                    |
|--------------|--------------|-------|------------------------------------------------|
| <b>Table</b> | <b>Table</b> |       | Comma-separated string, <i>Vin/Vout</i> pairs. |
|              | <b>IC</b>    | V     | Initial condition: output voltage.             |

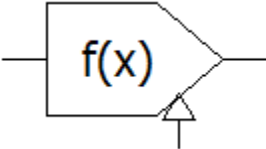
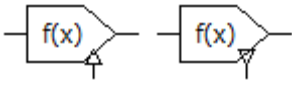
Look-up table. Function output is defined by **Table** parameter in the **csv** (comma separated values) format, as follows:

X1,Y1,X2,Y2,...,XN,YN

where Xi,Yi pair defines input value (X) and output value (Y). Output value between specified points is linearly interpolated. Output value below X1 is linearly extrapolated using X1...X2 interval data, output value above XN is linearly extrapolated using X(N-1)...XN interval data. Values X1...XN should be given in an ascending order.

See *Table model* chapter of NL5 Manual for more details.

## F – Function with clock

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Models                        | Signals                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Function<br>Pwr<br>Abs<br>Int | Lim<br>Table<br>SubCir |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                               |                        |
| <p>Function component with <b>clock</b> operates in <b>synchronized</b> mode: the function is calculated and applied to the output only on rising (or falling) edge of logical clock signal. As a result, this mode may provide faster simulation.</p> <p>For all models, when calculating DC operating point, and in AC analysis, output is set to specified output voltage <b>IC</b>. Please note that output voltage is always delayed by one calculation step. This may affect stability of the schematic with closed loop.</p> |                               |                        |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Parameter | Units | Description                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Function</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>f</b>  | V     | Output as function of the input.   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Arbitrary function <b>f</b> defines output voltage as a function of the following variables:</p> <p><b>x</b> – input voltage <i>Vin</i><br/> <b>t</b> - current time<br/> <b>V(name)</b> - voltage on the component <i>name</i><br/> <b>I(name)</b> - current through the component <i>name</i><br/> <b>P(name)</b> – power on the component <i>name</i><br/> <b>S(name)</b> – state of the component <i>name</i></p> <p>where <i>name</i> is the name of the component in the schematic. If <b>f</b> is blank, output is zero.</p> <p>Examples:</p> <p><math>f = x * x</math><br/> <math>f = x * \sin(t)</math><br/> <math>f = P(r1) + P(r2)</math></p> |           |       |                                    |

| Model      | Parameter    | Units | Description                        |
|------------|--------------|-------|------------------------------------|
| <b>Pwr</b> | <b>power</b> |       | Power.                             |
|            | <b>K</b>     | V/V   | Gain.                              |
|            | <b>IC</b>    | V     | Initial condition: output voltage. |

“Signed” power function:  $V = K * \text{pwr}(Vin, \text{power})$ .

The function is calculated as follows:

if **power** = 0:

- if  $Vin < 0$  . . . :  $V = -K$
- if  $Vin = 0$  . . . :  $V = 0$
- if  $Vin > 0$  . . . :  $V = K$

if **power**  $\neq 0$ :

- if  $Vin < 0$  . . . :  $V = -K * (-Vin)^{\text{power}}$
- if  $Vin = 0$  . . . :  $V = 0$
- if  $Vin > 0$  . . . :  $V = K * Vin^{\text{power}}$

| Model      | Parameter | Units | Description                        |
|------------|-----------|-------|------------------------------------|
| <b>Abs</b> | <b>K</b>  | V/V   | Gain.                              |
|            | <b>IC</b> | V     | Initial condition: output voltage. |

Absolute value:  $V = K * \text{abs}(Vin)$ .

| Model      | Parameter         | Units | Description                        |
|------------|-------------------|-------|------------------------------------|
| <b>Int</b> | <b>resolution</b> | V     | Resolution.                        |
|            | <b>K</b>          | V/V   | Gain.                              |
|            | <b>IC</b>         | V     | Initial condition: output voltage. |

Rounding function:  $V = K * \text{round}(Vin, \text{resolution})$ .

Round to the nearest multiple of **resolution**. If **resolution** = 1, round to the nearest integer.

| Model      | Parameter  | Units | Description                        |
|------------|------------|-------|------------------------------------|
| <b>Lim</b> | <b>Max</b> | V     | Maximum.                           |
|            | <b>Min</b> | V     | Minimum.                           |
|            | <b>IC</b>  | V     | Initial condition: output voltage. |

Limiting function is calculated as follows:

- if  $Vin < \text{Min}$  . . . :  $V = \text{Min}$
- if  $Vin > \text{Max}$  . . . :  $V = \text{Max}$
- Otherwise . . . :  $V = Vin$

| Model        | Parameter    | Units | Description                                    |
|--------------|--------------|-------|------------------------------------------------|
| <b>Table</b> | <b>Table</b> |       | Comma-separated string, <i>Vin/Vout</i> pairs. |
|              | <b>IC</b>    | V     | Initial condition: output voltage.             |

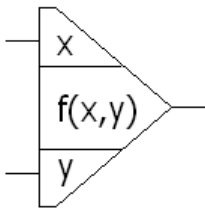
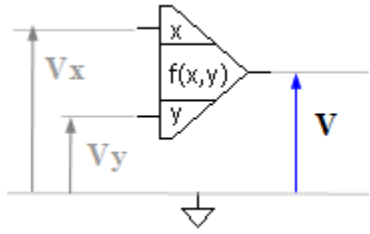
Look-up table. Function output is defined by **Table** parameter in the **csv** (comma separated values) format, as follows:

X1,Y1,X2,Y2,...,XN,YN

where Xi,Yi pair defines input value (X) and output value (Y). Output value between specified points is linearly interpolated. Output value below X1 is linearly extrapolated using X1...X2 interval data, output value above XN is linearly extrapolated using X(N-1)...XN interval data. Values X1...XN should be given in ascending order.

See *Table model* chapter of NL5 Manual for more details.

## F – Function-2

| Symbol                                                                            | Models                                             |                                                    | Signals                                                                             |
|-----------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Function<br>Mul<br>Div<br>Sum<br>Sub<br>Max<br>Min | Pwr<br>Mag<br>Phase<br>GT<br>LT<br>Table<br>SubCir |  |

The function is calculated and applied to the output on every calculation step.

For all models, when calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Please note that output voltage is always delayed by one calculation step. This may affect stability of the schematic with closed loop.

| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | V     | Output as function of the inputs.  |
|                 | <b>IC</b> | V     | Initial condition: output voltage. |

Arbitrary function **f** defines output voltage as a function of the following variables:

**x** – input voltage  $V_x$

**y** – input voltage  $V_y$

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

$f = \sqrt{x^2 + y^2}$

$f = x * y * \sin(t)$

$f = P(r1) + P(r2)$

| Model                                 | Parameter | Units | Description                        |
|---------------------------------------|-----------|-------|------------------------------------|
| <b>Mul</b>                            | <b>K</b>  | V/V   | Gain.                              |
|                                       | <b>IC</b> | V     | Initial condition: output voltage. |
| Multiplication: $V = K * V_x * V_y$ . |           |       |                                    |

| Model                                                     | Parameter | Units | Description                        |
|-----------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Div</b>                                                | <b>K</b>  | V/V   | Gain.                              |
|                                                           | <b>IC</b> | V     | Initial condition: output voltage. |
| Division.: $V = K * V_x / V_y$ . If $V_y = 0$ , $V = 0$ . |           |       |                                    |

| Model                             | Parameter | Units | Description                        |
|-----------------------------------|-----------|-------|------------------------------------|
| <b>Sum</b>                        | <b>K</b>  | V/V   | Gain.                              |
|                                   | <b>IC</b> | V     | Initial condition: output voltage. |
| Addition: $V = K * (V_x + V_y)$ . |           |       |                                    |

| Model                                | Parameter | Units | Description                        |
|--------------------------------------|-----------|-------|------------------------------------|
| <b>Sub</b>                           | <b>K</b>  | V/V   | Gain.                              |
|                                      | <b>IC</b> | V     | Initial condition: output voltage. |
| Subtraction: $V = K * (V_x - V_y)$ . |           |       |                                    |

| Model                                                                                                                | Parameter | Units | Description                        |
|----------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Max</b>                                                                                                           | <b>K</b>  | V/V   | Gain.                              |
|                                                                                                                      | <b>IC</b> | V     | Initial condition: output voltage. |
| Maximum: $V = K * \max(V_x, V_y)$ .<br>if $V_x \geq V_y$ . . : $V = K * V_x$<br>if $V_x < V_y$ . . . : $V = K * V_y$ |           |       |                                    |

| Model                                                                                                                | Parameter | Units | Description                        |
|----------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Min</b>                                                                                                           | <b>K</b>  | V/V   | Gain.                              |
|                                                                                                                      | <b>IC</b> | V     | Initial condition: output voltage. |
| Maximum: $V = K * \max(V_x, V_y)$ .<br>if $V_x \geq V_y$ . . : $V = K * V_x$<br>if $V_x < V_y$ . . . : $V = K * V_y$ |           |       |                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Parameter | Units | Description                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Pwr</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>K</b>  | V/V   | Gain.                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>“Signed” power function: <math>V = K * \text{pwr}(V_x, V_y)</math>.</p> <p>The function is calculated as follows:</p> <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p>if <math>V_y = 0</math>:</p> <p>if <math>V_x &lt; 0</math> . . . : <math>V = -K</math></p> <p>if <math>V_x = 0</math> . . . : <math>V = 0</math></p> <p>if <math>V_x &gt; 0</math> . . . : <math>V = K</math></p> </div> <div style="width: 45%;"> <p>if <math>V_y \neq 0</math>:</p> <p>if <math>V_x &lt; 0</math> . . . : <math>V = -K * (-V_x)^{V_y}</math></p> <p>if <math>V_x = 0</math> . . . : <math>V = 0</math></p> <p>if <math>V_x &gt; 0</math> . . . : <math>V = K * V_x^{V_y}</math></p> </div> </div> |           |       |                                    |

| Model                                                              | Parameter | Units | Description                        |
|--------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Mag</b>                                                         | <b>K</b>  | V/V   | Gain.                              |
|                                                                    | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Magnitude. <math>V = K * \text{sqrt}(V_x^2 + V_y^2)</math>.</p> |           |       |                                    |

| Model                                                                                                                                                                                                                       | Parameter | Units | Description                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Phase</b>                                                                                                                                                                                                                | <b>K</b>  | V/V   | Gain.                              |
|                                                                                                                                                                                                                             | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Phase: <math>V = K * \text{phase}(V_x, V_y)</math>.</p> <p>V in Volts is equal to phase of a vector <math>V_x + jV_y</math> in degrees.</p> <p>If <math>V_x = 0</math> and <math>V_y = 0</math>: <math>V = 0</math>.</p> |           |       |                                    |

| Model                                                                                                                                | Parameter | Units | Description                        |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>GT</b>                                                                                                                            | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Greater than: <math>V = V_x &gt; V_y</math> ? <b>High</b> : <b>Low</b>.</p> <p><b>High</b> and <b>Low</b> are logical levels.</p> |           |       |                                    |

| Model                                                                                                                             | Parameter | Units | Description                        |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>LT</b>                                                                                                                         | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Less than. <math>V = V_x &lt; V_y</math> ? <b>High</b> : <b>Low</b>.</p> <p><b>High</b> and <b>Low</b> are logical levels.</p> |           |       |                                    |

| Model        | Parameter    | Units | Description                                         |
|--------------|--------------|-------|-----------------------------------------------------|
| <b>Table</b> | <b>X</b>     |       | Comma-separated string, X (input values).           |
|              | <b>Y</b>     |       | Comma-separated string, Y (input values).           |
|              | <b>Table</b> |       | Comma-separated string, Table of Z (output values). |
|              | <b>IC</b>    | V     | Initial condition: output voltage.                  |

2D look-up table. **Table** parameter defines output Z as a function of X and Y inputs of the component in the following format:

Z11,Z12,...,Z1N,Z21,Z22,...,Z2N,...,ZM1,ZM2,...,ZMN

where:

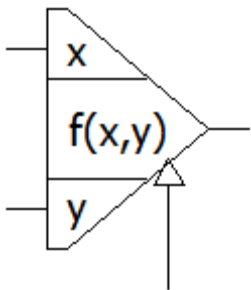
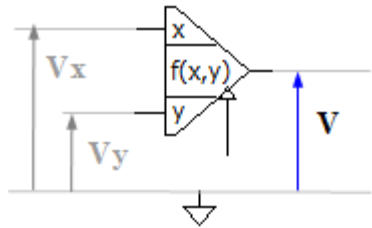
- Zij defines output of the function for input values Xi and Yj;
- N is total number of X input values, defined by **X** parameter;
- M is total number of Y input values, defined by **Y** parameter.

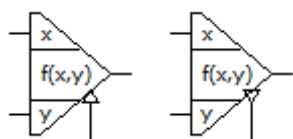
Output value between specified X and Y points is linearly interpolated on both coordinates. Output value below X1 is linearly extrapolated using X1...X2 interval data, output value above XN is linearly extrapolated using X(N-1)...XN interval data. The same rule is applied to Y coordinate

See *2D Table model* chapter of NL5 Manual for more details.



## F – Function-2 with clock

| Symbol                                                                            | Models                                                                                                                                                                                                                      | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <div>Function</div> <div>Mul</div> <div>Div</div> <div>Sum</div> <div>Sub</div> <div>Max</div> <div>Min</div> <div>Pwr</div> <div>Mag</div> <div>Phase</div> <div>GT</div> <div>LT</div> <div>Table</div> <div>SubCir</div> |  |



Function component with **clock** operates in **synchronized** mode: the function is calculated and applied to the output only on rising (or falling) edge of logical clock signal. As a result, this mode may provide faster simulation.

For all models, when calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Please note that output voltage is always delayed by one calculation step. This may affect stability of the schematic with closed loop.

| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | V     | Output as function of the inputs.  |
|                 | <b>IC</b> | V     | Initial condition: output voltage. |

Arbitrary function **f** defines output voltage as a function of the following variables:

**x** – input voltage  $V_x$

**y** – input voltage  $V_y$

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

$f = \sqrt{x^2 + y^2}$

$f = x * y * \sin(t)$

$f = P(r1) + P(r2)$

| Model                                 | Parameter | Units | Description                        |
|---------------------------------------|-----------|-------|------------------------------------|
| <b>Mul</b>                            | <b>K</b>  | V/V   | Gain.                              |
|                                       | <b>IC</b> | V     | Initial condition: output voltage. |
| Multiplication: $V = K * V_x * V_y$ . |           |       |                                    |

| Model                                                     | Parameter | Units | Description                        |
|-----------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Div</b>                                                | <b>K</b>  | V/V   | Gain.                              |
|                                                           | <b>IC</b> | V     | Initial condition: output voltage. |
| Division.: $V = K * V_x / V_y$ . If $V_y = 0$ , $V = 0$ . |           |       |                                    |

| Model                             | Parameter | Units | Description                        |
|-----------------------------------|-----------|-------|------------------------------------|
| <b>Sum</b>                        | <b>K</b>  | V/V   | Gain.                              |
|                                   | <b>IC</b> | V     | Initial condition: output voltage. |
| Addition: $V = K * (V_x + V_y)$ . |           |       |                                    |

| Model                                | Parameter | Units | Description                        |
|--------------------------------------|-----------|-------|------------------------------------|
| <b>Sub</b>                           | <b>K</b>  | V/V   | Gain.                              |
|                                      | <b>IC</b> | V     | Initial condition: output voltage. |
| Subtraction: $V = K * (V_x - V_y)$ . |           |       |                                    |

| Model                                                                                                                | Parameter | Units | Description                        |
|----------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Max</b>                                                                                                           | <b>K</b>  | V/V   | Gain.                              |
|                                                                                                                      | <b>IC</b> | V     | Initial condition: output voltage. |
| Maximum: $V = K * \max(V_x, V_y)$ .<br>if $V_x \geq V_y$ . . : $V = K * V_x$<br>if $V_x < V_y$ . . . : $V = K * V_y$ |           |       |                                    |

| Model                                                                                                                | Parameter | Units | Description                        |
|----------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Min</b>                                                                                                           | <b>K</b>  | V/V   | Gain.                              |
|                                                                                                                      | <b>IC</b> | V     | Initial condition: output voltage. |
| Maximum: $V = K * \max(V_x, V_y)$ .<br>if $V_x \geq V_y$ . . : $V = K * V_x$<br>if $V_x < V_y$ . . . : $V = K * V_y$ |           |       |                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Parameter | Units | Description                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Pwr</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>K</b>  | V/V   | Gain.                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>“Signed” power function: <math>V = K * \text{pwr}(V_x, V_y)</math>.</p> <p>The function is calculated as follows:</p> <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p>if <math>V_y = 0</math>:</p> <p>if <math>V_x &lt; 0</math> . . . : <math>V = -K</math></p> <p>if <math>V_x = 0</math> . . . : <math>V = 0</math></p> <p>if <math>V_x &gt; 0</math> . . . : <math>V = K</math></p> </div> <div style="width: 45%;"> <p>if <math>V_y \neq 0</math>:</p> <p>if <math>V_x &lt; 0</math> . . . : <math>V = -K * (-V_x)^{V_y}</math></p> <p>if <math>V_x = 0</math> . . . : <math>V = 0</math></p> <p>if <math>V_x &gt; 0</math> . . . : <math>V = K * V_x^{V_y}</math></p> </div> </div> |           |       |                                    |

| Model                                                              | Parameter | Units | Description                        |
|--------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Mag</b>                                                         | <b>K</b>  | V/V   | Gain.                              |
|                                                                    | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Magnitude. <math>V = K * \text{sqrt}(V_x^2 + V_y^2)</math>.</p> |           |       |                                    |

| Model                                                                                                                                                                                                                       | Parameter | Units | Description                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Phase</b>                                                                                                                                                                                                                | <b>K</b>  | V/V   | Gain.                              |
|                                                                                                                                                                                                                             | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Phase: <math>V = K * \text{phase}(V_x, V_y)</math>.</p> <p>V in Volts is equal to phase of a vector <math>V_x + jV_y</math> in degrees.</p> <p>If <math>V_x = 0</math> and <math>V_y = 0</math>: <math>V = 0</math>.</p> |           |       |                                    |

| Model                                                                                                                                | Parameter | Units | Description                        |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>GT</b>                                                                                                                            | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Greater than: <math>V = V_x &gt; V_y</math> ? <b>High</b> : <b>Low</b>.</p> <p><b>High</b> and <b>Low</b> are logical levels.</p> |           |       |                                    |

| Model                                                                                                                             | Parameter | Units | Description                        |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>LT</b>                                                                                                                         | <b>IC</b> | V     | Initial condition: output voltage. |
| <p>Less than. <math>V = V_x &lt; V_y</math> ? <b>High</b> : <b>Low</b>.</p> <p><b>High</b> and <b>Low</b> are logical levels.</p> |           |       |                                    |

| Model        | Parameter    | Units | Description                                         |
|--------------|--------------|-------|-----------------------------------------------------|
| <b>Table</b> | <b>X</b>     |       | Comma-separated string, X (input values).           |
|              | <b>Y</b>     |       | Comma-separated string, Y (input values).           |
|              | <b>Table</b> |       | Comma-separated string, Table of Z (output values). |
|              | <b>IC</b>    | V     | Initial condition: output voltage.                  |

2D look-up table. **Table** parameter defines output Z as a function of X and Y inputs of the component in the following format:

Z11,Z12,...,Z1N,Z21,Z22,...,Z2N,...,ZM1,ZM2,...,ZMN

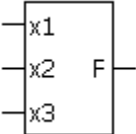
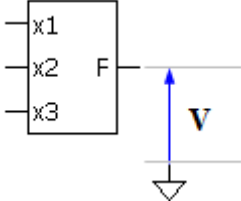
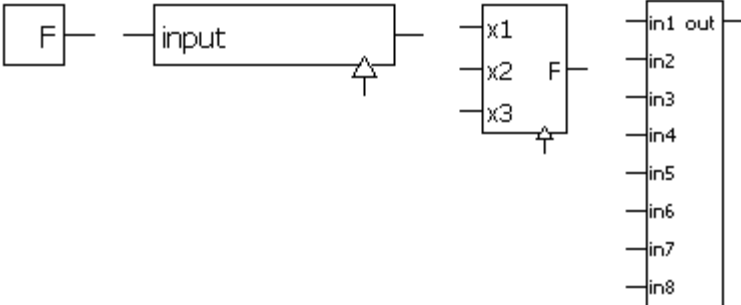
where:

- Zij defines output of the function for input values Xi and Yj;
- N is total number of X input values, defined by **X** parameter;
- M is total number of Y input values, defined by **Y** parameter.

Output value between specified X and Y points is linearly interpolated on both coordinates. Output value below X1 is linearly extrapolated using X1...X2 interval data, output value above XN is linearly extrapolated using X(N-1)...XN interval data. The same rule is applied to Y coordinate

See *2D Table model* chapter of NL5 Manual for more details.

F – Custom function

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Models   | Traces                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Function |  |
| <p>This is a customized component. It can be edited in the <b>Edit Component</b> dialog box. See <i>Customized component</i> chapter of NL5 Manual for instructions on editing a component.</p> <p>This component may have:</p> <ul style="list-style-type: none"><li>- arbitrary size up to 32(width) X 8(height),</li><li>- up to 8 inputs on the left side,</li><li>- one output on the right side,</li><li>- one or no clock pins on the bottom side,</li><li>- custom input and output names.</li></ul> <p>Examples of Custom function component:</p>  |          |                                                                                     |

| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | V     | Output as function of the inputs.  |
|                 | <b>IC</b> | V     | Initial condition: output voltage. |

Arbitrary function **f** defines output voltage as a function of the following variables:

**pin\_name** – input voltage on the input pin **pin\_name**.

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

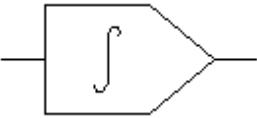
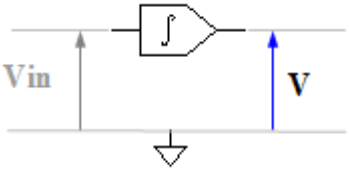
$f = \max(x1, x2, x3)$

$f = (in1+in2) * V(R1)$

If **clock** pin does not exist, the model operates in **continuous** mode: the function is calculated and applied to the output on every calculation step. If **clock** pin exists, the model operates in **synchronized** mode: the function is calculated and applied to the output only on rising (or falling) edge of logical clock signal. As a result, this mode may provide faster simulation than **continuous** mode.

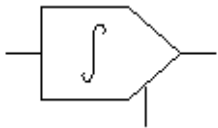
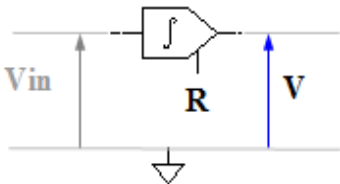
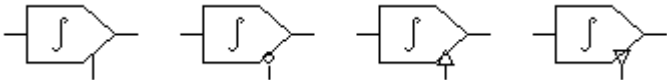
When calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Please note that input voltages and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. As a result, the output voltage is always delayed by one calculation step. This may affect stability of the schematic with closed loop.

F – Integral

| Symbol                                                                            | Models   | Signals                                                                             |
|-----------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------|
|  | Integral |  |

| Model                                                                                      | Parameter | Units | Description                        |
|--------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| Integral                                                                                   | K         | V/V   | Gain.                              |
|                                                                                            | IC        | V     | Initial condition: output voltage. |
| Integral: $V = K * \int Vin \, dt$ .                                                       |           |       |                                    |
| When calculating DC operating point, output is set to specified output voltage <b>IC</b> . |           |       |                                    |

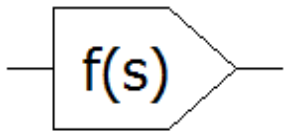
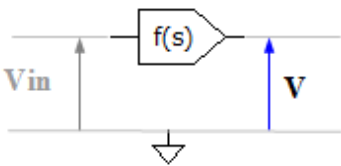
F – Integral with reset

| Symbol                                                                            | Models                                                                            | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Integral                                                                          |  |
| Views                                                                             |  |                                                                                     |

| Model                                                                                                                                                                                                                                                                                                                     | Parameter | Units | Description                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| Integral                                                                                                                                                                                                                                                                                                                  | K         | V/V   | Gain.                              |
|                                                                                                                                                                                                                                                                                                                           | IC        | V     | Initial condition: output voltage. |
| <p>Integral: <math>V = K * \int Vin \, dt</math>.</p> <p>If reset signal <b>R</b> is active, output is always zero. If rising or falling edge reset signal applied, output is set to zero and integration continues.</p> <p>When calculating DC operating point, output is set to specified output voltage <b>IC</b>.</p> |           |       |                                    |



F – s-function

| Symbol                                                                            | Models                                               |                                                   | Signals                                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Function<br>Poly<br>Poly1<br>Poly2<br>Poly3<br>Poly4 | Poly5<br>Roots<br>Table<br>File<br>Trace<br>Phase |  |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Parameter | Units | Description        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|--------------------|
| Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | f         | V/V   | Transfer function. |
| <p>Transfer function <b>f</b> defines transfer function in <b>s</b> domain. The following variables can represent frequency in the function:</p> <p><b>f</b> – current AC frequency, Hz<br/><b>w</b> – angular AC frequency, <b>w = 2πf</b><br/><b>s</b> or <b>p</b> – Laplace parameter, <b>s = p = j*2πf</b></p> <p>Example:</p> <p>f = 1/(1+s)<br/>f = exp(-R1*C1**s)</p> <p>Only operators and functions that support complex numbers can be used in this function. If <b>f</b> is blank, it is assumed to be <b>1</b>. If division by zero occurs during function calculation, this frequency is ignored, and data point is not added to the trace.</p> <p>At transient and DC operation point calculation for AC (if enabled), the component behaves as a buffer with infinite bandwidth and gain equal to <b>f(0)</b>.</p> |           |       |                    |

| Model       | Parameter   | Units | Description        |
|-------------|-------------|-------|--------------------|
| <b>Poly</b> | <b>Poly</b> |       | Polynomial string. |
|             | <b>IC</b>   |       | Initial condition. |

AC transfer function is a ratio of polynomials of Laplace parameter  $s$ :

$$f(s) = (b_0 + b_1*s + b_2*s^2 + \dots) / (a_0 + a_1*s + a_2*s^2 + \dots)$$

The model supports transient as well.

**Poly** parameter is a comma-separated string:

$$n, b_0, \dots, b_n, a_0, \dots, a_n$$

where **n** is maximum order of numerator and denominator ( $n = 0 \dots 5$ ), **bi** are numerator coefficients, and **ai** are denominator coefficients ( $i = 0 \dots n$ ). Please note that all **n** coefficients should be provided, even if they are zeroes. **Poly** parameter can be entered in the table in the **Poly window** of NL5 GUI.

Initial condition **IC** is a **csv** (comma separated values) string that keeps internal state of the component when transient is paused and **Save IC** command is performed. Modifying **IC** parameter manually is not recommended, except clearing it to blank (no **IC**).

If division by zero occurs during polynomial calculation, this frequency is ignored, and data point is not added to the trace.

At DC operation point calculation, **f(0)** is used.  
See *Poly model* chapter of NL5 Manual for more details.

| Model        | Parameter | Units | Description                            |
|--------------|-----------|-------|----------------------------------------|
| <b>Poly1</b> | <b>b0</b> |       | Numerator polynomial coefficients 0.   |
| <b>Poly2</b> | ...       |       | ...                                    |
| <b>Poly3</b> | <b>a0</b> |       | Denominator polynomial coefficients 0. |
| <b>Poly4</b> | ...       |       | ...                                    |
| <b>Poly5</b> | <b>IC</b> |       | Initial condition.                     |

AC transfer function is a ratio of polynomials of Laplace parameter  $s$ :

$$f(s) = (b_0 + b_1*s + b_2*s^2 + \dots) / (a_0 + a_1*s + a_2*s^2 + \dots)$$

These models support transient as well.

Initial condition **IC** is a **csv** (comma separated values) string that keeps internal state of the component when transient is paused and **Save IC** command is performed. Modifying **IC** parameter manually is not recommended, except clearing it to blank (no **IC**).

If division by zero occurs during polynomial calculation, this frequency is ignored, and data point is not added to the trace.

At DC operation point calculation, **f(0)** is used.

| Model        | Parameter    | Units | Description               |
|--------------|--------------|-------|---------------------------|
| <b>Roots</b> | <b>K</b>     |       | Gain.                     |
|              | <b>Roots</b> |       | Roots (zeroes and poles). |
|              | <b>IC</b>    |       | Initial condition.        |

AC transfer function is defines by zeroes and poles:

$$f(s) = K * (s-z1)*(s-z2).../(s-p1)/(s-p2)...$$

where **K** is gain, z1...zn are zeroes, p1...pN are poles.

Roots are defined by **Roots** parameter in the **csv** (comma-separated values) format, as follows:

Nz,Rez1,Imz1,...,Np,Rep1,Imp1,...

where:

Nz - number of zeroes

Rezi – real part of zi

Imzi – imaginary part of zi

Np - number of poles,

Repi – real part of pi

Impi – imaginary part of pi

There could be any number of zeroes and poles, however the resulting numerator and denominator polynomials order should not exceed 5. See *Roots model* chapter of NL5 Manual for details on entering/editing roots.

The model supports transient as well.

Initial condition **IC** is a **csv** (comma separated values) string that keeps internal state of the component when transient is paused and **Save IC** command is performed. Modifying **IC** parameter manually is not recommended, except clearing it to blank (no **IC**).

If division by zero occurs during calculation, this frequency is ignored, and data point is not added to the trace.

At DC operation point calculation, **f(0)** is used.

| Model        | Parameter    | Units | Description                         |
|--------------|--------------|-------|-------------------------------------|
| <b>Table</b> | <b>Table</b> |       | Comma-separated string.             |
|              | <b>K</b>     |       | Gain used for transient simulation. |

Look-up table. Transfer function is defined by **Table** parameter in the **csv** (comma separated values) format, as follows:

$$f1,mag1,phase1,f2,mag2,phase2,\dots,fN,magN,phaseN$$

where mag and phase are transfer function magnitude and phase at specified frequency. Output value between specified frequencies is interpolated using linear or logarithmic interpolation, depending on simulation scale. Below first and above last specified frequency, simulation result is not provided. Frequencies should be given in ascending order.

At transient simulation, constant gain **K** is used.

See *Table model* chapter of NL5 Manual for more details.

| Model       | Parameter   | Units | Description                         |
|-------------|-------------|-------|-------------------------------------|
| <b>File</b> | <b>File</b> |       | File name.                          |
|             | <b>K</b>    |       | Gain used for transient simulation. |

Transfer function is defined in the text file in the following format:

$$f1,mag1,phase1$$

$$f2,mag2,phase2$$

At transient simulation, constant gain **K** is used.

| Model        | Parameter    | Units | Description                         |
|--------------|--------------|-------|-------------------------------------|
| <b>Trace</b> | <b>Trace</b> |       | Trace name.                         |
|              | <b>K</b>     |       | Gain used for transient simulation. |

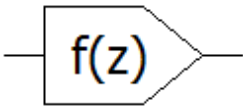
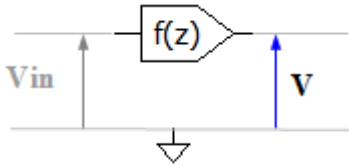
Transfer function is defined by an existing AC trace of **Data** type with “static” data: entered manually, duplicated, imported, pasted from the clipboard, etc.

**Trace** is a name of the existing trace shown in the Trace list.

At transient simulation, constant gain **K** is used.

| Model                                                                                                                                                                                                                                                  | Parameter    | Units   | Description |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------|-------------|
| <b>Phase</b>                                                                                                                                                                                                                                           | <b>Phase</b> | degrees | Phase.      |
| <p>Transfer function with unity magnitude and same phase for all frequencies:</p> $f = \exp(2\pi j * \text{Phase}/360)$ <p>At transient and DC operation point calculation for AC (if enabled), the component behaves as a buffer with unity gain.</p> |              |         |             |

F – z-function

| Symbol                                                                            | Models                             |                         | Signals                                                                             |
|-----------------------------------------------------------------------------------|------------------------------------|-------------------------|-------------------------------------------------------------------------------------|
|  | Function<br>Poly<br>Poly1<br>Poly2 | Poly3<br>Poly4<br>Poly5 |  |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Parameter | Units | Description        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|--------------------|
| Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | f         | V/V   | Transfer function. |
| <p>Transfer function <b>f</b> defines transfer function in <b>z</b> domain. The following variables can represent frequency in the function:</p> <p><b>f</b> – current AC frequency, Hz<br/><b>w</b> – angular AC frequency, <b>w = 2πf</b> .<br/><b>s</b> or <b>p</b> – Laplace parameter, <b>s = p = j*2πf</b>.<br/><b>z</b> – z-parameter.</p> <p>Definition of z-parameter is located in <i>AC settings/Advanced settings/AC</i> window. Please note that if T parameter is used in z-parameter formula - for example, exp(s*T) - it should be defined as a schematic variable.</p> <p>If <b>f</b> is blank, it is assumed to be <b>1</b>. If division by zero occurs during function calculation, this frequency is ignored, and data point is not added to the trace.</p> <p>At transient and DC operation point calculation for AC (if enabled), the component behaves as a buffer with unity gain.</p> |           |       |                    |

| Model | Parameter | Units | Description        |
|-------|-----------|-------|--------------------|
| Poly  | Poly      |       | Polynomial string. |

AC transfer function is a ratio of polynomials of z-parameter:

$$f(z) = (b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots) / (a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots)$$

**Poly** parameter is a comma-separated string:

**n,b0,...,bn,a0,...,an**

where **n** is maximum order of numerator and denominator ( $n = 0 \dots 5$ ), **bi** are numerator coefficients, and **ai** are denominator coefficients ( $i = 0 \dots n$ ). Please note that all **n** coefficients should be provided, even if they are zeroes. **Poly** parameter can be entered in the table in the **Poly window** of NL5 GUI.

If division by zero occurs during polynomial calculation, this frequency is ignored, and data point is not added to the trace.

At transient, and DC operation point calculation for AC (if enabled), the component behaves as a buffer with unity gain.

See *Poly model* chapter of NL5 Manual for more details.

| Model        | Parameter | Units | Description                            |
|--------------|-----------|-------|----------------------------------------|
| <b>Poly1</b> | <b>b0</b> |       | Numerator polynomial coefficients 0.   |
| <b>Poly2</b> | ...       |       | ...                                    |
| <b>Poly3</b> | <b>a0</b> |       | Denominator polynomial coefficients 0. |
| <b>Poly4</b> | ...       |       | ...                                    |
| <b>Poly5</b> |           |       |                                        |

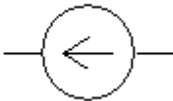
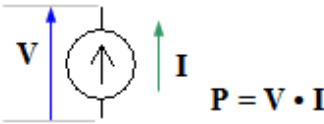
AC transfer function is a ratio of polynomials of z-parameter:

$$f(z) = (b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots) / (a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots)$$

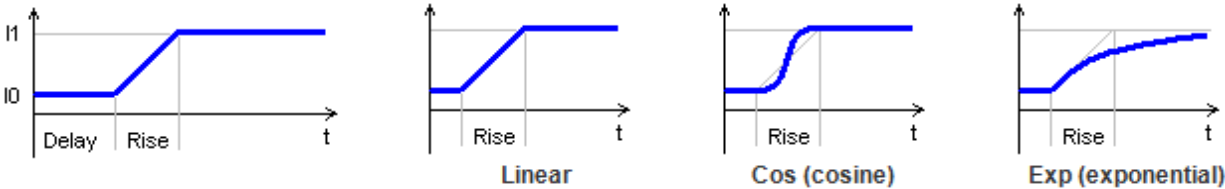
If division by zero occurs during polynomial calculation, this frequency is ignored, and data point is not added to the trace.

At transient, and DC operation point calculation for AC (if enabled), the component behaves as a buffer with unity gain.

I – Current source

| Symbol                                                                            | Models                                       |                                                      | Signals                                                                             |
|-----------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | I<br>Step<br>Single<br>Pulse<br>Clock<br>Sin | Sweep<br>Function<br>List<br>File<br>Trace<br>SubCir |  |

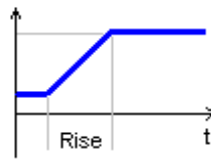
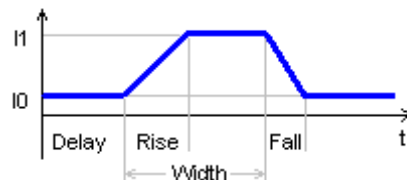
| Model                 | Parameter | Units | Description |
|-----------------------|-----------|-------|-------------|
| I                     | I         | A     | Current.    |
| Constant current = I. |           |       |             |

| Model                                                                                                | Parameter | Units | Description                |
|------------------------------------------------------------------------------------------------------|-----------|-------|----------------------------|
| Step                                                                                                 | I1        | A     | Step On current.           |
|                                                                                                      | I0        | A     | Step Off current.          |
|                                                                                                      | Slope     |       | Slope type: Linear/Cos/Exp |
|                                                                                                      | Rise      | s     | Step rise length.          |
|                                                                                                      | Delay     | s     | Delay before step starts.  |
| Step starts after <b>Delay</b> time. If <b>Rise</b> is non-zero, 3 <b>Slope</b> types are available. |           |       |                            |
|                  |           |       |                            |



| Model         | Parameter    | Units | Description                |
|---------------|--------------|-------|----------------------------|
| <b>Single</b> | <b>I1</b>    | A     | Pulse On current.          |
|               | <b>I0</b>    | A     | Pulse Off current.         |
|               | <b>Width</b> | s     | Pulse width.               |
|               | <b>Slope</b> |       | Slope type: Linear/Cos/Exp |
|               | <b>Rise</b>  | s     | Pulse rise length.         |
|               | <b>Fall</b>  | s     | Pulse fall length.         |
|               | <b>Delay</b> | s     | Delay before pulse starts. |

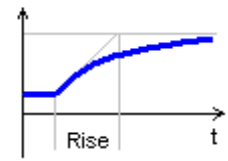
Single pulse starts after **Delay** time. **Rise** time is included into **Width**, **Fall** time is **not** included into **Width**. **Slope** type applies both to pulse rise and fall (if non-zero).



Linear



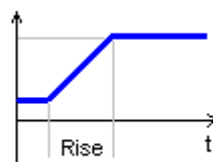
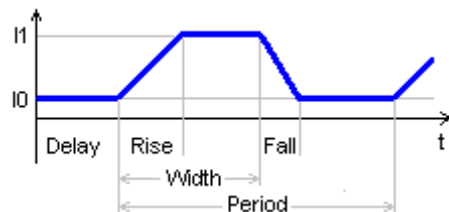
Cos (cosine)



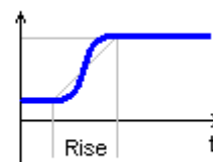
Exp (exponential)

| Model        | Parameter     | Units | Description                      |
|--------------|---------------|-------|----------------------------------|
| <b>Pulse</b> | <b>I1</b>     | A     | Pulse On current.                |
|              | <b>I0</b>     | A     | Pulse Off current.               |
|              | <b>Period</b> | s     | Period.                          |
|              | <b>Width</b>  | s     | Pulse width.                     |
|              | <b>Slope</b>  |       | Slope type: Linear/Cos/Exp       |
|              | <b>Rise</b>   | s     | Pulse rise length.               |
|              | <b>Fall</b>   | s     | Pulse fall length.               |
|              | <b>Delay</b>  | s     | Delay before first pulse starts. |

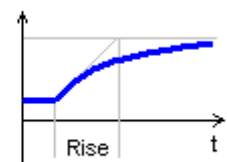
Pulses start after **Delay** time. **Rise** time is included into **Width**, **Fall** time is **not** included into **Width**. **Slope** type applies both to pulse rise and fall (if non-zero).



Linear



Cos (cosine)

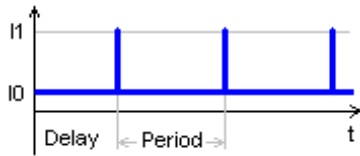


Exp (exponential)

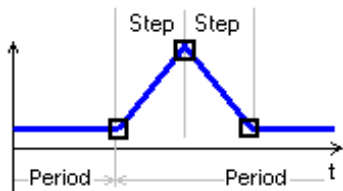
| Model | Parameter | Units | Description                       |
|-------|-----------|-------|-----------------------------------|
| Clock | I1        | A     | Pulse On current.                 |
|       | I0        | A     | Pulse Off current.                |
|       | Period    | s     | Period.                           |
|       | Step      | s     | Simulation step of rise and fall. |
|       | Delay     | s     | Delay before first pulse starts.  |

Periodic pulses with width of one simulation step. Pulses start after **Delay** time.

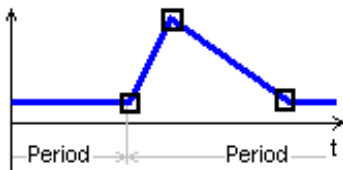
**Clock** model is recommended to produce a constant frequency clock signal. Unlike **Pulse** model, it won't force unnecessary step reduction at the end of the pulse, which may help to accelerate simulation.



If **Step** parameter is not zero **and** is less than current schematic simulation step, the step is adjusted to provide rise and fall of clock pulse to be equal to **Step**:

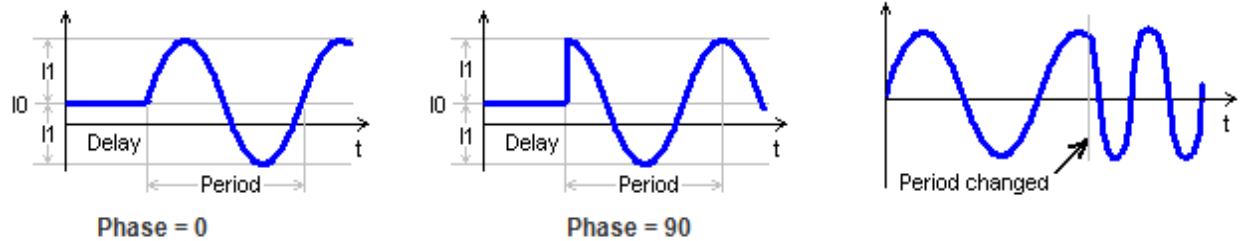


Otherwise, the clock pulse is created using current schematic simulation step, which depends on many factors, and cannot be easily predicted:



| Model | Parameter | Units | Description                      |
|-------|-----------|-------|----------------------------------|
| Sin   | I1        | A     | Current amplitude.               |
|       | I0        | A     | Current baseline.                |
|       | Period    | s     | Period.                          |
|       | Phase     | deg   | Phase.                           |
|       | Decay     | 1/s   | Decay constant                   |
|       | Delay     | s     | Delay before sine signal starts. |

Sine signal starts after **Delay** time. **Phase** is sine phase in degrees at the moment when signal starts. If transient is paused, sine period changed, and then transient is continued, the phase of the signal remains continuous, providing smooth sine signal of variable frequency. If **Decay** is not zero, the sine signal is exponentially dumped with time constant =  $1/\text{Decay}$ .



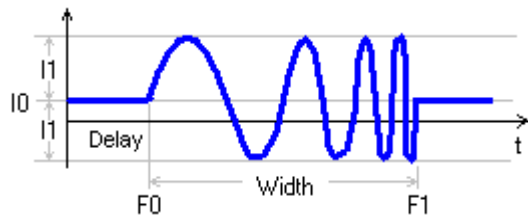
| Model | Parameter | Units | Description                 |
|-------|-----------|-------|-----------------------------|
| Sweep | I1        | V     | Current amplitude.          |
|       | I0        | V     | Current baseline.           |
|       | Width     | s     | Width of the signal.        |
|       | F0        | Hz    | Start frequency.            |
|       | F1        | Hz    | End frequency.              |
|       | Type      |       | Signal type: Linear/Exp.    |
|       | Delay     | s     | Delay before signal starts. |

Sinusoidal signal with variable frequency starts after **Delay** time. Signal frequency changes during **Width** interval from **F0** to **F1** linearly or exponentially, depending on specified **Type**.

If  $F0 = F1$ , then one period of frequency  $1/\text{Width}$  will be generated.

If lowest frequency is set to zero and **Type** = Exp, then lowest frequency  $0.01/\text{Width}$  will be used.

If needed, the highest frequency will be increased to provide integer number of signal periods, so that signal phase at the beginning and at the end of **Width** interval is exactly zero.



| Model           | Parameter | Units | Description |
|-----------------|-----------|-------|-------------|
| <b>Function</b> | <b>f</b>  | A     | Function    |

Arbitrary function **f** defines current as a function of the following variables:

- t** - current time
- V(name)** - voltage on the component **name**
- I(name)** - current through the component **name**
- P(name)** – power on the component **name**
- S(name)** – state of the component **name**

where **name** is the name of any component in the schematic. If **f** is blank, current is zero.

Example:

```
f = sin(t) * (1+cos(t*.01))
f = V(R1) * I(R1)
```

Please note that **V**, **I**, **P**, and **S** variables are taken at previous calculation step. This may affect stability of the schematic with closed loop.

| Model       | Parameter    | Units | Description               |
|-------------|--------------|-------|---------------------------|
| <b>List</b> | <b>List</b>  |       | Comma-separated string.   |
|             | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|             | <b>Delay</b> | s     | Delay.                    |

Piecewise linear signal is defined by **List** parameter in the **csv** (comma-separated values) format, as follows:

$$t_0, I_0, t_1, I_1, \dots, t_n, I_n$$

where all **t** and **I** can be numerical values or expressions.

If  $t < t_0$ , signal is  $I_0$ .

If  $t_0 < t < t_1$ , signal value is linearly interpolated between  $I_0$  and  $I_1$ , etc.

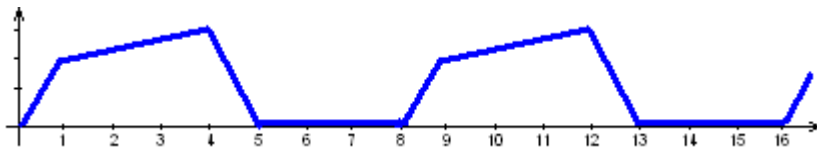
If  $t > t_n$ , and **Cycle** parameter is set to **No**, the signal value is  $I_n$ . Otherwise the signal defined in  $t_0 \dots t_n$  interval is repeated continuously.

Signal start is delayed by **Delay** time.

Example:

```
List = 0,0,1,2,4,3,5,0,8,0
```

If **Cycle = Yes**, **Delay = 0**, the following current will be generated:



See *List model* chapter of NL5 Manual for more details.

| Model       | Parameter    | Units | Description               |
|-------------|--------------|-------|---------------------------|
| <b>File</b> | <b>File</b>  |       | File name.                |
|             | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|             | <b>Delay</b> | s     | Delay.                    |

Piecewise linear signal is defined in the text file. If **File** parameter does not have a full path, NL5 will search for the file in the directory where schematic file is located, then in the Library directories, and then in the Library directories relative to schematic file directory (see NL5 Manual, *Schematic Properties*).

Signal is defined in the **csv** (comma-separated values) format, as follows:

```
<if first line does not start with a number, it is ignored>
t0,I0
t1,I1
.....
tn,In
```

where all  $t$  and  $I$  can be numerical values or expressions.

If  $t < t_0$ , signal is  $v_0$ .

If  $t_0 < t < t_1$ , signal value is linearly interpolated between  $I_0$  and  $I_1$ , etc.

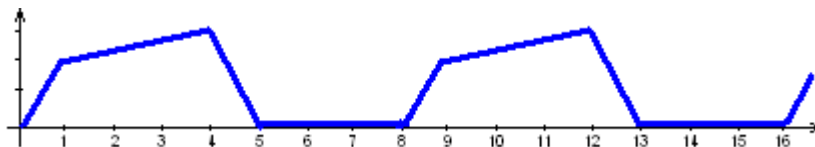
If  $t > t_n$ , and **Cycle** parameter is set to **No**, the signal value is  $I_n$ . Otherwise the signal defined in  $t_0 \dots t_n$  interval is repeated continuously.

Signal start is delayed by **Delay** time.

Example:

```
0,0
1,2
4,3
5,0
8,0
```

If **Cycle = Yes**, **Delay = 0**, the following current will be generated:



| Model        | Parameter    | Units | Description               |
|--------------|--------------|-------|---------------------------|
| <b>Trace</b> | <b>Trace</b> |       | Trace name.               |
|              | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|              | <b>Delay</b> | s     | Delay.                    |

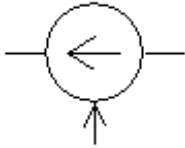
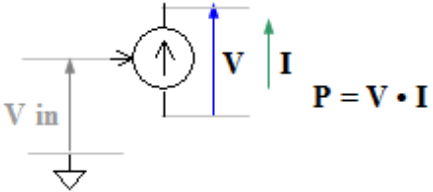
Signal is defined by an existing trace of **Data** type with “static” data: entered manually, duplicated, imported, pasted from the clipboard, etc. Signal value at required simulation point is obtained by linear interpolation between existing data points of the trace.

**Trace** is a name of the existing trace shown in the Trace list.

If **Cycle** parameter is set to **Yes**, the signal is repeated continuously.

Signal start is delayed by **Delay** time.

## I – Logic controlled current source

| Symbol                                                                            | Models                                                                                                                                                                                                                      | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <div>I</div> <div>One-shot</div> <div>Step</div> <div>Single</div> <div>Pulse</div> <div>Clock</div> <div>Sin</div> <div>Sweep</div> <div>Function</div> <div>List</div> <div>File</div> <div>Trace</div> <div>SubCir</div> |  |

| Model                 | Parameter | Units | Description |
|-----------------------|-----------|-------|-------------|
| I                     | I         | A     | Current.    |
| Constant current = I. |           |       |             |

| Model                                                                                                                                                                                                                                                                                                                                                          | Parameter | Units | Description        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|--------------------|
| One-shot                                                                                                                                                                                                                                                                                                                                                       | I1        | A     | Pulse On current.  |
|                                                                                                                                                                                                                                                                                                                                                                | I0        | A     | Pulse Off current. |
|                                                                                                                                                                                                                                                                                                                                                                | Width     | s     | Pulse width.       |
| <p>One-shot pulse generator. When increasing input voltage <math>V_{in}</math> crosses logical threshold, current pulse of <b>Width</b> duration is generated. <b>I0</b> is pulse Off level, <b>I1</b> is pulse On level.</p> <p>If increasing <math>V_{in}</math> crosses logical threshold value while pulse is being generated, the pulse is restarted.</p> |           |       |                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                           | Parameter | Units | Description                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|----------------------------|
| Step                                                                                                                                                                                                                                                                                                                                                                                            | I1        | A     | Step On current.           |
|                                                                                                                                                                                                                                                                                                                                                                                                 | I0        | A     | Step Off current.          |
|                                                                                                                                                                                                                                                                                                                                                                                                 | Slope     |       | Slope type: Linear/Cos/Exp |
|                                                                                                                                                                                                                                                                                                                                                                                                 | Rise      | s     | Step rise length.          |
|                                                                                                                                                                                                                                                                                                                                                                                                 | Delay     | s     | Delay before step starts.  |
| <p>When control signal <math>V_{in}</math> is below logical threshold, output is in Off state. When increasing control signal <math>V_{in}</math> crosses logical threshold, a signal similar to <b>Step</b> model of <b>Current source</b> component is generated. When decreasing control signal <math>V_{in}</math> drops below logical threshold, output goes to Off state immediately.</p> |           |       |                            |

| Model         | Parameter    | Units | Description                |
|---------------|--------------|-------|----------------------------|
| <b>Single</b> | <b>I1</b>    | A     | Pulse On current.          |
|               | <b>I0</b>    | A     | Pulse Off current.         |
|               | <b>Width</b> | s     | Pulse width.               |
|               | <b>Slope</b> |       | Slope type: Linear/Cos/Exp |
|               | <b>Rise</b>  | s     | Pulse rise length.         |
|               | <b>Fall</b>  | s     | Pulse fall length.         |
|               | <b>Delay</b> | s     | Delay before pulse starts. |

When control signal *Vin* is below logical threshold, output is in Off state. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Single** model of **Current source** component is generated. When decreasing control signal *Vin* drops below logical threshold, output goes to Off state immediately.

| Model        | Parameter     | Units | Description                      |
|--------------|---------------|-------|----------------------------------|
| <b>Pulse</b> | <b>I1</b>     | A     | Pulse On current.                |
|              | <b>I0</b>     | A     | Pulse Off current.               |
|              | <b>Period</b> | s     | Period.                          |
|              | <b>Width</b>  | s     | Pulse width.                     |
|              | <b>Slope</b>  |       | Slope type: Linear/Cos/Exp       |
|              | <b>Rise</b>   | s     | Pulse rise length.               |
|              | <b>Fall</b>   | s     | Pulse fall length.               |
|              | <b>Delay</b>  | s     | Delay before first pulse starts. |

When control signal *Vin* is below logical threshold, output is in Off state. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Pulse** model of **Current source** component is generated. When decreasing control signal *Vin* drops below logical threshold, output goes to Off state immediately.

| Model        | Parameter     | Units | Description                       |
|--------------|---------------|-------|-----------------------------------|
| <b>Clock</b> | <b>I1</b>     | A     | Pulse On current.                 |
|              | <b>I0</b>     | A     | Pulse Off current.                |
|              | <b>Period</b> | s     | Period.                           |
|              | <b>Step</b>   | s     | Simulation step of rise and fall. |
|              | <b>Delay</b>  | s     | Delay before first pulse starts.  |

When control signal *Vin* is below logical threshold, output is in Off state. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Clock** model of **Current source** component is generated. When decreasing control signal *Vin* drops below logical threshold, output goes to Off state immediately.

| Model                                                                                                                                                                                                                                                                                                                                                                    | Parameter     | Units | Description                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|----------------------------------|
| <b>Sin</b>                                                                                                                                                                                                                                                                                                                                                               | <b>I1</b>     | A     | Current amplitude.               |
|                                                                                                                                                                                                                                                                                                                                                                          | <b>I0</b>     | A     | Current baseline.                |
|                                                                                                                                                                                                                                                                                                                                                                          | <b>Period</b> | s     | Period.                          |
|                                                                                                                                                                                                                                                                                                                                                                          | <b>Phase</b>  | deg   | Phase.                           |
|                                                                                                                                                                                                                                                                                                                                                                          | <b>Decay</b>  | 1/s   | Decay constant                   |
|                                                                                                                                                                                                                                                                                                                                                                          | <b>Delay</b>  | s     | Delay before sine signal starts. |
| <p>When control signal <i>Vin</i> is below logical threshold, output current is <b>I0</b>. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>Sin</b> model of <b>Current source</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to <b>I0</b> immediately.</p> |               |       |                                  |

| Model                                                                                                                                                                                                                                                                                                                                                                      | Parameter    | Units | Description                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|-----------------------------|
| <b>Sweep</b>                                                                                                                                                                                                                                                                                                                                                               | <b>I1</b>    | V     | Current amplitude.          |
|                                                                                                                                                                                                                                                                                                                                                                            | <b>I0</b>    | V     | Current baseline.           |
|                                                                                                                                                                                                                                                                                                                                                                            | <b>Width</b> | s     | Width of the signal.        |
|                                                                                                                                                                                                                                                                                                                                                                            | <b>F0</b>    | Hz    | Start frequency.            |
|                                                                                                                                                                                                                                                                                                                                                                            | <b>F1</b>    | Hz    | End frequency.              |
|                                                                                                                                                                                                                                                                                                                                                                            | <b>Type</b>  |       | Signal type: Linear/Exp.    |
|                                                                                                                                                                                                                                                                                                                                                                            | <b>Delay</b> | s     | Delay before signal starts. |
| <p>When control signal <i>Vin</i> is below logical threshold, output current is <b>I0</b>. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>Sweep</b> model of <b>Current source</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to <b>I0</b> immediately.</p> |              |       |                             |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Parameter | Units | Description |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-------------|
| <b>Function</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>f</b>  | A     | Function    |
| <p>When control signal <i>Vin</i> is below logical threshold, output is zero. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>Function</b> model of <b>Current source</b> component is generated. If the function is using current time variable <i>t</i>, this moment will be considered as <i>t</i>=0. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to zero immediately</p> |           |       |             |

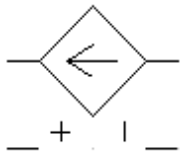
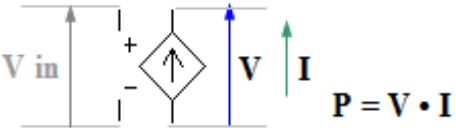
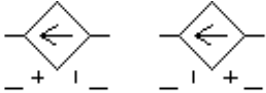
| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Parameter    | Units | Description               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>List</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>List</b>  |       | Comma-separated string.   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <i>Vin</i> is below logical threshold, output is equal to <b>I0</b> value of <b>List</b> signal. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>List</b> model of <b>Current source</b> component is generated. This moment is also considered as <i>t</i>=0 for the <b>List</b> signal. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to <b>I0</b> immediately.</p> |              |       |                           |



| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Parameter    | Units | Description               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>File</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>File</b>  |       | File name.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <math>V_{in}</math> is below logical threshold, output is equal to <b>I0</b> value specified in the <b>File</b>. When increasing control signal <math>V_{in}</math> crosses logical threshold, a signal similar to <b>File</b> model of <b>Current source</b> component is generated. This moment is also considered as <math>t=0</math> for the <b>File</b> signal. When decreasing control signal <math>V_{in}</math> drops below logical threshold, output goes to <b>I0</b> immediately.</p> |              |       |                           |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Parameter    | Units | Description               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>Trace</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Trace</b> |       | Trace name.               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <math>V_{in}</math> is below logical threshold, output is zero. When increasing control signal <math>V_{in}</math> crosses logical threshold, a signal similar to <b>Trace</b> model of <b>Current source</b> component is generated. This moment is also considered as <math>t=0</math> for the <b>Trace</b> signal. When decreasing control signal <math>V_{in}</math> drops below logical threshold, output goes to zero immediately.</p> |              |       |                           |

I – Voltage controlled current source

| Symbol                                                                            | Models                                                                            | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Linear<br>I<br>Function<br>PWL<br><br>VCO<br>One-shot<br>PWM<br>SubCir            |  |
| Views                                                                             |  |                                                                                     |

| Model                                                        | Parameter | Units | Description |
|--------------------------------------------------------------|-----------|-------|-------------|
| Linear                                                       | K         | A/V   | Gain        |
| Linear voltage controlled current source: $I = K * V_{in}$ . |           |       |             |

| Model                 | Parameter | Units | Description |
|-----------------------|-----------|-------|-------------|
| I                     | I         | A     | Current.    |
| Constant current = I. |           |       |             |

| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | A     | Output as function of the input.   |
|                 | <b>IC</b> | A     | Initial condition: output current. |

Arbitrary function **f** defines output current as a function of the following variables:

- x** – input voltage *V<sub>in</sub>*
- t** - current time
- V(name)** - voltage on the component **name**
- I(name)** - current through the component **name**
- P(name)** – power on the component **name**
- S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

```
f = x*x
f = x * sin(t)
f = P(r1) + P(r2)
```

When calculating DC operating point, and in AC analysis, output is set to specified output current **IC**. Please note that variable **x** (input voltage *V<sub>in</sub>*) and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. This may affect stability of the schematic with closed loop.

| Model      | Parameter  | Units | Description                                        |
|------------|------------|-------|----------------------------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, K( <i>V<sub>in</sub></i> ) |

Piecewise linear voltage controlled current source. **pwl** string defines gain as a function of input voltage K(*V<sub>in</sub>*). The transfer function of the source I(*V<sub>in</sub>*) is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that K(*V<sub>in</sub>*) is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

| Model      | Parameter    | Units | Description                                |
|------------|--------------|-------|--------------------------------------------|
| <b>VCO</b> | <b>I1</b>    | A     | Current amplitude.                         |
|            | <b>I0</b>    | A     | Current baseline or Off level.             |
|            | <b>dFdV</b>  | Hz/V  | Gain.                                      |
|            | <b>Type</b>  |       | Signal type: Sin/Square/Triangle/Sawtooth. |
|            | <b>Phase</b> | deg   | Phase.                                     |

Voltage controlled oscillator. Output current is a signal with frequency equal to:

$$f(\text{Hz}) = \text{dFdV} * V_{in}.$$

For **Sine** signal, **I0** is baseline, and **I1** is amplitude. For **Square**, **Triangle**, and **Sawtooth** signals, **I0** is Off level, **I1** is On level. **Phase** is additional phase of the signal, in degrees.

| Model           | Parameter        | Units | Description        |
|-----------------|------------------|-------|--------------------|
| <b>One-shot</b> | <b>I1</b>        | A     | Pulse On current.  |
|                 | <b>I0</b>        | A     | Pulse Off current. |
|                 | <b>Width</b>     | s     | Pulse width.       |
|                 | <b>Threshold</b> | V     | Voltage threshold. |

One-shot pulse generator. When increasing input voltage  $V_{in}$  crosses **Threshold** value, current pulse of **Width** duration is generated. **I0** is pulse Off level, **I1** is pulse On level.

If increasing  $V_{in}$  crosses **Threshold** value while pulse is being generated, the pulse is restarted.

| Model      | Parameter    | Units | Description                               |
|------------|--------------|-------|-------------------------------------------|
| <b>PWM</b> | <b>I1</b>    | A     | Pulse On current.                         |
|            | <b>I0</b>    | A     | Pulse Off current.                        |
|            | <b>F</b>     | Hz    | Frequency.                                |
|            | <b>Vmax</b>  | V     | Input voltage corresponding to 100% duty. |
|            | <b>Phase</b> | deg   | Phase.                                    |

Voltage controlled Pulse-Width Modulator. Output current is a pulse signal of frequency **F** shifted by **Phase**. Input voltage  $V_{in}$  is sampled at the beginning of each cycle of the signal, and width of the output pulse during this cycle is calculated according to the equation:

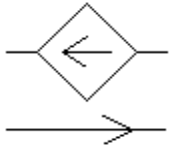
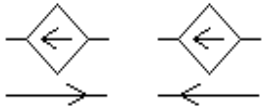
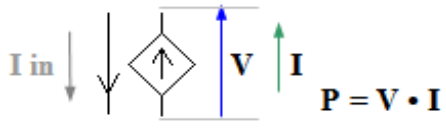
$$\text{width} = 1/F * (V_{in} / V_{\text{max}})$$

or

$$\text{duty} = 100\% * (V_{in} / V_{\text{max}});$$

If the width is equal or less than zero, a short On pulse with the width equal to the minimum calculation step at that moment will be generated. If the width is equal or greater than period of frequency **F**, a short Off pulse at the end of the period will be generated. As a result, the frequency of the output signal is always **F**.

## I – Current controlled current source

| Symbol                                                                                     | Models                         | Signals                                                                             |
|--------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------|
|           | Linear<br>I<br>Function<br>PWL | CCO<br>One-shot<br>PWM<br>SubCir                                                    |
| Views<br> |                                |  |

| Model                                                        | Parameter | Units | Description |
|--------------------------------------------------------------|-----------|-------|-------------|
| <b>Linear</b>                                                | <b>K</b>  | A/A   | Gain        |
| Linear current controlled current source: $I = K * I_{in}$ . |           |       |             |

| Model                 | Parameter | Units | Description |
|-----------------------|-----------|-------|-------------|
| <b>I</b>              | <b>I</b>  | A     | Current.    |
| Constant current = I. |           |       |             |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Parameter | Units | Description                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Function</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>f</b>  | A     | Output as function of the input.   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>IC</b> | A     | Initial condition: output current. |
| <p>Arbitrary function <b>f</b> defines output current as a function of the following variables:</p> <ul style="list-style-type: none"> <li><b>x</b> – input current <i>I<sub>in</sub></i></li> <li><b>t</b> - current time</li> <li><b>V(name)</b> - voltage on the component <b>name</b></li> <li><b>I(name)</b> - current through the component <b>name</b></li> <li><b>P(name)</b> – power on the component <b>name</b></li> <li><b>S(name)</b> – state of the component <b>name</b></li> </ul> <p>where <b>name</b> is the name of the component in the schematic. If <b>f</b> is blank, output is zero.</p> <p>Example:</p> <pre>f = x*x f = x * sin(t) f = P(r1) + P(r2)</pre> <p>When calculating DC operating point, and in AC analysis, output is set to specified output current <b>IC</b>. Please note that variable <b>x</b> (input current <i>I<sub>in</sub></i>) and variables <b>V</b>, <b>I</b>, <b>P</b>, and <b>S</b> are taken at previous calculation step. This may affect stability of the schematic with closed loop.</p> |           |       |                                    |

| Model      | Parameter  | Units | Description                             |
|------------|------------|-------|-----------------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, K( <i>lin</i> ) |

Piecewise linear current controlled current source. **pwl** string defines gain as a function of input current K(*lin*). The transfer function of the source I(*lin*) is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that K(*lin*) is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

| Model      | Parameter    | Units | Description                                |
|------------|--------------|-------|--------------------------------------------|
| <b>CCO</b> | <b>I1</b>    | A     | Current amplitude.                         |
|            | <b>I0</b>    | A     | Current baseline or Off level.             |
|            | <b>dFdl</b>  | Hz/A  | Gain.                                      |
|            | <b>Type</b>  |       | Signal type: Sin/Square/Triangle/Sawtooth. |
|            | <b>Phase</b> | deg   | Phase.                                     |

Current controlled oscillator. Output current is a signal with frequency equal to:

$$f(\text{Hz}) = \text{dFdl} * \text{lin}.$$

For **Sine** signal, **I0** is baseline, and **I1** is amplitude. For **Square**, **Triangle**, and **Sawtooth** signals, **I0** is Off level, **I1** is On level. **Phase** is additional phase of the signal, in degrees.

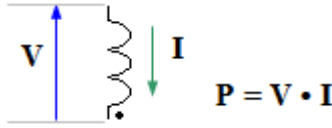
| Model           | Parameter        | Units | Description        |
|-----------------|------------------|-------|--------------------|
| <b>One-shot</b> | <b>I1</b>        | A     | Pulse On current.  |
|                 | <b>I0</b>        | A     | Pulse Off current. |
|                 | <b>Width</b>     | s     | Pulse width.       |
|                 | <b>Threshold</b> | A     | Current threshold. |

One-shot pulse generator. When increasing input current *lin* crosses **Threshold** value, current pulse of **Width** duration is generated. **I0** is pulse Off level, **I1** is pulse On level.

If increasing *lin* crosses **Threshold** value while pulse is being generated, the pulse is restarted.

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Parameter              | Units | Description                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------|-------------------------------------------|
| <b>PWM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>I1</b>              | A     | Pulse On current.                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>I0</b>              | A     | Pulse Off current.                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>F</b>               | Hz    | Frequency.                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>I<sub>max</sub></b> | A     | Input current corresponding to 100% duty. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Phase</b>           | deg   | Phase.                                    |
| <p>Current controlled Pulse-Width Modulator. Output current is a pulse signal of frequency <b>F</b> shifted by <b>Phase</b>. Input current <i>I<sub>in</sub></i> is sampled at the beginning of each cycle of the signal, and width of the output pulse during this cycle is calculated according to the equation:</p> $\text{width} = 1/F * (I_{in} / I_{max})$ <p>or</p> $\text{duty} = 100\% * (I_{in} / I_{max});$ <p>If the width is equal or less than zero, a short On pulse with the width equal to the minimum calculation step at that moment will be generated. If the width is equal or greater than period of frequency <b>F</b>, a short Off pulse at the end of the period will be generated. As a result, the frequency of the output signal is always <b>F</b>.</p> |                        |       |                                           |

## L – Inductor

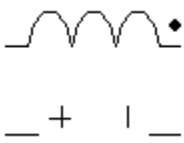
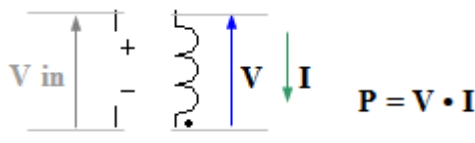
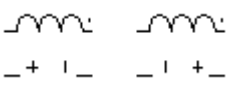
| Symbol                                                                            | Models             | Signals                                                                             |
|-----------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|
|  | L<br>PWL<br>SubCir |  |

| Model                                                                                                                                                                                                                                                                                                                                                                                        | Parameter | Units | Description                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------------------------------|
| <b>L</b>                                                                                                                                                                                                                                                                                                                                                                                     | <b>L</b>  | H     | Inductance                                                 |
|                                                                                                                                                                                                                                                                                                                                                                                              | <b>IC</b> | A     | Initial condition: current. Leave blank if IC not defined. |
| <p>Linear inductor, <math>V = L \cdot di/dt</math>.</p> <p>When calculating DC operating point, if <b>IC</b> is defined, inductor is replaced with current source equal to <b>IC</b>. Otherwise, inductor is temporarily replaced by short circuit, DC operating point is calculated, and then the current through the short circuit is assigned to the inductor as its initial current.</p> |           |       |                                                            |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Parameter  | Units | Description                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|------------------------------------------------------------|
| <b>PWL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>pwl</b> |       | Comma-separated string, L(I)                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>IC</b>  | A     | Initial condition: current. Leave blank if IC not defined. |
| <p>Piecewise constant inductor: <b>pwl</b> string defines inductance as a function of current through the inductor L(I). H(I) is piecewise linear function, and it always goes through the origin (0,0). See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that L(I) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> <p>When calculating DC operating point, if <b>IC</b> is defined, inductor is replaced with current source equal to <b>IC</b>. Otherwise, inductor is temporarily replaced by short circuit, DC operating point is calculated, and then the current through the short circuit is assigned to the inductor as its initial current.</p> |            |       |                                                            |

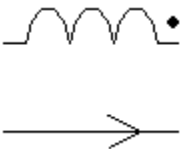
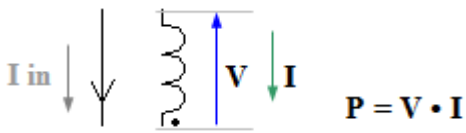
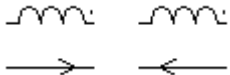


L – Voltage controlled inductor

| Symbol                                                                            | Models                                                                            | Signals                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | PWL                                                                               |  |
| Views                                                                             |  |                                                                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Parameter | Units | Description                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------------------------------|
| PWL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | pwl       |       | Comma-separated string, L(Vin)                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | IC        | A     | Initial condition: current. Leave blank if IC not defined. |
| <p>Piecewise constant voltage controlled inductor. <b>pwl</b> string defines inductance as a function of control voltage L(Vin). At any moment:</p> $V = L(Vin) * di/dt.$ <p>See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that L(Vin) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> <p>When calculating DC operating point, if <b>IC</b> is defined, inductor is replaced with current source equal to <b>IC</b>. Otherwise, inductor is temporarily replaced by short circuit, DC operating point is calculated, and then the current through the short circuit is assigned to the inductor as its initial current.</p> |           |       |                                                            |

L – Current controlled inductor

| Symbol                                                                                     | Models | Signals                                                                            |
|--------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------|
|           | PWL    |  |
| Views<br> |        |                                                                                    |

| Model | Parameter | Units | Description                                                |
|-------|-----------|-------|------------------------------------------------------------|
| PWL   | pwl       |       | Comma-separated string, L(lin)                             |
|       | IC        | A     | Initial condition: current. Leave blank if IC not defined. |

Piecewise constant current controlled inductor. **pwl** string defines inductance as a function of control current C(lin). At any moment:

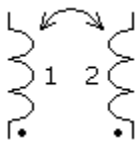
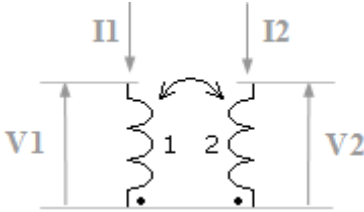
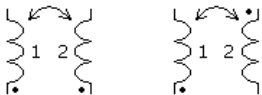
$$V = L(lin) * dl/dt.$$

See *PWL model* chapter of NL5 Manual for details.

Please note that L(lin) is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

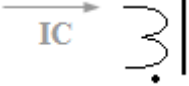
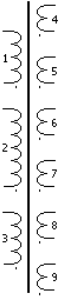
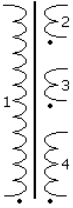
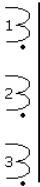
When calculating DC operating point, if **IC** is defined, inductor is replaced with current source equal to **IC**. Otherwise, inductor is temporarily replaced by short circuit, DC operating point is calculated, and then the current through the short circuit is assigned to the inductor as its initial current.

L – Coupled inductors

| Symbol                                                                            | Models                                                                            | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | L                                                                                 |  |
| Views                                                                             |  |                                                                                     |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                 | Parameter | Units | Description                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|----------------------------------------------------------------|
| L                                                                                                                                                                                                                                                                                                                                                                                                                     | L1        | H     | L1 inductance                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | L2        | H     | L2 inductance                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | K         |       | Coupling coefficient (-1...1)                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | IC1       | A     | L1 initial condition: current. Leave blank if IC1 not defined. |
|                                                                                                                                                                                                                                                                                                                                                                                                                       | IC2       | A     | L2 initial condition: current. Leave blank if IC2 not defined. |
| <p>Coupled linear inductors.</p> $V1 = L1 * dI1/dt + M * dI2/dt$ $V2 = M * dI1/dt + L2 * dI2/dt$ <p>Where <math>M = K * \text{sqrt}(L1 * L2)</math> is mutual inductance.</p> <p>When calculating DC operating point, initial conditions <b>IC1</b> and <b>IC2</b> are independently applied to corresponding inductors <b>L1</b> and <b>L2</b>, similar to how it is done for the component <b>L</b> (inductor).</p> |           |       |                                                                |

L – Custom coupled inductors

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Models      | Signals                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | L<br>SubCir |  |
| <p>This is a customized component. A component can be edited in the <b>Edit Component</b> dialog box. See <i>Customized component</i> chapter of NL5 Manual for instructions on editing a component.</p> <p>This component may have:</p> <ul style="list-style-type: none"><li>- height from 2 to 32,</li><li>- up to 32 windings (total),</li><li>- arbitrary length of a winding.</li></ul> <p>Examples of Custom coupled inductors component:</p> <div></div> |             |                                                                                     |

| Model    | Parameter      | Units | Description                                                    |
|----------|----------------|-------|----------------------------------------------------------------|
| <b>L</b> | <b>L1</b>      | H     | L1 inductance                                                  |
|          | <b>...</b>     | H     | <b>...</b>                                                     |
|          | <b>LN</b>      | H     | LN inductance                                                  |
|          | <b>K12</b>     |       | L1-L2 coupling coefficient (-1...1)                            |
|          | <b>...</b>     |       | <b>...</b>                                                     |
|          | <b>K(N-1)N</b> |       | L(N-1)-LN coupling coefficient (-1...1)                        |
|          | <b>IC1</b>     | A     | L1 initial condition: current. Leave blank if IC1 not defined. |
|          | <b>...</b>     | A     | <b>...</b>                                                     |
|          | <b>ICN</b>     | A     | LN initial condition: current. Leave blank if ICN not defined. |

Custom coupled inductors.

$$V1 = L1 * dI1/dt + M12 * dI2/dt + ... + M1N * dIN/dt$$

$$V2 = M12 * dI1/dt + L2 * dI2/dt + ... + M2N * dIN/dt$$

...

$$VN = M1N * dI1/dt + M2N * dI2/dt + ... + LN * dIN/dt$$

Where  $M_{ij} = K_{ij} * \sqrt{L_i * L_j}$  is mutual inductance,  $M_{ij} = M_{ji}$ .

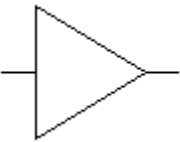
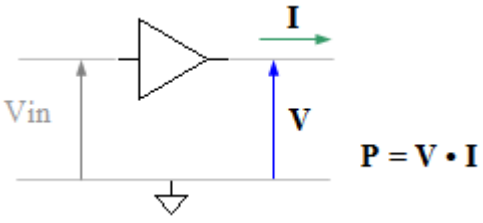
When calculating DC operating point, initial conditions **ICN** are independently applied to corresponding inductors **LN**, similar to how it is done for the component **L** (inductor).

If only one winding is defined, a component behaves exactly as a linear inductor **L**.

Please be aware that coupling coefficients **Kij** should be properly specified within allowable range (-1...1) in order to represent a "physically-realizable" system. If all coupling coefficients are equal to 1 (or -1), using **Winding** components **W** with one magnetizing inductor may give better performance and more stable solution.

If number of windings is more than 9, coupling coefficient parameters **Kij** are shown with underscore '\_' between numbers **i** and **j**: for example, **K6\_24**. For number of windings 9 and less, the old notation is used for backward compatibility.

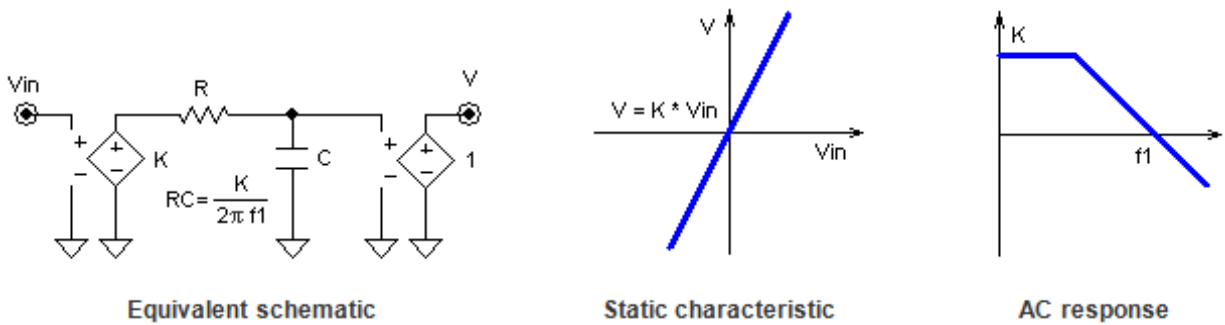
O – Amplifier

| Symbol                                                                            | Models                                                     | Signals                                                                            |
|-----------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | Linear<br>OpAmp<br>Comparator<br>Function<br>PWL<br>SubCir |  |

| Model  | Parameter | Units | Description                        |
|--------|-----------|-------|------------------------------------|
| Linear | K         | V/V   | Gain                               |
|        | f1        | Hz    | Unit gain frequency.               |
|        | IC        | V     | Initial condition: output voltage. |

Linear amplifier. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**).

When calculating DC operating point, if **f1** is not infinity, and **IC** is defined, amplifier output is set to specified output voltage **IC**. If **IC** is blank, static characteristic is used.

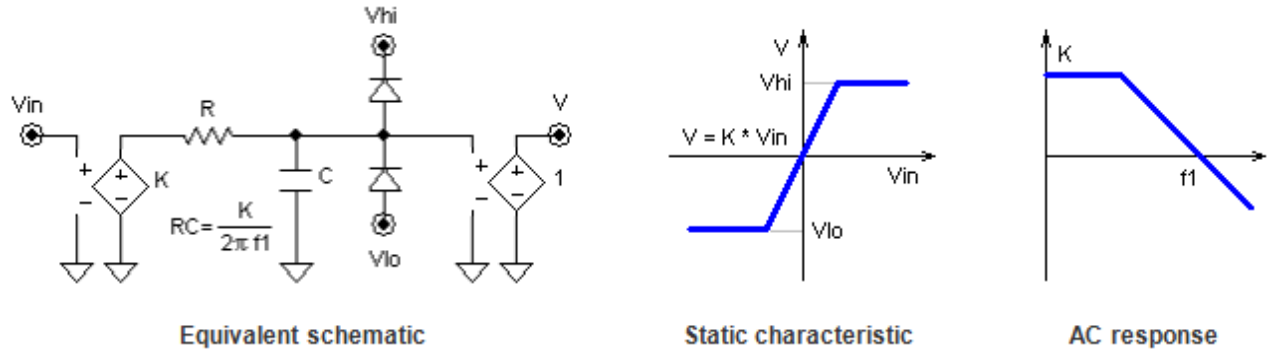


| Model | Parameter  | Units | Description                        |
|-------|------------|-------|------------------------------------|
| OpAmp | <b>K</b>   | V/V   | Gain                               |
|       | <b>f1</b>  | Hz    | Unit gain frequency.               |
|       | <b>Vhi</b> | V     | Max output voltage.                |
|       | <b>Vlo</b> | V     | Min output voltage.                |
|       | <b>IC</b>  | V     | Initial condition: output voltage. |

Linear amplifier with output limiter. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**). Output voltage is limiting between **Vlo** and **Vhi**.

When calculating DC operating point, amplifier output is set to specified output voltage **IC**.

If both **K** and **f1** are set to infinity, the model may experience convergence problem: use **Comparator** model instead.



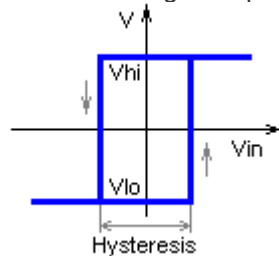
| Model      | Parameter         | Units | Description                  |
|------------|-------------------|-------|------------------------------|
| Comparator | <b>Hysteresis</b> | V     | Hysteresis                   |
|            | <b>Vhi</b>        | V     | Max output voltage.          |
|            | <b>Vlo</b>        | V     | Min output voltage.          |
|            | <b>Delay</b>      | s     | Output delay.                |
|            | <b>IC</b>         |       | Initial condition: Low/High. |

Comparator with hysteresis. Comparator output is set to **Vhi** or **Vlo** using following rules:

$V_{in} > \text{Hysteresis}/2 \dots V = V_{hi}$   
 $V_{in} < -\text{Hysteresis}/2 \dots V = V_{lo}$   
Otherwise .....  $V = \text{previous state}$

The output is delayed by **Delay** time. Input pulses shorter than **Delay** will not pass through and will not affect output.

When calculating DC operating point comparator, output is set to **Vlo** or to **Vhi**, according to selected **IC**.



| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | V     | Output as function of the input.   |
|                 | <b>IC</b> | V     | Initial condition: output voltage. |

Arbitrary function **f** defines output voltage as a function of the following variables:

**x** – input voltage  $V_{in}$

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

$$f = x * x$$

$$f = x * \sin(t)$$

$$f = P(r1) + P(r2)$$

When calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Please note that variable **x** (input voltage  $V_{in}$ ) and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. This may affect stability of the schematic with closed loop.

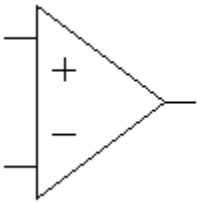
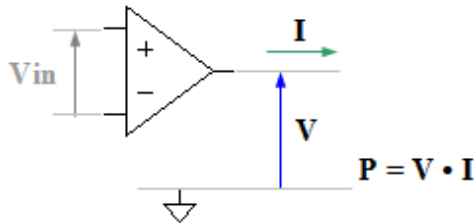
| Model      | Parameter  | Units | Description                         |
|------------|------------|-------|-------------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, $K(V_{in})$ |

Piecewise linear amplifier. **pwl** string defines gain as a function of input voltage  $K(V_{in})$ . Amplifier transfer function is  $V(V_{in})$  is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that  $K(V_{in})$  is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.



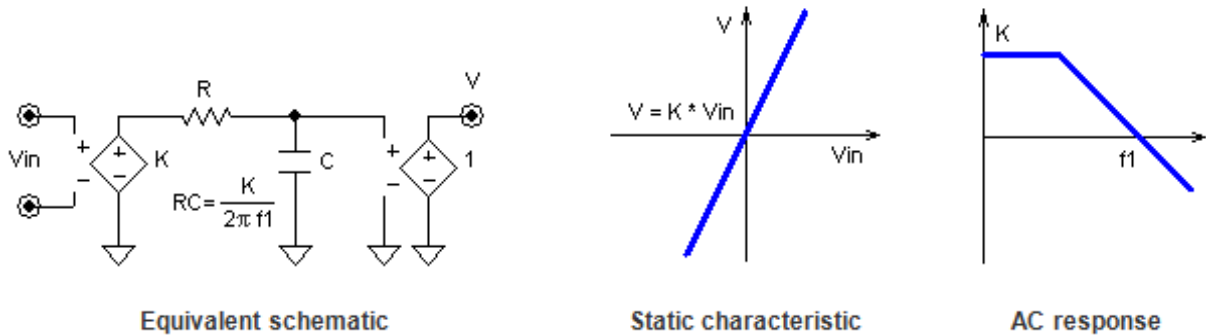
O – Differential amplifier

| Symbol                                                                            | Models                                                     | Signals                                                                            |
|-----------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | Linear<br>OpAmp<br>Comparator<br>Function<br>PWL<br>SubCir |  |

| Model  | Parameter | Units | Description                        |
|--------|-----------|-------|------------------------------------|
| Linear | K         | V/V   | Gain                               |
|        | f1        | Hz    | Unit gain frequency.               |
|        | IC        | V     | Initial condition: output voltage. |

Linear differential amplifier. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**).

When calculating DC operating point, if **f1** is not infinity, and **IC** is defined, amplifier output is set to specified output voltage **IC**. If **IC** is blank, static characteristic is used.

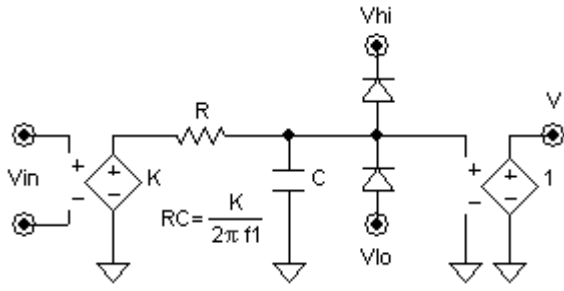


| Model | Parameter | Units | Description                        |
|-------|-----------|-------|------------------------------------|
| OpAmp | K         | V/V   | Gain                               |
|       | f1        | Hz    | Unit gain frequency.               |
|       | Vhi       | V     | Max output voltage.                |
|       | Vlo       | V     | Min output voltage.                |
|       | IC        | V     | Initial condition: output voltage. |

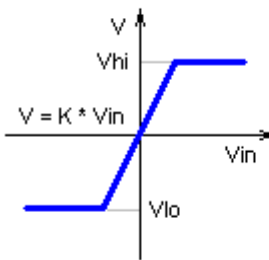
Linear amplifier with output limiter. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**). Output voltage is limiting between **Vlo** and **Vhi**.

When calculating DC operating point, amplifier output is set to specified output voltage **IC**.

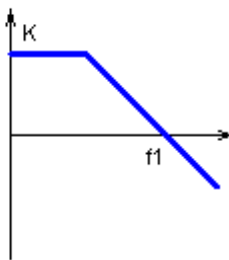
If both **K** and **f1** are set to infinity, the model may experience convergence problem: use **Comparator** model instead.



Equivalent schematic



Static characteristic



AC response

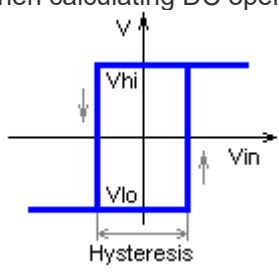
| Model      | Parameter  | Units | Description                  |
|------------|------------|-------|------------------------------|
| Comparator | Hysteresis | V     | Hysteresis                   |
|            | Vhi        | V     | Max output voltage.          |
|            | Vlo        | V     | Min output voltage.          |
|            | Delay      | s     | Output delay.                |
|            | IC         |       | Initial condition: Low/High. |

Comparator with hysteresis. Comparator output is set to **Vhi** or **Vlo** using following rules:

$V_{in} > \text{Hysteresis}/2 \dots V = V_{hi}$   
 $V_{in} < -\text{Hysteresis}/2 \dots V = V_{lo}$   
Otherwise .....  $V = \text{previous state}$

The output is delayed by **Delay** time. Input pulses shorter than **Delay** will not pass through and will not affect output.

When calculating DC operating point comparator, output is set to **Vlo** or to **Vhi**, according to selected **IC**.



| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | V     | Output as function of the input.   |
|                 | <b>IC</b> | V     | Initial condition: output voltage. |

Arbitrary function **f** defines output voltage as a function of the following variables:

**x** – input voltage  $V_{in}$

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

$$f = x * x$$

$$f = x * \sin(t)$$

$$f = P(r1) + P(r2)$$

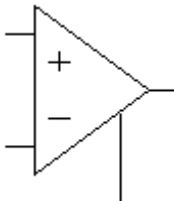
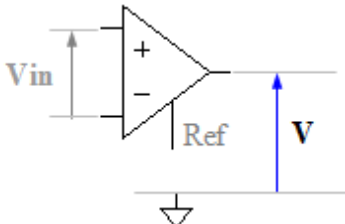
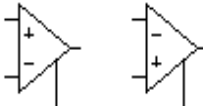
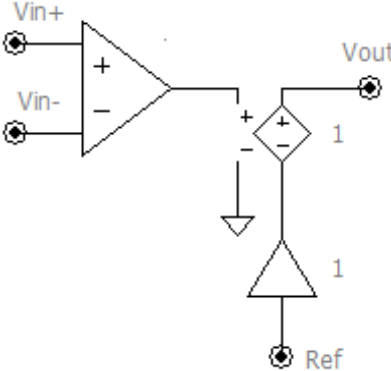
When calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Please note that variable **x** (input voltage  $V_{in}$ ) and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. This may affect stability of the schematic with closed loop.

| Model      | Parameter  | Units | Description                         |
|------------|------------|-------|-------------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, $K(V_{in})$ |

Piecewise linear amplifier. **pwl** string defines gain as a function of input voltage  $K(V_{in})$ . Amplifier transfer function is  $V(V_{in})$  is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that  $K(V_{in})$  is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

O – Differential amplifier with reference

| Symbol                                                                                                                                                                                                                                                                                                                                                                | Models                                                                            | Signals                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                      | Linear<br>OpAmp<br>Comparator<br>Function<br>PWL<br>SubCir                        |  |
| Views                                                                                                                                                                                                                                                                                                                                                                 |  |                                                                                     |
| <p>Equivalentent schematic :</p>                                                                                                                                                                                                                                                    |                                                                                   |                                                                                     |
| <p>All models are similar to the models of <b>Differential amplifier</b> component. Model parameters apply to the differential amplifier shown on the equivalent schematic. For example, <b>Vhi</b> and <b>Vlo</b> parameters of the <b>OpAmp</b> model will limit output of the differential amplifier, rather than voltage at the component output <i>Vout</i>.</p> |                                                                                   |                                                                                     |

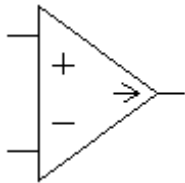
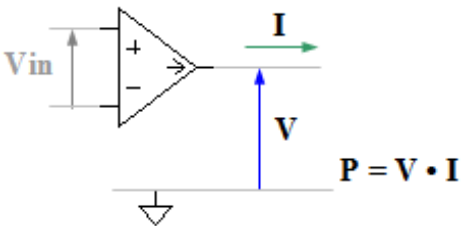
O – Fully differential amplifier

| Symbol | Models                                                     | Signals |
|--------|------------------------------------------------------------|---------|
|        | Linear<br>OpAmp<br>Comparator<br>Function<br>PWL<br>SubCir |         |
| Views  |                                                            |         |

Equivalent schematic:

All models are similar to the models of **Differential amplifier** component. Model parameters apply to the differential amplifier shown on the equivalent schematic. For example, **Vhi** and **Vlo** parameters of the **OpAmp** model will limit output of the differential amplifier, rather than voltages at the component outputs *Vout+* and *Vout-*.

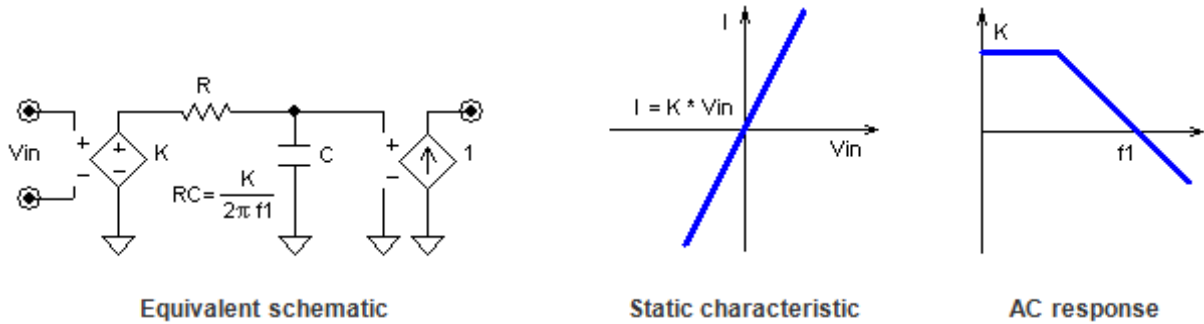
O – Differential amplifier with current output

| Symbol                                                                            | Models                                                     | Signals                                                                            |
|-----------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | Linear<br>OpAmp<br>Comparator<br>Function<br>PWL<br>SubCir |  |

| Model  | Parameter | Units | Description                        |
|--------|-----------|-------|------------------------------------|
| Linear | K         | A/V   | Gain                               |
|        | f1        | Hz    | Unit gain frequency.               |
|        | IC        | A     | Initial condition: output current. |

Linear differential amplifier. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**).

When calculating DC operating point, if **f1** is not infinity, and **IC** is defined, amplifier output is set to specified output current **IC**. If **IC** is blank, static characteristic is used.

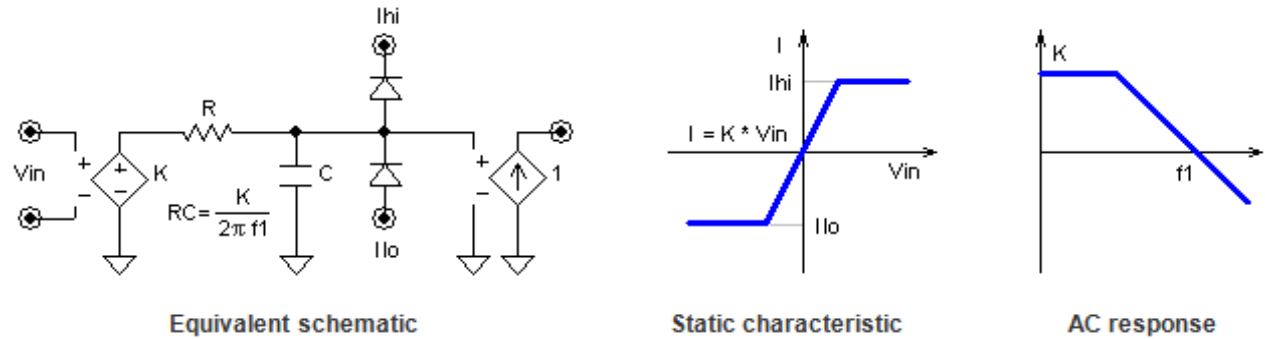


| Model | Parameter | Units | Description                        |
|-------|-----------|-------|------------------------------------|
| OpAmp | K         | V/A   | Gain                               |
|       | f1        | Hz    | Unit gain frequency.               |
|       | Ihi       | A     | Max output current.                |
|       | Ilo       | A     | Min output current.                |
|       | IC        | A     | Initial condition: output current. |

Linear amplifier with output limiter. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**). Output current is limiting between **Ilo** and **Ihi**.

When calculating DC operating point, amplifier output is set to specified output current **IC**.

If both **K** and **f1** are set to infinity, the model may experience convergence problem: use **Comparator** model instead.



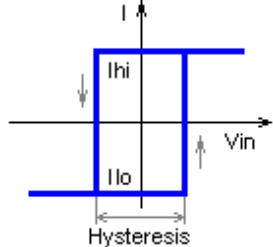
| Model      | Parameter  | Units | Description                  |
|------------|------------|-------|------------------------------|
| Comparator | Hysteresis | V     | Hysteresis                   |
|            | Ihi        | A     | Max output current.          |
|            | Ilo        | A     | Min output current.          |
|            | Delay      | s     | Output delay.                |
|            | IC         |       | Initial condition: Low/High. |

Comparator with hysteresis. Comparator output is set to **Ihi** or **Ilo** using following rules:

$V_{in} > \text{Hysteresis}/2 \dots I = I_{hi}$   
 $V_{in} < -\text{Hysteresis}/2 \dots I = I_{lo}$   
Otherwise .....  $I = \text{previous state}$

The output is delayed by **Delay** time. Input pulses shorter than **Delay** will not pass through and will not affect output.

When calculating DC operating point comparator, output is set to **Ilo** or to **Ihi**, according to selected **IC**.



| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | A     | Output as function of the input.   |
|                 | <b>IC</b> | A     | Initial condition: output current. |

Arbitrary function **f** defines output current as a function of the following variables:

**x** – input voltage  $V_{in}$

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

$$f = x * x$$

$$f = x * \sin(t)$$

$$f = P(r1) + P(r2)$$

When calculating DC operating point, and in AC analysis, output is set to specified output current **IC**. Please note that variable **x** (input voltage  $V_{in}$ ) and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. This may affect stability of the schematic with closed loop.

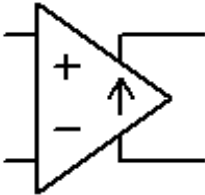
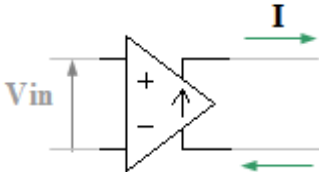
| Model      | Parameter  | Units | Description                         |
|------------|------------|-------|-------------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, $K(V_{in})$ |

Piecewise linear amplifier. **pwl** string defines gain as a function of input voltage  $K(V_{in})$ . Amplifier transfer function is  $I(V_{in})$  is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that  $K(V_{in})$  is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.



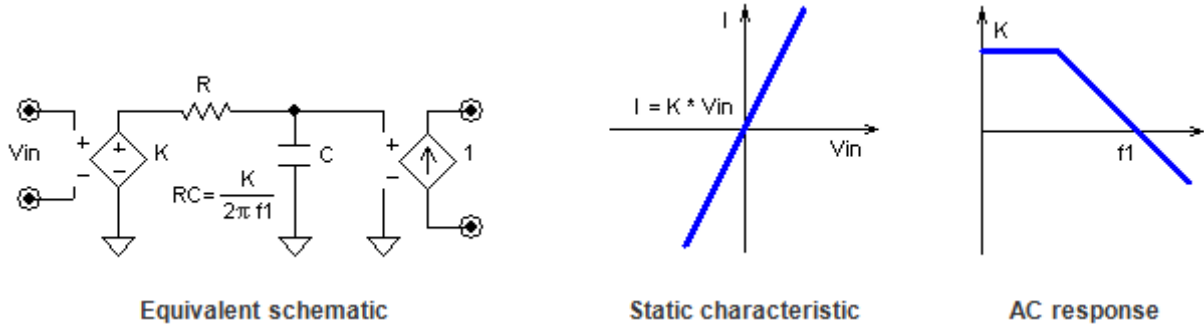
O – Differential amplifier with differential current output

| Symbol                                                                            | Models                                                     | Signals                                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Linear<br>OpAmp<br>Comparator<br>Function<br>PWL<br>SubCir |  |

| Model  | Parameter | Units | Description                        |
|--------|-----------|-------|------------------------------------|
| Linear | K         | A/V   | Gain                               |
|        | f1        | Hz    | Unit gain frequency.               |
|        | IC        | A     | Initial condition: output current. |

Linear differential amplifier. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**).

When calculating DC operating point, if **f1** is not infinity, and **IC** is defined, amplifier output is set to specified output current **IC**. If **IC** is blank, static characteristic is used.

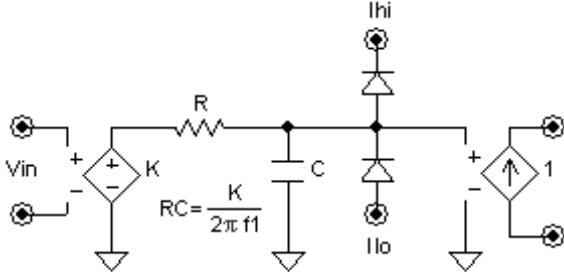


| Model | Parameter | Units | Description                        |
|-------|-----------|-------|------------------------------------|
| OpAmp | K         | V/A   | Gain                               |
|       | f1        | Hz    | Unit gain frequency.               |
|       | Ihi       | A     | Max output current.                |
|       | Ilo       | A     | Min output current.                |
|       | IC        | A     | Initial condition: output current. |

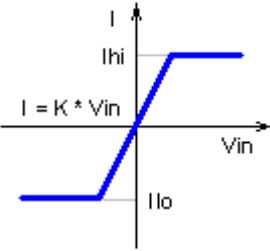
Linear amplifier with output limiter. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**). Output current is limiting between **Ilo** and **Ihi**.

When calculating DC operating point, amplifier output is set to specified output current **IC**.

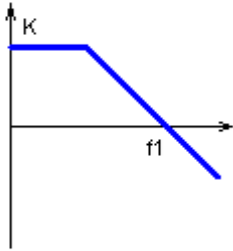
If both **K** and **f1** are set to infinity, the model may experience convergence problem: use **Comparator** model instead.



Equivalent schematic



Static characteristic



AC response

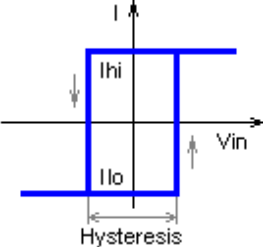
| Model      | Parameter  | Units | Description                  |
|------------|------------|-------|------------------------------|
| Comparator | Hysteresis | V     | Hysteresis                   |
|            | Ihi        | A     | Max output current.          |
|            | Ilo        | A     | Min output current.          |
|            | Delay      | s     | Output delay.                |
|            | IC         |       | Initial condition: Low/High. |

Comparator with hysteresis. Comparator output is set to **Ihi** or **Ilo** using following rules:

$V_{in} > \text{Hysteresis}/2 \dots I = I_{hi}$   
 $V_{in} < -\text{Hysteresis}/2 \dots I = I_{lo}$   
Otherwise .....  $I = \text{previous state}$

The output is delayed by **Delay** time. Input pulses shorter than **Delay** will not pass through and will not affect output.

When calculating DC operating point comparator, output is set to **Ilo** or to **Ihi**, according to selected **IC**.



| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | A     | Output as function of the input.   |
|                 | <b>IC</b> | A     | Initial condition: output current. |

Arbitrary function **f** defines output current as a function of the following variables:

**x** – input voltage  $V_{in}$

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

$$f = x * x$$

$$f = x * \sin(t)$$

$$f = P(r1) + P(r2)$$

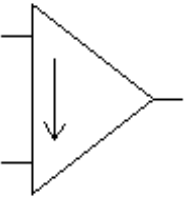
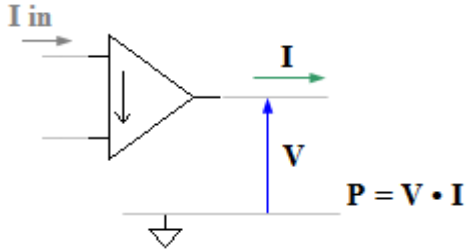
When calculating DC operating point, and in AC analysis, output is set to specified output current **IC**. Please note that variable **x** (input voltage  $V_{in}$ ) and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. This may affect stability of the schematic with closed loop.

| Model      | Parameter  | Units | Description                         |
|------------|------------|-------|-------------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, $K(V_{in})$ |

Piecewise linear amplifier. **pwl** string defines gain as a function of input voltage  $K(V_{in})$ . Amplifier transfer function is  $I(V_{in})$  is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that  $K(V_{in})$  is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

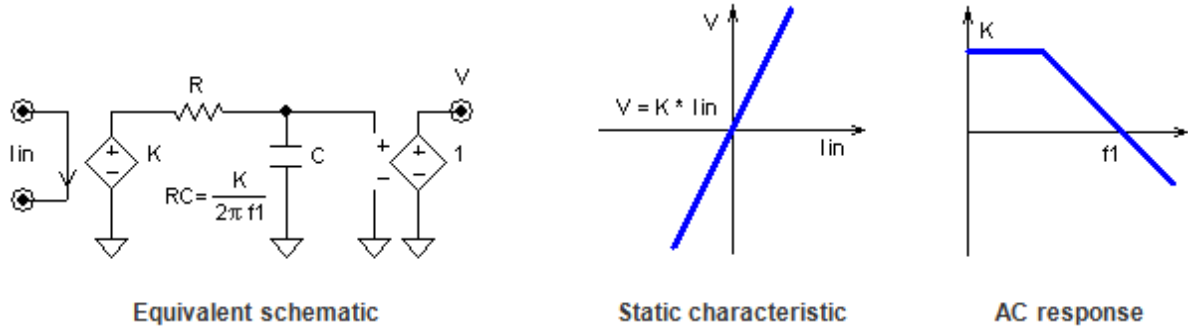
O – Current amplifier

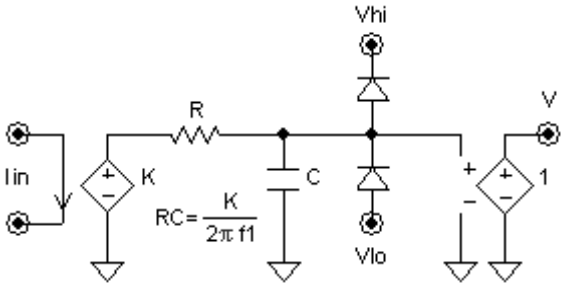
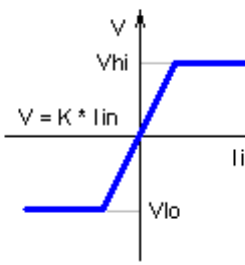
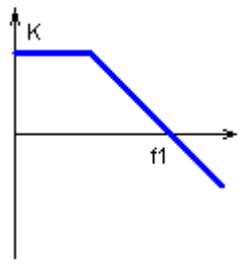
| Symbol                                                                            | Models                                                     | Signals                                                                            |
|-----------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | Linear<br>OpAmp<br>Comparator<br>Function<br>PWL<br>SubCir |  |

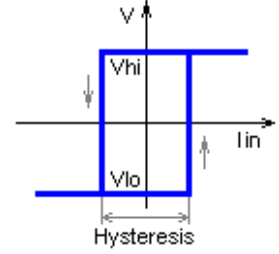
| Model  | Parameter | Units | Description                        |
|--------|-----------|-------|------------------------------------|
| Linear | K         | V/A   | Gain                               |
|        | f1        | Hz    | Unit gain frequency.               |
|        | IC        | V     | Initial condition: output voltage. |

Linear differential amplifier. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**).

When calculating DC operating point, if **f1** is not infinity, and **IC** is defined, amplifier output is set to specified output voltage **IC**. If **IC** is blank, static characteristic is used.



| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Parameter | Units | Description                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| OpAmp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | K         | V/A   | Gain                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | f1        | Hz    | Unit gain frequency.               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Vhi       | V     | Max output voltage.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Vlo       | V     | Min output voltage.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | IC        | V     | Initial condition: output voltage. |
| <p>Linear amplifier with output limiter. <b>K</b> is open loop gain. Frequency response consists of one pole, <b>f1</b> is unit gain frequency. <b>K</b> and <b>f1</b> can be set to infinity (<b>inf</b>). Output voltage is limiting between <b>Vlo</b> and <b>Vhi</b>.</p> <p>When calculating DC operating point, amplifier output is set to specified output voltage <b>IC</b>.</p> <p>If both <b>K</b> and <b>f1</b> are set to infinity, the model may experience convergence problem: use <b>Comparator</b> model instead.</p> <div><div><p>Equivalent schematic</p></div><div><p>Static characteristic</p></div><div><p>AC response</p></div></div> |           |       |                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Parameter  | Units | Description                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|------------------------------|
| Comparator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hysteresis | A     | Hysteresis                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Vhi        | V     | Max output voltage.          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Vlo        | V     | Min output voltage.          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Delay      | s     | Output delay.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | IC         |       | Initial condition: Low/High. |
| <p>Comparator with hysteresis. Comparator output is set to <b>Vhi</b> or <b>Vlo</b> using following rules:</p> <p><math>I_{in} &gt; \text{Hysteresis}/2 \dots : V = V_{hi}</math><br/><math>I_{in} &lt; -\text{Hysteresis}/2 \dots : V = V_{lo}</math><br/>Otherwise .....: <math>V = \text{previous state}</math></p> <p>The output is delayed by <b>Delay</b> time. Input pulses shorter than <b>Delay</b> will not pass through and will not affect output.</p> <p>When calculating DC operating point comparator, output is set to <b>Vlo</b> or to <b>Vhi</b>, according to selected <b>IC</b>.</p> <div></div> |            |       |                              |

| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | V     | Output as function of the input.   |
|                 | <b>IC</b> | V     | Initial condition: output voltage. |

Arbitrary function **f** defines output voltage as a function of the following variables:

**x** – input current *lin*

**t** - current time

**V(name)** - voltage on the component **name**

**I(name)** - current through the component **name**

**P(name)** – power on the component **name**

**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

Example:

$$f = x * x$$

$$f = x * \sin(t)$$

$$f = P(r1) + P(r2)$$

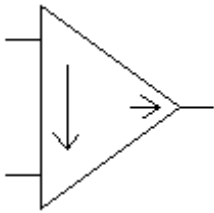
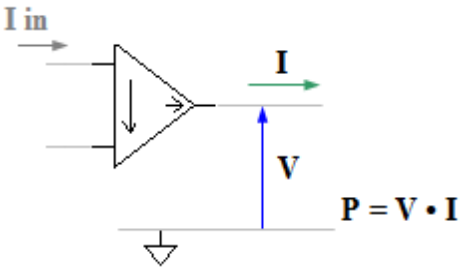
When calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Please note that variable **x** (input current *lin*) and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. This may affect stability of the schematic with closed loop.

| Model      | Parameter  | Units | Description                    |
|------------|------------|-------|--------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, K(Vin) |

Piecewise linear amplifier. **pwl** string defines gain as a function of input current  $K(lin)$ . Amplifier transfer function is  $V(lin)$  is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that  $K(lin)$  is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

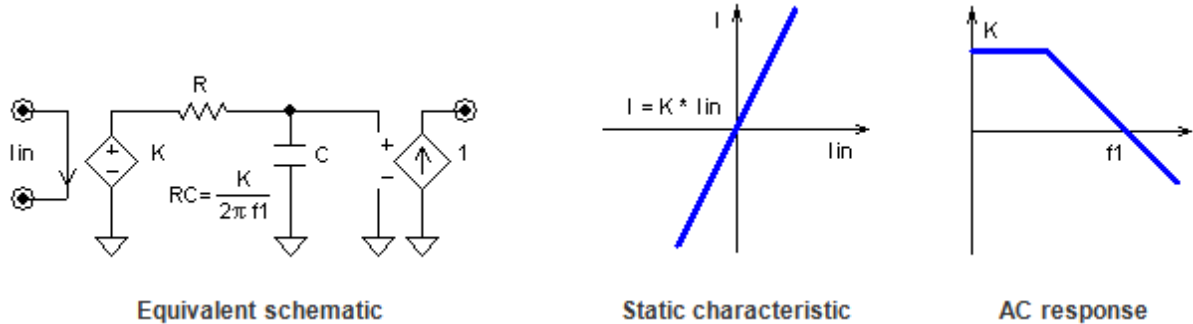
O – Current amplifier with current output

| Symbol                                                                            | Models                                                     | Signals                                                                            |
|-----------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | Linear<br>OpAmp<br>Comparator<br>Function<br>PWL<br>SubCir |  |

| Model  | Parameter | Units | Description                        |
|--------|-----------|-------|------------------------------------|
| Linear | K         | A/A   | Gain                               |
|        | f1        | Hz    | Unit gain frequency.               |
|        | IC        | A     | Initial condition: output current. |

Linear differential amplifier. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**).

When calculating DC operating point, if **f1** is not infinity, and **IC** is defined, amplifier output is set to specified output current **IC**. If **IC** is blank, static characteristic is used.

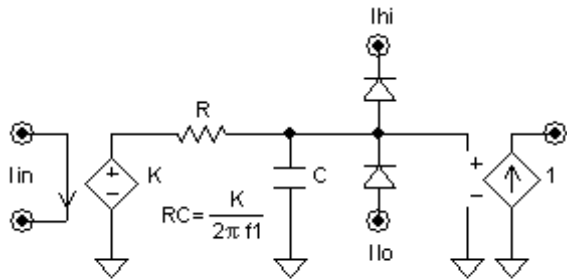


| Model | Parameter | Units | Description                        |
|-------|-----------|-------|------------------------------------|
| OpAmp | K         | A/A   | Gain                               |
|       | f1        | Hz    | Unit gain frequency.               |
|       | Ihi       | A     | Max output current.                |
|       | Ilo       | A     | Min output current.                |
|       | IC        | A     | Initial condition: output current. |

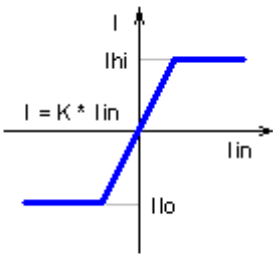
Linear amplifier with output limiter. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**). Output current is limiting between **Ilo** and **Ihi**.

When calculating DC operating point, amplifier output is set to specified output current **IC**.

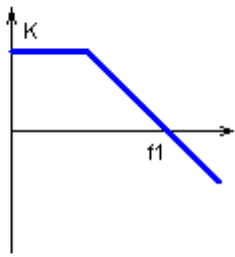
If both **K** and **f1** are set to infinity, the model may experience convergence problem: use **Comparator** model instead.



Equivalent schematic



Static characteristic



AC response



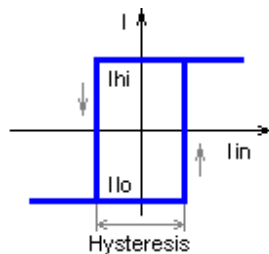
| Model             | Parameter         | Units | Description                  |
|-------------------|-------------------|-------|------------------------------|
| <b>Comparator</b> | <b>Hysteresis</b> | A     | Hysteresis                   |
|                   | <b>Ihi</b>        | A     | Max output current.          |
|                   | <b>Ilo</b>        | A     | Min output current.          |
|                   | <b>Delay</b>      | s     | Output delay.                |
|                   | <b>IC</b>         |       | Initial condition: Low/High. |

Comparator with hysteresis. Comparator output is set to **Ihi** or **Ilo** using following rules:

$I_{in} > \text{Hysteresis}/2 \dots : I = I_{hi}$   
 $I_{in} < -\text{Hysteresis}/2 \dots : I = I_{lo}$   
 Otherwise . . . . . :  $I = \text{previous state}$

The output is delayed by **Delay** time. Input pulses shorter than **Delay** will not pass through and will not affect output.

When calculating DC operating point comparator, output is set to **Ilo** or to **Ihi**, according to selected **IC**.



| Model           | Parameter | Units | Description                        |
|-----------------|-----------|-------|------------------------------------|
| <b>Function</b> | <b>f</b>  | A     | Output as function of the input.   |
|                 | <b>IC</b> | A     | Initial condition: output current. |

Arbitrary function **f** defines output current as a function of the following variables:

**x** – input current  $I_{in}$   
**t** - current time  
**V(name)** - voltage on the component **name**  
**I(name)** - current through the component **name**  
**P(name)** – power on the component **name**  
**S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

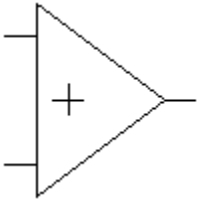
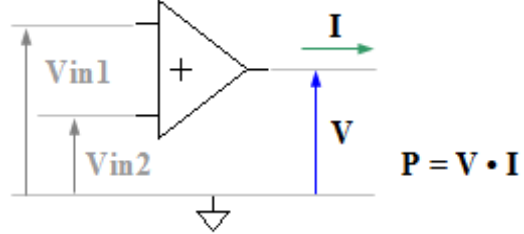
Example:

$f = x * x$   
 $f = x * \sin(t)$   
 $f = P(r1) + P(r2)$

When calculating DC operating point, and in AC analysis, output is set to specified output current **IC**. Please note that variable **x** (input current  $I_{in}$ ) and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. This may affect stability of the schematic with closed loop.

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Parameter  | Units | Description                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|----------------------------------|
| <b>PWL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>pwl</b> |       | Comma-separated string, $K(lin)$ |
| <p>Piecewise linear amplifier. <b>pwl</b> string defines gain as a function of input current <math>K(lin)</math>. Amplifier transfer function is <math>I(lin)</math> is piecewise linear function, and it always goes through the origin (0,0). See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that <math>K(lin)</math> is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |            |       |                                  |

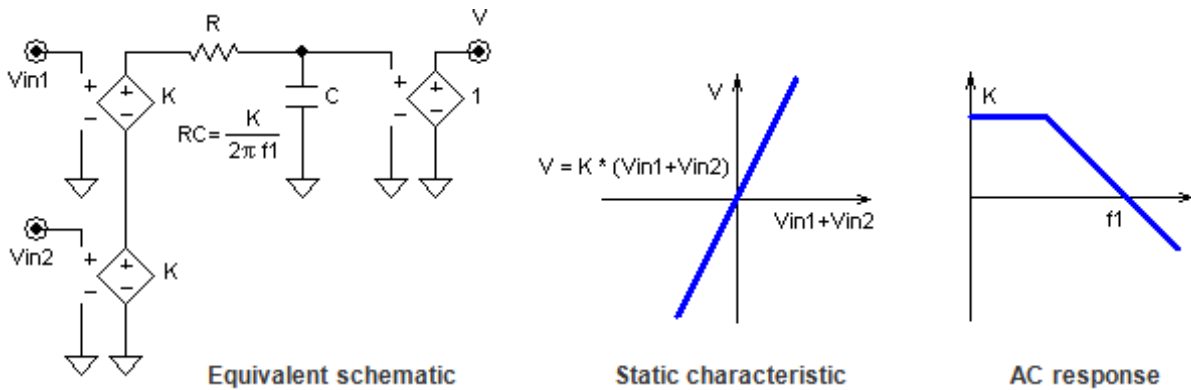
O – Summing amplifier

| Symbol                                                                            | Models                                       | Signals                                                                            |
|-----------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------|
|  | Linear<br>OpAmp<br>Function<br>PWL<br>SubCir |  |

| Model  | Parameter | Units | Description                        |
|--------|-----------|-------|------------------------------------|
| Linear | K         | V/V   | Gain                               |
|        | f1        | Hz    | Unit gain frequency.               |
|        | IC        | V     | Initial condition: output voltage. |

Linear summing amplifier. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**).

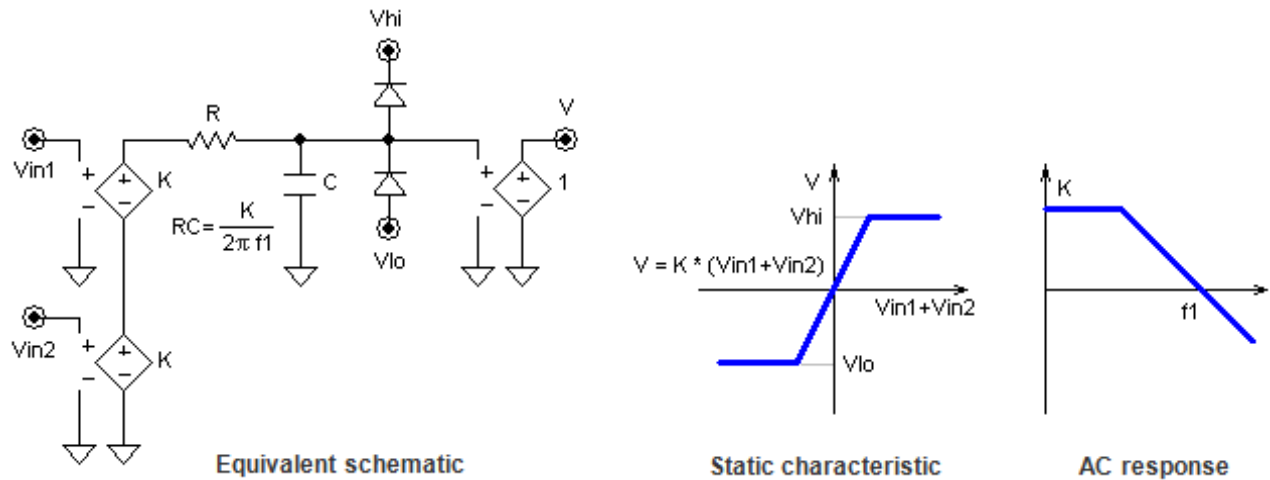
When calculating DC operating point, if **f1** is not infinity, and **IC** is defined, amplifier output is set to specified output voltage **IC**. If **IC** is blank, static characteristic is used.



| Model | Parameter | Units | Description                        |
|-------|-----------|-------|------------------------------------|
| OpAmp | K         | V/V   | Gain                               |
|       | f1        | Hz    | Unit gain frequency.               |
|       | Vhi       | V     | Max output voltage.                |
|       | Vlo       | V     | Min output voltage.                |
|       | IC        | V     | Initial condition: output voltage. |

Linear amplifier with output limiter. **K** is open loop gain. Frequency response consists of one pole, **f1** is unit gain frequency. **K** and **f1** can be set to infinity (**inf**). Output voltage is limiting between **Vlo** and **Vhi**.

When calculating DC operating point, amplifier output is set to specified output voltage **IC**.



| Model    | Parameter | Units | Description                        |
|----------|-----------|-------|------------------------------------|
| Function | f         | V     | Output as function of the input.   |
|          | IC        | V     | Initial condition: output voltage. |

Arbitrary function **f** defines output voltage as a function of the following variables:

- x** – voltage  $V_{in1} + V_{in2}$
- t** - current time
- V(name)** - voltage on the component **name**
- I(name)** - current through the component **name**
- P(name)** – power on the component **name**
- S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, output is zero.

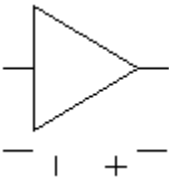
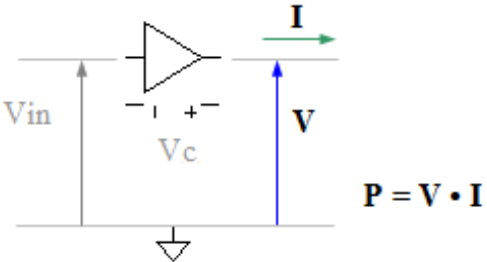
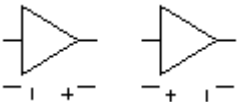
Example:

- $f = x * x$
- $f = x * \sin(t)$
- $f = P(r1) + P(r2)$

When calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Please note that variable **x** (input voltage  $V_{in1} + V_{in2}$ ) and variables **V**, **I**, **P**, and **S** are taken at previous calculation step. This may affect stability of the schematic with closed loop.

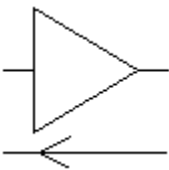
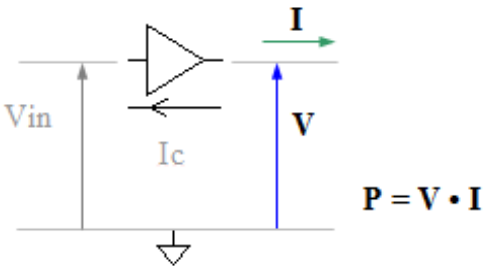
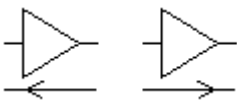
| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Parameter  | Units | Description                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|----------------------------------------------|
| <b>PWL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>pwl</b> |       | Comma-separated string, $K(V_{in1}+V_{in2})$ |
| <p>Piecewise linear amplifier. <b>pwl</b> string defines gain as a function of sum of input voltages <math>K(V_{in1}+V_{in2})</math>. Amplifier transfer function is <math>V(V_{in1}+V_{in2})</math> is piecewise linear function, and it always goes through the origin (0,0). See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that <math>K(V_{in1}+V_{in2})</math> is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |            |       |                                              |

O – Voltage controlled amplifier

| Symbol                                                                            | Models                                                                            | Signals                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | PWL                                                                               |  |
| Views                                                                             |  |                                                                                    |

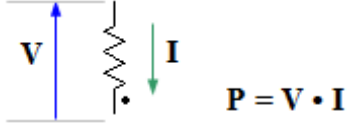
| Model                                                                                                                                                                                                                                                                                                                                                                                     | Parameter | Units | Description                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-------------------------------|
| PWL                                                                                                                                                                                                                                                                                                                                                                                       | pwl       |       | Comma-separated string, K(Vc) |
| <p>Piecewise constant voltage controlled amplifier. <b>pwl</b> string defines gain as a function of control voltage K(Vc). At any moment:</p> $V = K(Vc) * Vin.$ <p>See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that K(Vc) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |           |       |                               |

O – Current controlled amplifier

| Symbol                                                                            | Models                                                                            | Signals                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | PWL                                                                               |  |
| Views                                                                             |  |                                                                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                     | Parameter | Units | Description                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-------------------------------|
| PWL                                                                                                                                                                                                                                                                                                                                                                                       | pwl       |       | Comma-separated string, K(Ic) |
| <p>Piecewise constant current controlled amplifier. <b>pwl</b> string defines gain as a function of control current K(Ic). At any moment:</p> $V = K(Ic) * Vin.$ <p>See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that K(Ic) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |           |       |                               |

## R – Resistor

| Symbol                                                                            | Models                      | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------|
|  | R<br>PWL<br>PWL-I<br>SubCir |  |

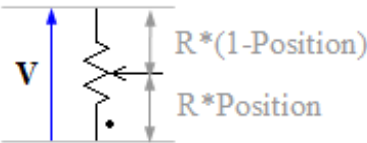
| Model                          | Parameter | Units | Description |
|--------------------------------|-----------|-------|-------------|
| <b>R</b>                       | <b>R</b>  | Ohm   | Resistance  |
| Linear resistor: $V = R * I$ . |           |       |             |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Parameter  | Units | Description                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|------------------------------|
| <b>PWL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>pwl</b> |       | Comma-separated string, R(V) |
| <p>Piecewise constant resistor. <b>pwl</b> string defines resistance as a function of voltage across the resistor R(V). Current through the resistor I(V) is piecewise linear function, and it always goes through the origin (0,0). See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that R(V) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |            |       |                              |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Parameter  | Units | Description                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|------------------------------|
| <b>PWL-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>pwl</b> |       | Comma-separated string, R(I) |
| <p>Piecewise constant resistor. <b>pwl</b> string defines resistance as a function of current through the resistor R(I). Voltage across the resistor V(I) is piecewise linear function, and it always goes through the origin (0,0). See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that R(I) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |            |       |                              |

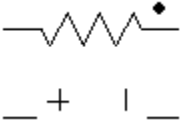
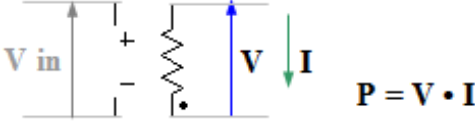
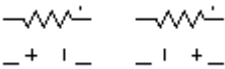


R – Potentiometer

| Symbol                                                                            | Models        | Signals                                                                             |
|-----------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------|
|  | Potentiometer |  |

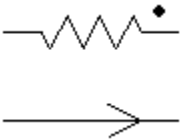
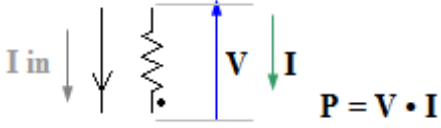
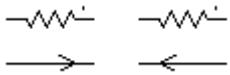
| Model                                                                                                                                                                       | Parameter | Units | Description                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-------------------------------|
| Potentiometer                                                                                                                                                               | R         | Ohm   | Resistance                    |
|                                                                                                                                                                             | Position  |       | Position of the wiper (0...1) |
| <p>Position of the wiper is referenced to the terminal with dot:</p> <p>0 – wiper is connected to the terminal with dot<br/>1 – wiper is connected to another terminal.</p> |           |       |                               |

R – Voltage controlled resistor

| Symbol                                                                            | Models                                                                            | Signals                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | PWL                                                                               |  |
| Views                                                                             |  |                                                                                    |

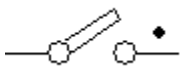
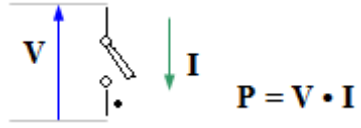
| Model                                                                                                                                                                                                                                                                                                                                                                                           | Parameter | Units | Description                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|--------------------------------|
| PWL                                                                                                                                                                                                                                                                                                                                                                                             | pwl       |       | Comma-separated string, R(Vin) |
| <p>Piecewise constant voltage controlled resistor. <b>pwl</b> string defines resistance as a function of control voltage R(Vin). At any moment:</p> $V = R(Vin) * I.$ <p>See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that R(Vin) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |           |       |                                |

R – Current controlled resistor

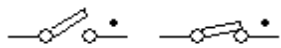
| Symbol                                                                            | Models                                                                            | Signals                                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  | PWL                                                                               |  |
| Views                                                                             |  |                                                                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                           | Parameter | Units | Description                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|--------------------------------|
| PWL                                                                                                                                                                                                                                                                                                                                                                                             | pwl       |       | Comma-separated string, R(lin) |
| <p>Piecewise constant current controlled resistor. <b>pwl</b> string defines resistance as a function of control current R(lin). At any moment:</p> $V = R(lin) * I.$ <p>See <i>PWL model</i> chapter of NL5 Manual for details.</p> <p>Please note that R(lin) is <b>piecewise constant</b> function, although the model and parameter are still called <b>pwl</b> for historical reasons.</p> |           |       |                                |

S – Switch

| Symbol                                                                            | Models                               |                                 | Signals                                                                             |
|-----------------------------------------------------------------------------------|--------------------------------------|---------------------------------|-------------------------------------------------------------------------------------|
|  | Off<br>On<br>Step<br>Single<br>Pulse | Clock<br>List<br>File<br>SubCir |  |

For **Off/On** models and models with **Active** parameter, switch symbol indicates switch position in non-active state:

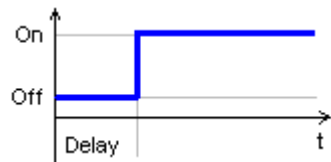


| Model                                         | Parameter      | Units | Description |
|-----------------------------------------------|----------------|-------|-------------|
| Off                                           | No parameters. |       |             |
| Switch is always in Off state (open circuit). |                |       |             |

| Model                                         | Parameter      | Units | Description |
|-----------------------------------------------|----------------|-------|-------------|
| On                                            | No parameters. |       |             |
| Switch is always in On state (short circuit). |                |       |             |

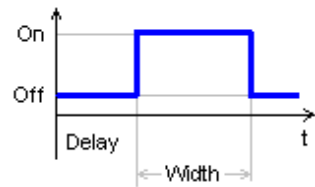
| Model       | Parameter     | Units | Description                  |
|-------------|---------------|-------|------------------------------|
| <b>Step</b> | <b>Delay</b>  | s     | Delay before active state.   |
|             | <b>Active</b> |       | Active switch state: Off/On. |

Switch goes to **Active** state after **Delay** time. Switching diagram for **Active** = On:



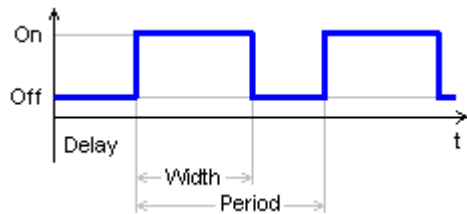
| Model  | Parameter | Units | Description                      |
|--------|-----------|-------|----------------------------------|
| Single | Width     | s     | Pulse width.                     |
|        | Delay     | s     | Delay before first pulse starts. |
|        | Active    |       | Active switch state: Off/On.     |

Single pulse starts after **Delay** time. Switching diagram is shown for **Active** = On:



| Model | Parameter | Units | Description                      |
|-------|-----------|-------|----------------------------------|
| Pulse | Period    | s     | Period.                          |
|       | Width     | s     | Pulse width.                     |
|       | Delay     | s     | Delay before first pulse starts. |
|       | Active    |       | Active switch state: Off/On.     |

Pulses start after **Delay** time. Switching diagram is shown for **Active** = On:

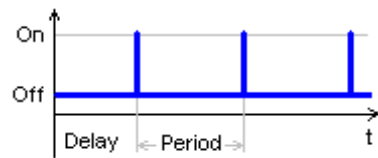


| Model | Parameter | Units | Description                       |
|-------|-----------|-------|-----------------------------------|
| Clock | Period    | s     | Period.                           |
|       | Step      | s     | Simulation step of rise and fall. |
|       | Delay     | s     | Delay before first pulse starts.  |
|       | Active    |       | Active switch state: Off/On.      |

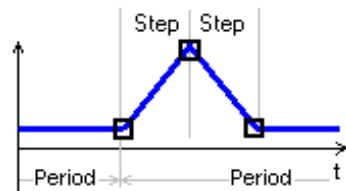
Periodic pulses with width of one simulation step. Pulses start after **Delay** time. Switch goes to **Active** state for one simulation step only.

Unlike **Pulse** model, **Clock** model won't force unnecessary step reduction at the end of the pulse, which may help to accelerate simulation.

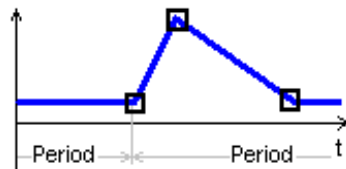
Switching diagram is shown for **Active** = On:



If **Step** parameter is not zero **and** is less than current schematic simulation step, the step is adjusted to provide rise and fall of clock pulse to be equal to **Step**:



Otherwise, the clock pulse is created using current schematic simulation step, which depends on many factors, and cannot be easily predicted:



| Model | Parameter | Units | Description               |
|-------|-----------|-------|---------------------------|
| List  | List      |       | Comma-separated string.   |
|       | Cycle     |       | Cycling (repeat): No/Yes. |
|       | Delay     | s     | Delay.                    |

Switching sequence is defined in the **List** parameter in the **csv** (comma-separated values) format, as follows:

`t0,s0,t1,s1,...,tn,sn`

`s0...sn` defines switch state: positive number corresponds to On state, zero or negative number - Off state.

If  $t < t_0$ , switch is in `s0` state.

At  $t_0$  switch is set to `s0` state, at  $t_1$  switch is set to `s1` state, and so on.

If  $t > t_n$ , and **Cycle** parameter is set to **No**, switch remains in state `sn`. Otherwise the sequence is repeated continuously.

Switching start is delayed by **Delay** time.

Example:

`List = 0,0,3,1,4,0,5,1,8,0`

If **Cycle** = **Yes**, **Delay** = 0, the following sequence will be generated:



See *List model* chapter of NL5 Manual for more details.

| Model       | Parameter    | Units | Description               |
|-------------|--------------|-------|---------------------------|
| <b>File</b> | <b>File</b>  |       | File name.                |
|             | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|             | <b>Delay</b> | s     | Delay.                    |

Switching sequence defined in the text file. If **File** parameter does not have a full path, NL5 will search for the file in the directory where schematic file is located, then in the Library directories, and then in the Library directories relative to schematic file directory (see NL5 Manual, *Schematic Properties*).

Switching sequence is defined in the **csv** (comma-separated values) format, as follows:

```
<if first line does not start with a number, it is ignored>
t0,s0
t1,s1
.....
tn,sn
```

s0...sn defines switch state: positive number corresponds to On state, zero or negative number - Off state.  
If  $t < t_0$ , switch is in s0 state.

At  $t_0$  switch is set to s0 state, at  $t_1$  switch is set to s1 state, and so on.

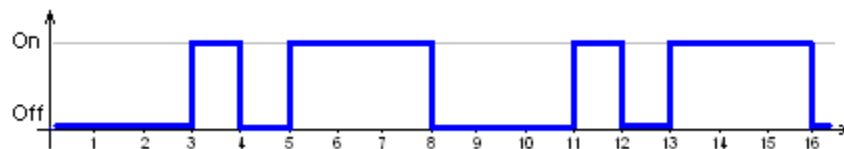
If  $t > t_n$ , and **Cycle** parameter is set to **No**, switch remains in state sn. Otherwise the sequence is repeated continuously.

Switching start delayed by **Delay** time.

Example:

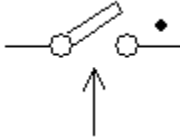
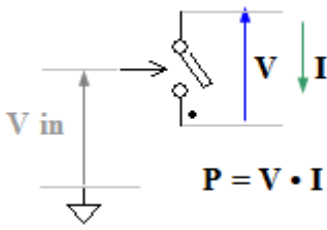
```
0,0
3,1
4,0
5,1
8,0
```

If **Cycle** = **Yes**, **Delay** = 0, the following sequence will be generated:

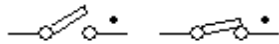




## S – Logic controlled switch

| Symbol                                                                            | Models                                            | Signals                                                                             |
|-----------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Switch<br>Off<br>On<br>One-shot<br>Step<br>Single |  |

For **Off/On** models and models with **Active** parameter, switch symbol indicates switch position in non-active state:



| Model         | Parameter     | Units | Description                |
|---------------|---------------|-------|----------------------------|
| <b>Switch</b> | <b>Active</b> |       | Active state: Off/On.      |
|               | <b>IC</b>     |       | Initial condition: Off/On. |

Logic controlled switch. Switch is set to active or non-active state using following rules:

$V_{in} > \text{logical threshold} \dots : \text{active}$   
 $V_{in} < \text{logical threshold} \dots : \text{non-active}$

When calculating DC operating point, switch is set to the state defined in **IC**.

| Model                                         | Parameter      | Units | Description |
|-----------------------------------------------|----------------|-------|-------------|
| Off                                           | No parameters. |       |             |
| Switch is always in Off state (open circuit). |                |       |             |

| Model                                         | Parameter      | Units | Description |
|-----------------------------------------------|----------------|-------|-------------|
| On                                            | No parameters. |       |             |
| Switch is always in On state (short circuit). |                |       |             |

| Model           | Parameter     | Units | Description           |
|-----------------|---------------|-------|-----------------------|
| <b>One-shot</b> | <b>Width</b>  | s     | Pulse width.          |
|                 | <b>Active</b> |       | Active state: Off/On. |

One-shot switch. When increasing control signal  $V_{in}$  crosses logical threshold, switch is set to **Active** state for **Width** time interval.

If increasing  $V_{in}$  crosses logical threshold value while switch is in active state, the **Active** interval is restarted.

| Model       | Parameter     | Units | Description                  |
|-------------|---------------|-------|------------------------------|
| <b>Step</b> | <b>Delay</b>  | s     | Delay before active state.   |
|             | <b>Active</b> |       | Active switch state: Off/On. |

When control signal  $V_{in}$  is below logical threshold, switch is in non-active state. When increasing control signal  $V_{in}$  crosses logical threshold, a switching sequence similar to **Step** model of **Switch** component is generated. When decreasing control signal  $V_{in}$  drops below logical threshold, switch goes to non-active state immediately.

| Model         | Parameter     | Units | Description                      |
|---------------|---------------|-------|----------------------------------|
| <b>Single</b> | <b>Width</b>  | s     | Pulse width.                     |
|               | <b>Delay</b>  | s     | Delay before first pulse starts. |
|               | <b>Active</b> |       | Active switch state: Off/On.     |

When control signal  $V_{in}$  is below logical threshold, switch is in non-active state. When increasing control signal  $V_{in}$  crosses logical threshold, a switching sequence similar to **Single** model of **Switch** component is generated. When decreasing control signal  $V_{in}$  drops below logical threshold, switch goes to non-active state immediately.

| Model        | Parameter     | Units | Description                      |
|--------------|---------------|-------|----------------------------------|
| <b>Pulse</b> | <b>Period</b> | s     | Period.                          |
|              | <b>Width</b>  | s     | Pulse width.                     |
|              | <b>Delay</b>  | s     | Delay before first pulse starts. |
|              | <b>Active</b> |       | Active switch state: Off/On.     |

When control signal  $V_{in}$  is below logical threshold, switch is in non-active state. When increasing control signal  $V_{in}$  crosses logical threshold, a switching sequence similar to **Pulse** model of **Switch** component is generated. When decreasing control signal  $V_{in}$  drops below logical threshold, switch goes to non-active state immediately.

| Model                                                                                                                                                                                                                                                                                                                                                                                   | Parameter     | Units | Description                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|-----------------------------------|
| <b>Clock</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>Period</b> | s     | Period.                           |
|                                                                                                                                                                                                                                                                                                                                                                                         | <b>Step</b>   | s     | Simulation step of rise and fall. |
|                                                                                                                                                                                                                                                                                                                                                                                         | <b>Delay</b>  | s     | Delay before first pulse starts.  |
|                                                                                                                                                                                                                                                                                                                                                                                         | <b>Active</b> |       | Active switch state: Off/On.      |
| <p>When control signal <i>Vin</i> is below logical threshold, switch is in non-active state. When increasing control signal <i>Vin</i> crosses logical threshold, a switching sequence similar to <b>Pulse</b> model of <b>Switch</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, switch goes to non-active state immediately.</p> |               |       |                                   |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                     | Parameter    | Units | Description               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>List</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <b>List</b>  |       | Comma-separated string.   |
|                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <i>Vin</i> is below logical threshold, switch is in <b>s0</b> state defined in the <b>List</b> parameter. When increasing control signal <i>Vin</i> crosses logical threshold, a switching sequence similar to <b>List</b> model of <b>Switch</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, switch goes to <b>s0</b> state immediately.</p> |              |       |                           |

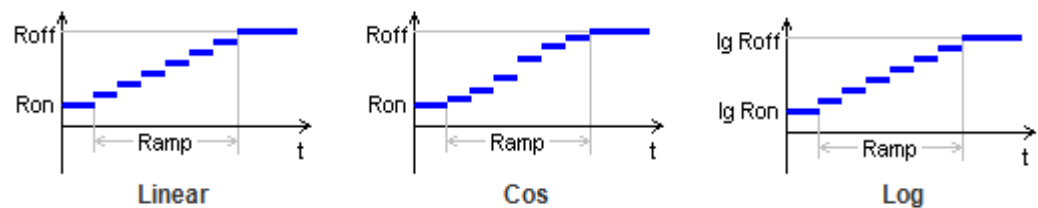
| Model                                                                                                                                                                                                                                                                                                                                                                                                              | Parameter    | Units | Description               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>File</b>                                                                                                                                                                                                                                                                                                                                                                                                        | <b>File</b>  |       | File name.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <i>Vin</i> is below logical threshold, switch is in <b>s0</b> state specified in the <b>File</b>. When increasing control signal <i>Vin</i> crosses logical threshold, a switching sequence similar to <b>File</b> model of <b>Switch</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, switch goes to non-active state immediately.</p> |              |       |                           |

| Model | Parameter | Units | Description                                |
|-------|-----------|-------|--------------------------------------------|
| Steps | Roff      | Ohm   | Off state resistance.                      |
|       | Ron       | Ohm   | On state resistance.                       |
|       | Slope     |       | Type of resistance change: Linear/Cos/Log. |
|       | Ramp      | s     | Resistance ramp time.                      |
|       | Steps     |       | Number of resistance steps in the ramp.    |
|       | IC        |       | Initial condition: Off/On.                 |

Switch with resistance ramping. When increasing input voltage  $V_{in}$  crosses logical threshold, switch resistance starts ramping from **Roff** to **Ron**. When decreasing input voltage  $V_{in}$  crosses logical threshold, switch resistance starts ramping from **Ron** to **Roff**.

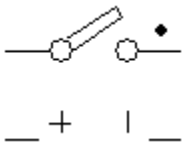
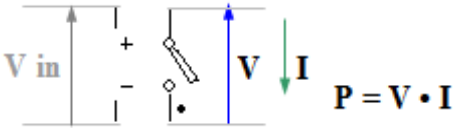
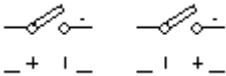
Resistance is changing during **Ramp** time interval, with number of steps specified by **Steps** parameter. If **Steps** = 0, resistance is changed instantly.

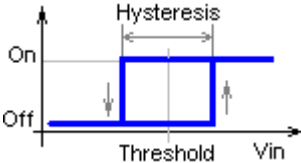
**Slope** parameter specifies how resistance is changing during the ramp:



When calculating DC operating point switch is set to the state specified in **IC**.

S – Voltage controlled switch

| Symbol                                                                                                                                                                                                              | Models                                                 | Signals                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------|
|                                                                                                                                    | Switch<br>Off<br>On<br><br>One-shot<br>Steps<br>SubCir |  |
| Views<br>                                                                                                                          |                                                        |                                                                                    |
| For <b>Off/On</b> models and models with <b>Active</b> parameter, switch symbol indicates switch position in non-active state:<br> |                                                        |                                                                                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Parameter         | Units | Description                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------|----------------------------|
| <b>Switch</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Threshold</b>  | V     | Voltage threshold.         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Hysteresis</b> | V     | Hysteresis.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Active</b>     |       | Active state: Off/On.      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>IC</b>         |       | Initial condition: Off/On. |
| Voltage controlled switch. Switch is set to active or non-active state using following rules:<br><br>$V_{in} > \text{Threshold} + \text{Hysteresis}/2 \dots$ : active<br>$V_{in} < \text{Threshold} - \text{Hysteresis}/2 \dots$ : non-active<br>Otherwise ..... : previous state<br><br>When calculating DC operating point switch is set to the state defined in <b>IC</b> .<br><br>Switching diagram for <b>Active</b> = On:<br><br> |                   |       |                            |

| Model                                         | Parameter      | Units | Description |
|-----------------------------------------------|----------------|-------|-------------|
| Off                                           | No parameters. |       |             |
| Switch is always in Off state (open circuit). |                |       |             |

| Model                                         | Parameter      | Units | Description |
|-----------------------------------------------|----------------|-------|-------------|
| On                                            | No parameters. |       |             |
| Switch is always in On state (short circuit). |                |       |             |

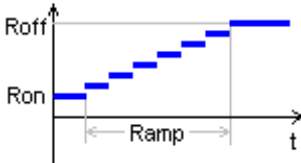
| Model                                                                                                                                                                 | Parameter | Units | Description           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-----------------------|
| One-shot                                                                                                                                                              | Width     | s     | Pulse width.          |
|                                                                                                                                                                       | Threshold | V     | Voltage threshold.    |
|                                                                                                                                                                       | Active    |       | Active state: Off/On. |
| One-shot pulse generator. When increasing input voltage $V_{in}$ crosses <b>Threshold</b> value, switch is set to <b>Active</b> state for <b>Width</b> time interval. |           |       |                       |
| If increasing $V_{in}$ crosses logical threshold value while switch is in active state, the <b>Active</b> interval is restarted.                                      |           |       |                       |

| Model | Parameter  | Units | Description                                |
|-------|------------|-------|--------------------------------------------|
| Steps | Threshold  | V     | Voltage threshold.                         |
|       | Hysteresis | V     | Hysteresis.                                |
|       | Roff       | Ohm   | Off state resistance.                      |
|       | Ron        | Ohm   | On state resistance.                       |
|       | Slope      |       | Type of resistance change: Linear/Cos/Log. |
|       | Ramp       | s     | Resistance ramp time.                      |
|       | Steps      |       | Number of resistance steps in the ramp.    |
|       | IC         |       | Initial condition: Off/On.                 |

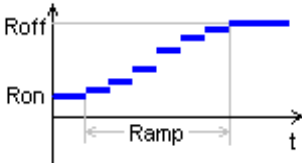
Switch with resistance ramping. When increasing input voltage  $V_{in}$  crosses **Threshold + Hysteresis/2** value, switch resistance starts ramping from **Roff** to **Ron**. When decreasing input voltage  $V_{in}$  crosses **Threshold - Hysteresis/2** value, switch resistance starts ramping from **Ron** to **Roff**.

Resistance is changing during **Ramp** time interval, with number of steps specified by **Steps** parameter. If **Steps** = 0, resistance is changed instantly.

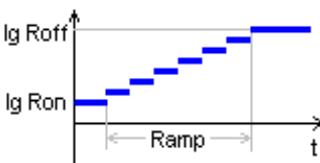
**Slope** parameter specifies how resistance is changing during the ramp:



Linear



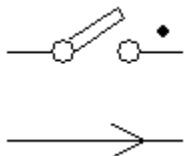
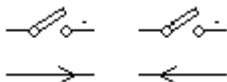
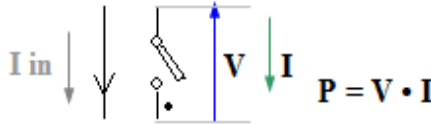
Cos

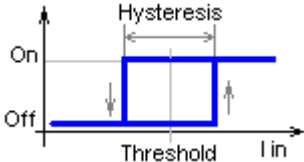


Log

When calculating DC operating point switch is set to the state specified in **IC**.

S – Current controlled switch

| Symbol                                                                                                                                                                                                              | Models              | Signals                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|
|                                                                                                                                    | Switch<br>Off<br>On | One-shot<br>Steps<br>SubCir                                                         |
| Views<br>                                                                                                                          |                     |  |
| For <b>Off/On</b> models and models with <b>Active</b> parameter, switch symbol indicates switch position in non-active state:<br> |                     |                                                                                     |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Parameter  | Units | Description                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|----------------------------|
| Switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Threshold  | A     | Current threshold.         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Hysteresis | A     | Hysteresis.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Active     |       | Active state: Off/On.      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | IC         |       | Initial condition: Off/On. |
| <p>Current controlled switch. Switch is set to active or non-active state using following rules:</p> <p><math>I_{in} &gt; \text{Threshold} + \text{Hysteresis}/2 \dots</math> : active</p> <p><math>I_{in} &lt; \text{Threshold} - \text{Hysteresis}/2 \dots</math> : non-active</p> <p>Otherwise ..... : previous state</p> <p>When calculating DC operating point switch is set to the state defined in <b>IC</b>.</p> <p>Switching diagram for <b>Active</b> = On:</p>  |            |       |                            |

| Model                                         | Parameter      | Units | Description |
|-----------------------------------------------|----------------|-------|-------------|
| Off                                           | No parameters. |       |             |
| Switch is always in Off state (open circuit). |                |       |             |

| Model                                         | Parameter      | Units | Description |
|-----------------------------------------------|----------------|-------|-------------|
| On                                            | No parameters. |       |             |
| Switch is always in On state (short circuit). |                |       |             |

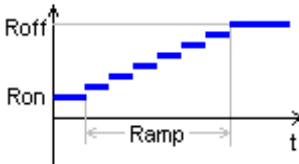
| Model                                                                                                                                                                 | Parameter | Units | Description           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-----------------------|
| One-shot                                                                                                                                                              | Width     | s     | Pulse width.          |
|                                                                                                                                                                       | Threshold | A     | Current threshold.    |
|                                                                                                                                                                       | Active    |       | Active state: Off/On. |
| One-shot pulse generator. When increasing input current $I_{in}$ crosses <b>Threshold</b> value, switch is set to <b>Active</b> state for <b>Width</b> time interval. |           |       |                       |
| If increasing $I_{in}$ crosses logical threshold value while switch is in active state, the <b>Active</b> interval is restarted.                                      |           |       |                       |

| Model | Parameter  | Units | Description                                |
|-------|------------|-------|--------------------------------------------|
| Steps | Threshold  | A     | Current threshold.                         |
|       | Hysteresis | A     | Hysteresis.                                |
|       | Roff       | Ohm   | Off state resistance.                      |
|       | Ron        | Ohm   | On state resistance.                       |
|       | Slope      |       | Type of resistance change: Linear/Cos/Log. |
|       | Ramp       | s     | Resistance ramp time.                      |
|       | Steps      |       | Number of resistance steps in the ramp.    |
|       | IC         |       | Initial condition: Off/On.                 |

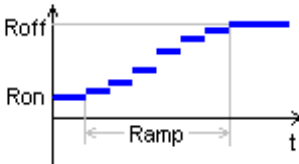
Switch with resistance ramping. When increasing input current  $I_{in}$  crosses **Threshold + Hysteresis/2** value, switch resistance starts ramping from **Roff** to **Ron**. When decreasing input current  $I_{in}$  crosses **Threshold - Hysteresis/2** value, switch resistance starts ramping from **Ron** to **Roff**.

Resistance is changing during **Ramp** time interval, with number of steps specified by **Steps** parameter. If **Steps** = 0, resistance is changed instantly.

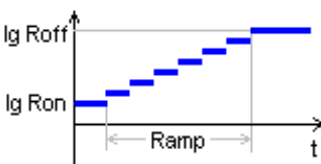
**Slope** parameter specifies how resistance is changing during the ramp:



Linear



Cos

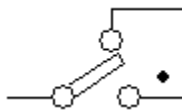
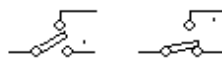
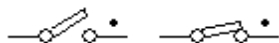


Log

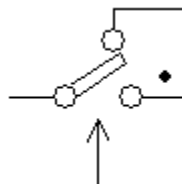
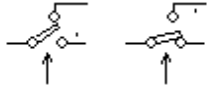
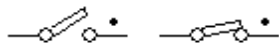
When calculating DC operating point switch is set to the state specified in **IC**.



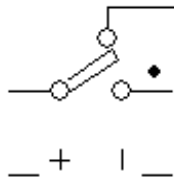
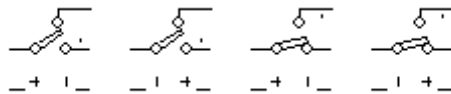
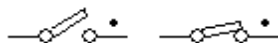
S – SPDT switch

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                   | Models                               |                                 | Signals |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------|---------------------------------|---------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                   | Off<br>On<br>Step<br>Single<br>Pulse | Clock<br>List<br>File<br>SubCir |         |
| Views                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                                      |                                 |         |
| <p>All models and operation of the SPDT (single pole, double throw) switch are similar to single pole <b>Switch</b>, with Off and On states defined as follows:</p> <p>Off state: “common to pin with dot” - open, “common to another pin” - short.<br/>On state: “common to pin with dot” - short, “common to another pin” - open.</p> <p>For <b>Off/On</b> models and models with <b>Active</b> parameter, switch symbol indicates switch position in non-active state:</p>  |                                                                                   |                                      |                                 |         |

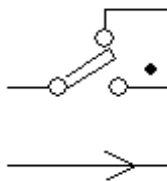
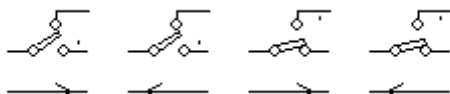
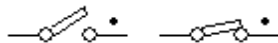
S – SPDT logic controlled switch

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                     | Models                                            |                                                   | Signals |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                     | Switch<br>Off<br>On<br>One-shot<br>Step<br>Single | Pulse<br>Clock<br>List<br>File<br>Steps<br>SubCir |         |
| Views                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                                                   |                                                   |         |
| <p>All models and operation of the SPDT (single pole, double throw) logic controlled switch are similar to single pole <b>Logic controlled switch</b>, with Off and On states defined as follows:</p> <p>Off state: “common to pin with dot” - open, “common to another pin” - short.<br/>On state: “common to pin with dot” - short, “common to another pin” - open.</p> <p>For <b>Off/On</b> models and models with <b>Active</b> parameter, switch symbol indicates switch position in non-active state:</p>  |                                                                                     |                                                   |                                                   |         |

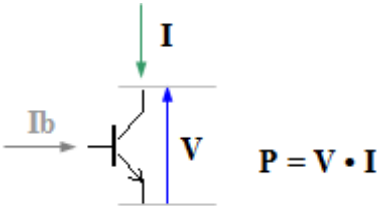
S – SPDT voltage controlled switch

| Symbol                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Models              |                             | Signals |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------|---------|
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Switch<br>Off<br>On | One-shot<br>Steps<br>SubCir |         |
| Views                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |                             |         |
|                                                                                   | <p>All models and operation of the SPDT (single pole, double throw) voltage controlled switch are similar to single pole <b>Voltage controlled switch</b>, with Off and On states defined as follows:</p> <p>Off state: “common to pin with dot” - open, “common to another pin” - short.<br/>On state: “common to pin with dot” - short, “common to another pin” - open.</p> <p>For <b>Off/On</b> models and models with <b>Active</b> parameter, switch symbol indicates switch position in non-active state:</p>  |                     |                             |         |

S – SPDT current controlled switch

| Symbol                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Models              |                             | Signals |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------|---------|
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Switch<br>Off<br>On | One-shot<br>Steps<br>SubCir |         |
| Views                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |                             |         |
|                                                                                     | <p>All models and operation of SPDT (single pole, double throw) current controlled switch are similar to single pole <b>Current controlled switch</b> component, with Off and On states defined as follows:</p> <p>Off state: “common to pin with dot” - open, “common to another pin” - short.<br/>On state: “common to pin with dot” - short, “common to another pin” - open.</p> <p>For <b>Off/On</b> models and models with <b>Active</b> parameter, switch symbol indicates switch position in non-active state:</p>  |                     |                             |         |

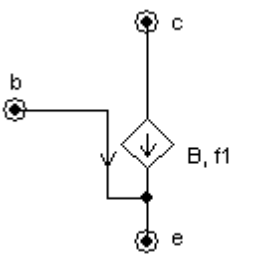
T – NPN transistor

| Symbol                                                                            | Models                                   | Signals                                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|
|  | Linear<br>Switch<br>Transistor<br>SubCir |  |

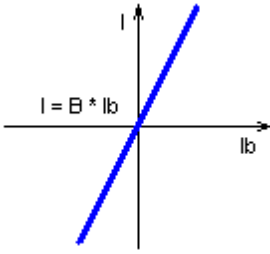
| Model  | Parameter | Units | Description                           |
|--------|-----------|-------|---------------------------------------|
| Linear | <b>B</b>  | A/A   | Gain (beta)                           |
|        | <b>f1</b> | Hz    | Unit gain frequency.                  |
|        | <b>IC</b> | A     | Initial condition: collector current. |

Linear BJT transistor. Current controlled current source with specified bandwidth. **B** is open loop gain (beta). Frequency response consists of one pole, **f1** is unit gain frequency. **B** and **f1** can be set to infinity (**inf**).

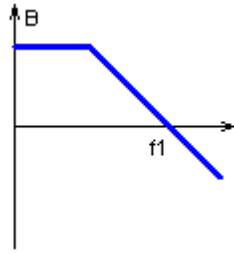
When calculating DC operating point, if **f1** is not infinity and **IC** is defined, collector current is set to specified output current **IC**. If **IC** is blank, static characteristic is used.



Equivalent schematic



Static characteristic

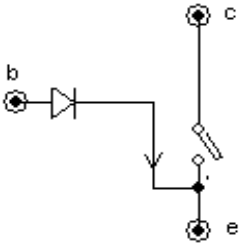


AC response

| Model  | Parameter  | Units | Description                                      |
|--------|------------|-------|--------------------------------------------------|
| Switch | <b>Vbe</b> | V     | Forward voltage drop of base-emitter diode.      |
|        | <b>IC</b>  |       | Initial condition of base-emitter diode: Off/On. |

BJT transistor switch. Current controlled switch with a base-emitter diode. Switch is closed if diode current is non-zero.

When calculating DC operating point, the diode is set to the state specified in **IC**.



Equivalent schematic

| Model      | Parameter | Units | Description                                        |
|------------|-----------|-------|----------------------------------------------------|
| Transistor | B         | A/A   | Gain (beta)                                        |
|            | f1        | Hz    | Unit gain frequency.                               |
|            | Vbe       | V     | Forward voltage drop of base-emitter diode.        |
|            | Vsat      | V     | Collector-emitter saturation voltage drop.         |
|            | IC        | A     | Initial condition: collector current.              |
|            | ICbe      |       | Initial condition of base-emitter diode: Off/On.   |
|            | ICbc      |       | Initial condition of base-collector diode: Off/On. |

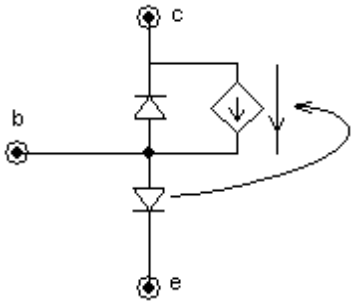
BJT transistor. Simplified Ebers-Moll BJT transistor model with saturation. It consists of two diodes (base-emitter and base-collector), and current source controlled by current through base-emitter diode with gain “alpha”:

$$\alpha = \frac{\beta}{1 + \beta}$$

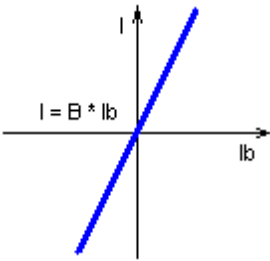
If collector-emitter voltage is higher than **Vsat**, base-collector diode is open, transistor is not saturated, and behaves as **Linear** model (current controlled current source with specified bandwidth). **B** is open loop gain (beta). Low signal frequency response consists of one pole, **f1** is unit gain frequency. **B** and **f1** can be set to infinity (**inf**).

If collector voltage drops below **Vsat**, base-collector diode is closed, and transistor is saturated: collector-emitter voltage is equal to **Vsat**.

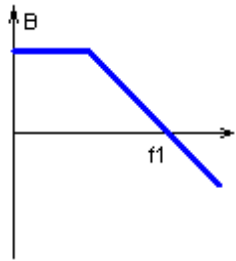
When calculating DC operating point, if **f1** is not infinity and **IC** is defined, collector current is set to specified output current **IC**. If **IC** is blank, static characteristic is used. Base-emitter diode is set to the state specified in **ICbe**, Base-collector diode is set to the state specified in **ICbc**.



Equivalent schematic

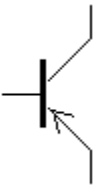
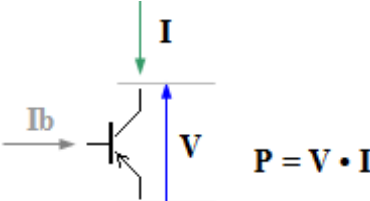


Non-saturated static characteristic



Low signal AC response

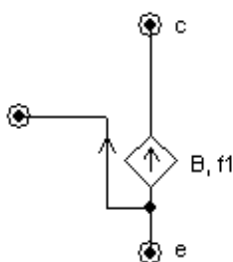
T – PNP transistor

| Symbol                                                                            | Models                                   | Signals                                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|
|  | Linear<br>Switch<br>Transistor<br>SubCir |  |

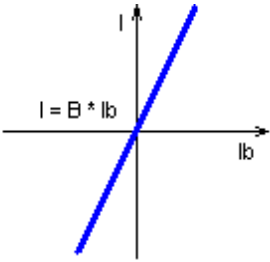
| Model  | Parameter | Units | Description                           |
|--------|-----------|-------|---------------------------------------|
| Linear | <b>B</b>  | A/A   | Gain (beta)                           |
|        | <b>f1</b> | Hz    | Unit gain frequency.                  |
|        | <b>IC</b> | A     | Initial condition: collector current. |

Linear BJT transistor. Current controlled current source with specified bandwidth. **B** is open loop gain (beta). Frequency response consists of one pole, **f1** is unit gain frequency. **B** and **f1** can be set to infinity (**inf**).

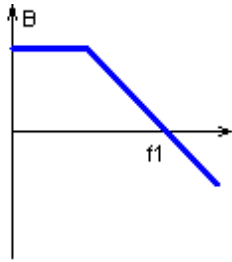
When calculating DC operating point, if **f1** is not infinity and **IC** is defined, collector current is set to specified output current **IC**. If **IC** is blank, static characteristic is used.



Equivalent schematic



Static characteristic

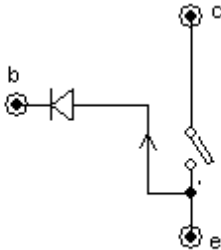


AC response

| Model  | Parameter  | Units | Description                                      |
|--------|------------|-------|--------------------------------------------------|
| Switch | <b>Vbe</b> | V     | Forward voltage drop of base-emitter diode.      |
|        | <b>IC</b>  |       | Initial condition of base-emitter diode: Off/On. |

BJT transistor switch. Current controlled switch with a base-emitter diode. Switch is closed if diode current is non-zero.

When calculating DC operating point, the diode is set to the state specified in **IC**.



Equivalent schematic

| Model      | Parameter | Units | Description                                        |
|------------|-----------|-------|----------------------------------------------------|
| Transistor | B         | A/A   | Gain (beta)                                        |
|            | f1        | Hz    | Unit gain frequency.                               |
|            | Vbe       | V     | Forward voltage drop of base-emitter diode.        |
|            | Vsat      | V     | Collector-emitter saturation voltage drop.         |
|            | IC        | A     | Initial condition: collector current.              |
|            | ICbe      |       | Initial condition of base-emitter diode: Off/On.   |
|            | ICbc      |       | Initial condition of base-collector diode: Off/On. |

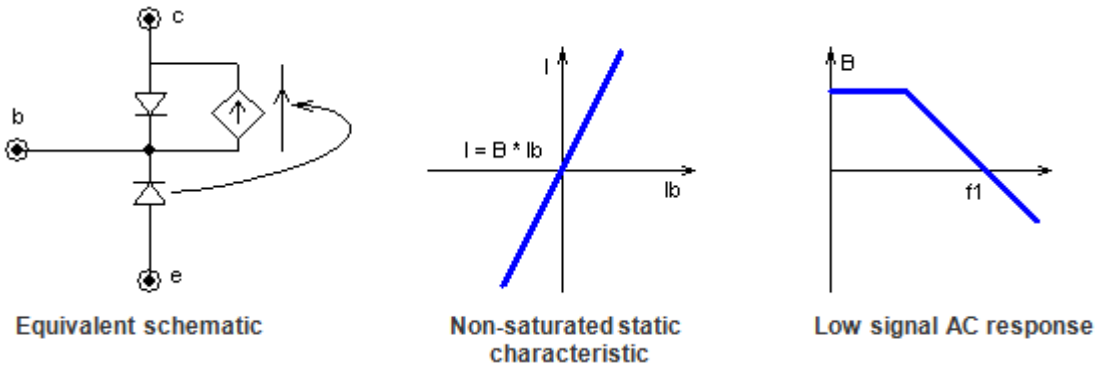
BJT transistor. Simplified Ebers-Moll BJT transistor model with saturation. It consists of two diodes (base-emitter and base-collector), and current source controlled by current through base-emitter diode with gain “alpha”:

$$\alpha = \frac{\beta}{1 + \beta}$$

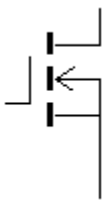
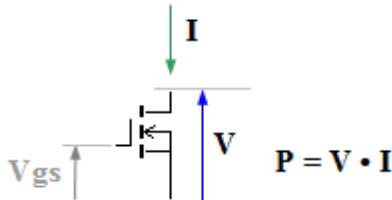
If collector-emitter voltage is negative, and less than **-Vsat**, base-collector diode is open, transistor is not saturated, and behaves as **Linear** model (current controlled current source with specified bandwidth). **B** is open loop gain (beta). Low signal frequency response consists of one pole, **f1** is unit gain frequency. **B** and **f1** can be set to infinity (**inf**).

If collector voltage is higher than **-Vsat**, base-collector diode is closed, and transistor is saturated: collector-emitter voltage is equal to **-Vsat**.

When calculating DC operating point, if **f1** is not infinity and **IC** is defined, collector current is set to specified output current **IC**. If **IC** is blank, static characteristic is used. Base-emitter diode is set to the state specified in **ICbe**, Base-collector diode is set to the state specified in **ICbc**.



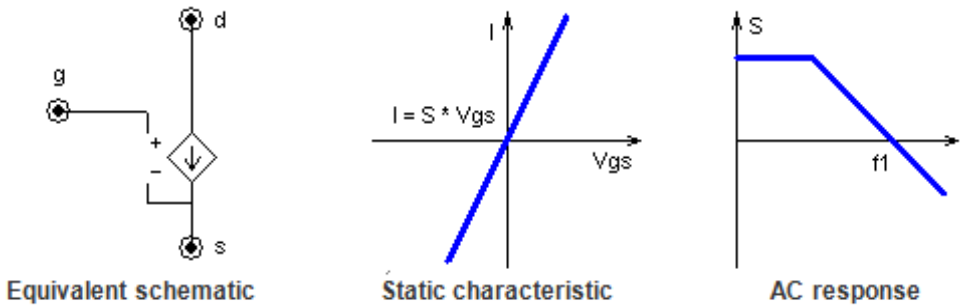
T – N-FET

| Symbol                                                                            | Models                                                 | Signals                                                                             |
|-----------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Linear<br>Switch<br>Switch-D<br>FET<br>FET-D<br>SubCir |  |

| Model  | Parameter | Units | Description                       |
|--------|-----------|-------|-----------------------------------|
| Linear | S         | A/V   | Slope                             |
|        | f1        | Hz    | Unit gain frequency.              |
|        | IC        | A     | Initial condition: drain current. |

Linear FET transistor. Voltage controlled current source with specified bandwidth. **S** is open loop slope. Frequency response consists of one pole, **f1** is unit gain frequency. **S** and **f1** can be set to infinity (**inf**).

When calculating DC operating point, if **f1** is not infinity and **IC** is defined, drain current is set to specified output current **IC**. If **IC** is blank, static characteristic is used.



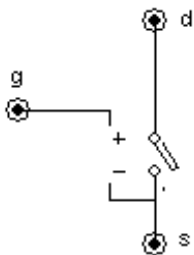
This model of FET transistor does not have a **body diode**. If you need a body diode, you should add it as an external component.

| Model  | Parameter | Units | Description                              |
|--------|-----------|-------|------------------------------------------|
| Switch | Vth       | V     | Threshold.                               |
|        | IC        |       | Initial condition of the switch: Off/On. |

FET switch. Voltage controlled switch. Switch is closed if gate-source voltage exceeds threshold **Vth**.

When calculating DC operating point switch is set to the state specified in **IC**.

This model of FET transistor does not have a **body diode**. If you need a body diode, you should add it as an external component, or use **Switch-D** model..

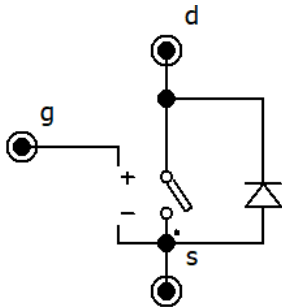


Equivalent schematic

| Model                                                                               | Parameter | Units | Description                     |
|-------------------------------------------------------------------------------------|-----------|-------|---------------------------------|
|  | Vth       | V     | Threshold.                      |
|                                                                                     | Vd        | V     | Body diode forward voltage drop |
|                                                                                     | IC        |       | Initial condition: Off/On/Diode |
|                                                                                     |           |       |                                 |

FET switch with body diode. The model is the same as Switch model with one addition: a body diode connected between drain and source.

When calculating DC operating point switch is set to the state specified in **IC**. “Diode” state means diode is closed.



Equivalent schematic

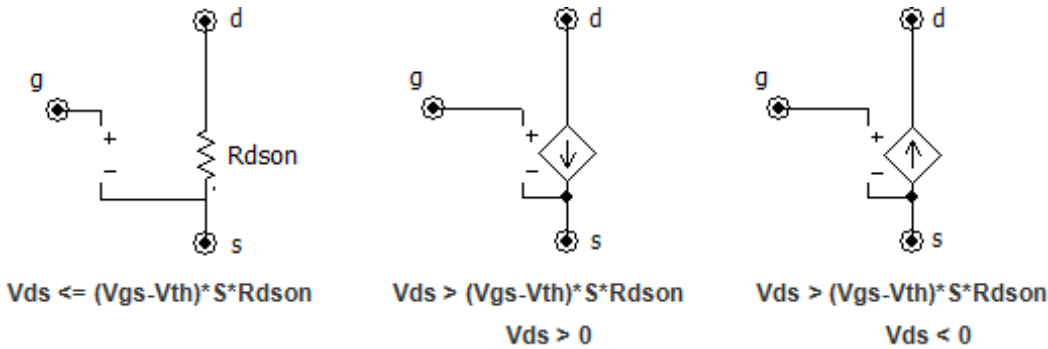


| Model | Parameter | Units | Description                         |
|-------|-----------|-------|-------------------------------------|
| FET   | S         | A/V   | Slope.                              |
|       | Vth       | V     | Threshold.                          |
|       | Rdson     | Ohm   | Rdson resistance.                   |
|       | IC        |       | Initial condition: Off/R/Plus/Minus |

FET transistor. The model has 3 modes of operation.

1.  $V_{gs} \leq V_{th}$  .....:  $I = 0$  (open)
2.  $V_{gs} > V_{th}$ ,  $V_{ds} \leq (V_{gs} - V_{th}) * S * R_{dson}$  ..:  $V = I * R_{dson}$  (resistor)
3.  $V_{gs} > V_{th}$ ,  $V_{ds} > (V_{gs} - V_{th}) * S * R_{dson}$  ..:  $I = (V_{gs} - V_{th}) * S$  (current source)

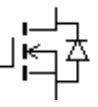
FET works similar for positive and negative drain-source voltage, current direction changes accordingly.  
Equivalent schematics ( $V_{gs} > V_{th}$ ):



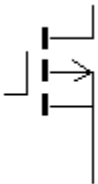
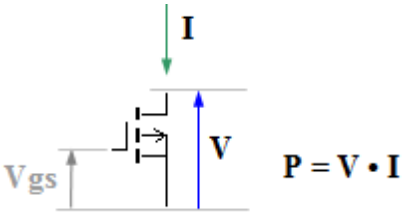
When calculating DC operating point, transistor is set to an initial state specified by IC parameter as follows:

- Off . . . :  $I = 0$  (open)
- R . . . . :  $V = I * R_{dson}$  (resistor)
- Plus . . :  $V_{ds} > 0$ ,  $I = (V_{gs} - V_{th}) * S$  (“positive” current source)
- Minus. :  $V_{ds} < 0$ ,  $I = (V_{gs} - V_{th}) * S$  (“negative” current source)

This model of FET transistor does not have a **body diode**. If you need a body diode, you should add it as an external component.

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Parameter    | Units | Description                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|-------------------------------------------|
| <b>FET-D</b><br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>S</b>     | A/V   | Slope.                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Vth</b>   | V     | Threshold.                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Rdson</b> | Ohm   | Rdson resistance.                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Vd</b>    | V     | Body diode forward voltage drop           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>IC</b>    |       | Initial condition: Off/R/Plus/Minus/Diode |
| <p>FET transistor with body diode. The model is the same as <b>FET</b> model with one addition: a body diode connected between drain and source. In each mode of operation, the diode may be open or close, depending on external conditions.</p> <p>When calculating DC operating point, transistor is set to an initial state specified by <b>IC</b> as follows:</p> <p><b>Off</b> . . . : <math>I = 0</math> (open)</p> <p><b>R</b> . . . . : <math>V = I * \mathbf{Rdson}</math> (resistor)</p> <p><b>Plus</b> . . : <math>V_{ds} &gt; 0, I = (V_{gs} - \mathbf{Vth}) * \mathbf{S}</math> ("positive" current source)</p> <p><b>Minus</b> . : <math>V_{ds} &lt; 0, I = (V_{gs} - \mathbf{Vth}) * \mathbf{S}</math> ("negative" current source)</p> <p><b>Diode</b> . : The body diode is conducting.</p> |              |       |                                           |

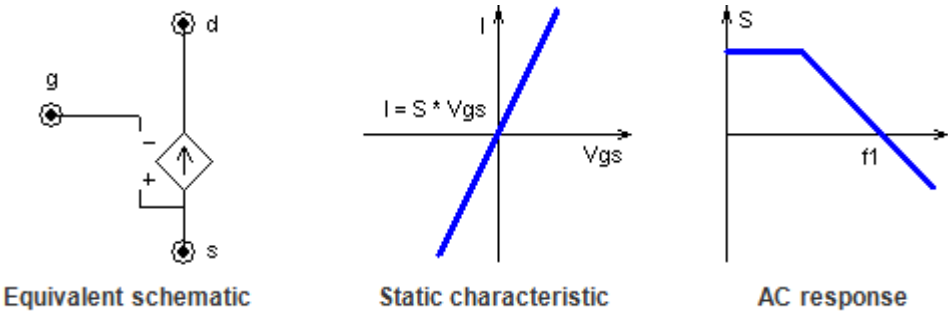
T – P-FET

| Symbol                                                                            | Models                                                 | Signals                                                                             |
|-----------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Linear<br>Switch<br>Switch-D<br>FET<br>FET-D<br>SubCir |  |

| Model  | Parameter | Units | Description                       |
|--------|-----------|-------|-----------------------------------|
| Linear | S         | A/V   | Slope                             |
|        | f1        | Hz    | Unit gain frequency.              |
|        | IC        | A     | Initial condition: drain current. |

Linear FET transistor. Voltage controlled current source with specified bandwidth. **S** is open loop slope. Frequency response consists of one pole, **f1** is unit gain frequency. **S** and **f1** can be set to infinity (**inf**).

When calculating DC operating point, if **f1** is not infinity and **IC** is defined, drain current is set to specified output current **IC**. If **IC** is blank, static characteristic is used.



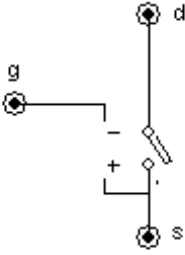
This model of FET transistor does not have a **body diode**. If you need a body diode, you should add it as an external component.

| Model  | Parameter | Units | Description                              |
|--------|-----------|-------|------------------------------------------|
| Switch | Vth       | V     | Threshold.                               |
|        | IC        |       | Initial condition of the switch: Off/On. |

**FET switch.** Voltage controlled switch. Switch is closed if gate-source voltage is less than threshold **Vth**.

When calculating DC operating point switch is set to the state specified in **IC**.

This model of FET transistor does not have a **body diode**. If you need a body diode, you should add it as an external component.

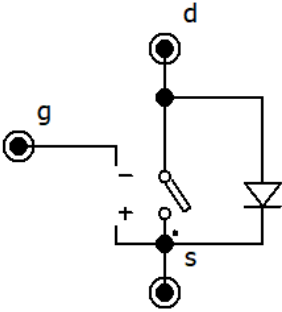


Equivalent schematic

| Model                                                                               | Parameter | Units | Description                     |
|-------------------------------------------------------------------------------------|-----------|-------|---------------------------------|
|  | Vth       | V     | Threshold.                      |
|                                                                                     | Vd        | V     | Body diode forward voltage drop |
|                                                                                     | IC        |       | Initial condition: Off/On/Diode |
|                                                                                     |           |       |                                 |

FET switch with body diode. The model is the same as Switch model with one addition: a body diode connected between drain and source.

When calculating DC operating point switch is set to the state specified in **IC**. “Diode” state means diode is closed.



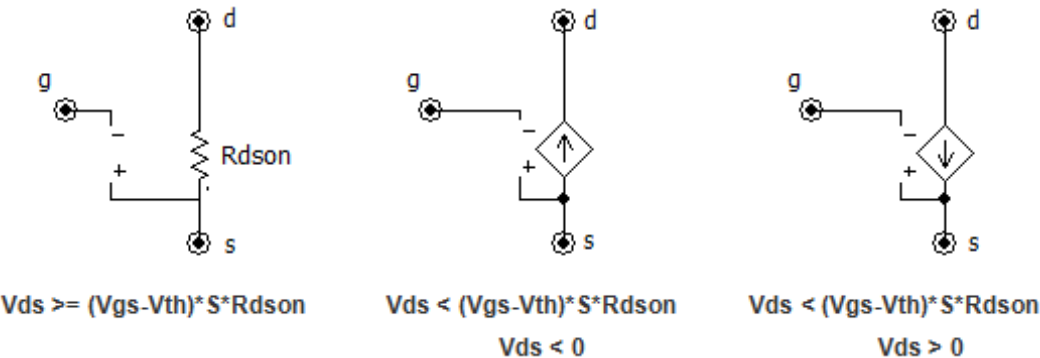
Equivalent schematic

| Model | Parameter | Units | Description                         |
|-------|-----------|-------|-------------------------------------|
| FET   | S         | A/V   | Slope.                              |
|       | Vth       | V     | Threshold.                          |
|       | Rdson     | Ohm   | Rdson resistance.                   |
|       | IC        |       | Initial condition: Off/R/Plus/Minus |

FET transistor. The model has 3 modes of operation.

- 1.  $V_{gs} \geq V_{th}$  .....:  $I = 0$  (open)
- 2.  $V_{gs} < V_{th}$ ,  $V_{ds} \geq (V_{gs} - V_{th}) * S * R_{dson}$  ...:  $V = I * R_{dson}$  (resistor)
- 3.  $V_{gs} < V_{th}$ ,  $V_{ds} < (V_{gs} - V_{th}) * S * R_{dson}$  ...:  $I = (V_{gs} - V_{th}) * S$  (current source)

FET works similar for positive and negative drain-source voltage, current direction changes accordingly.  
Equivalent schematics ( $V_{gs} < V_{th}$ ):



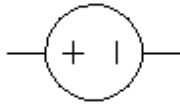
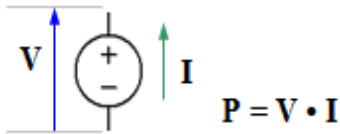
When calculating DC operating point, transistor is set to an initial state specified by **IC** parameter as follows:

- Off** ...:  $I = 0$  (open)
- R** ...:  $V = I * R_{dson}$  (resistor)
- Plus** ...:  $V_{ds} < 0$ ,  $I = (V_{gs} - V_{th}) * S$  ("positive" current source)
- Minus** ...:  $V_{ds} > 0$ ,  $I = (V_{gs} - V_{th}) * S$  ("negative" current source)

This model of FET transistor does not have a **body diode**. If you need a body diode, you should add it as an external component.

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Parameter    | Units | Description                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|-------------------------------------------|
| <b>FET-D</b><br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>S</b>     | A/V   | Slope.                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Vth</b>   | V     | Threshold.                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Rdson</b> | Ohm   | Rdson resistance.                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Vd</b>    | V     | Body diode forward voltage drop           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>IC</b>    |       | Initial condition: Off/R/Plus/Minus/Diode |
| <p>FET transistor with body diode. The model is the same as <b>FET</b> model with one addition: a body diode connected between drain and source. In each mode of operation, the diode may be open or close, depending on external conditions.</p> <p>When calculating DC operating point, transistor is set to an initial state specified by <b>IC</b> as follows:</p> <p><b>Off</b> . . . : <math>I = 0</math> (open)</p> <p><b>R</b> . . . . : <math>V = I * \mathbf{Rdson}</math> (resistor)</p> <p><b>Plus</b> . . : <math>V_{ds} &lt; 0</math>, <math>I = (V_{gs} - \mathbf{Vth}) * \mathbf{S}</math> ("positive" current source)</p> <p><b>Minus</b> . : <math>V_{ds} &gt; 0</math>, <math>I = (V_{gs} - \mathbf{Vth}) * \mathbf{S}</math> ("negative" current source)</p> <p><b>Diode</b> . : The body diode is conducting.</p> |              |       |                                           |

V – Voltage source

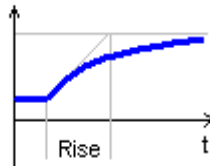
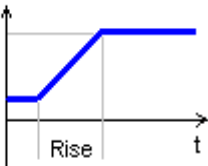
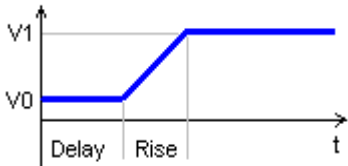
| Symbol                                                                            | Models                                       |                                                      | Signals                                                                             |
|-----------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | V<br>Step<br>Single<br>Pulse<br>Clock<br>Sin | Sweep<br>Function<br>List<br>File<br>Trace<br>SubCir |  |

| Model | Parameter | Units | Description |
|-------|-----------|-------|-------------|
| V     | V         | V     | Voltage.    |

Constant voltage = V.

| Model | Parameter | Units | Description                |
|-------|-----------|-------|----------------------------|
| Step  | V1        | V     | Step On voltage.           |
|       | V0        | V     | Step Off voltage.          |
|       | Slope     |       | Slope type: Linear/Cos/Exp |
|       | Rise      | s     | Step rise length.          |
|       | Delay     | s     | Delay before step starts.  |

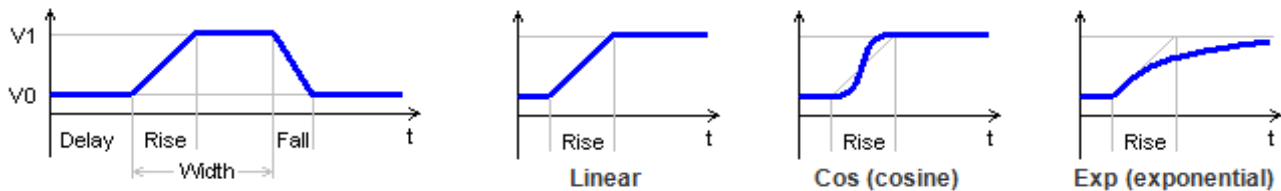
Step starts after **Delay** time. If **Rise** is non-zero, 3 **Slope** types are available.



Linear      Cos (cosine)      Exp (exponential)

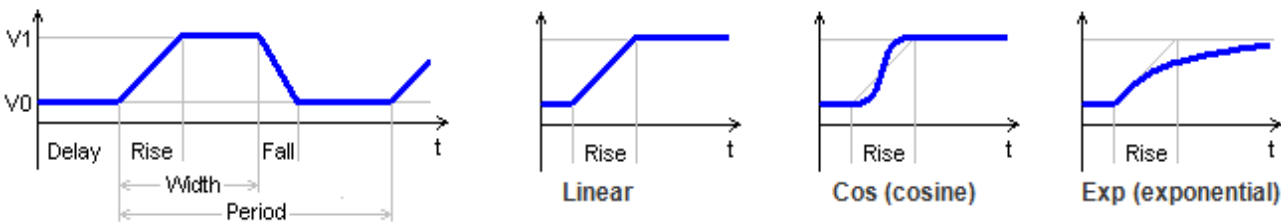
| Model  | Parameter | Units | Description                |
|--------|-----------|-------|----------------------------|
| Single | V1        | V     | Pulse On voltage.          |
|        | V0        | V     | Pulse Off voltage.         |
|        | Width     | s     | Pulse width.               |
|        | Slope     |       | Slope type: Linear/Cos/Exp |
|        | Rise      | s     | Pulse rise length.         |
|        | Fall      | s     | Pulse fall length.         |
|        | Delay     | s     | Delay before pulse starts. |

Single pulse starts after **Delay** time. **Rise** time is included into **Width**, **Fall** time is **not** included into **Width**. **Slope** type applies both to pulse rise and fall (if non-zero).



| Model | Parameter | Units | Description                      |
|-------|-----------|-------|----------------------------------|
| Pulse | V1        | V     | Pulse On voltage.                |
|       | V0        | V     | Pulse Off voltage.               |
|       | Period    | s     | Period.                          |
|       | Width     | s     | Pulse width.                     |
|       | Slope     |       | Slope type: Linear/Cos/Exp       |
|       | Rise      | s     | Pulse rise length.               |
|       | Fall      | s     | Pulse fall length.               |
|       | Delay     | s     | Delay before first pulse starts. |

**Periodic pulse source.** Pulses start after **Delay** time. **Rise** time is included into **Width**, **Fall** time is **not** included into **Width**. **Slope** type applies both to pulse rise and fall (if non-zero).

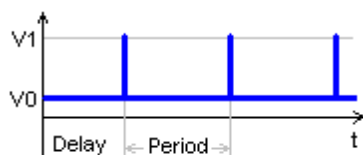




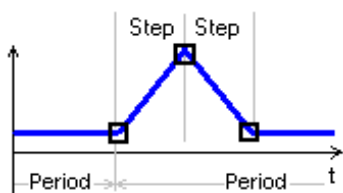
| Model        | Parameter     | Units | Description                       |
|--------------|---------------|-------|-----------------------------------|
| <b>Clock</b> | <b>V1</b>     | V     | Pulse On voltage.                 |
|              | <b>V0</b>     | V     | Pulse Off voltage.                |
|              | <b>Period</b> | s     | Period.                           |
|              | <b>Step</b>   | s     | Simulation step of rise and fall. |
|              | <b>Delay</b>  | s     | Delay before first pulse starts.  |

Periodic pulses with width of one simulation step. Pulses start after **Delay** time.

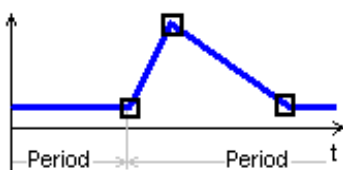
**Clock** model is recommended to produce a constant frequency clock signal for C-code, DLL, logical components, etc. Unlike **Pulse** model, it won't force unnecessary step reduction at the end of the pulse, which may help to accelerate simulation.



If **Step** parameter is not zero **and** is less than current schematic simulation step, the step is adjusted to provide rise and fall of clock pulse to be equal to **Step**:

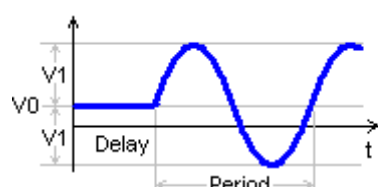


Otherwise, the clock pulse is created using current schematic simulation step, which depends on many factors, and cannot be easily predicted:

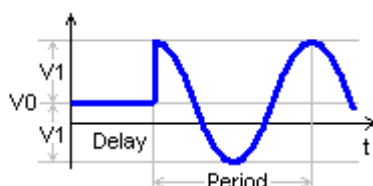


| Model      | Parameter     | Units | Description                      |
|------------|---------------|-------|----------------------------------|
| <b>Sin</b> | <b>V1</b>     | V     | Voltage amplitude.               |
|            | <b>V0</b>     | V     | Voltage baseline.                |
|            | <b>Period</b> | s     | Period.                          |
|            | <b>Phase</b>  | deg   | Phase.                           |
|            | <b>Decay</b>  | 1/s   | Decay constant                   |
|            | <b>Delay</b>  | s     | Delay before sine signal starts. |

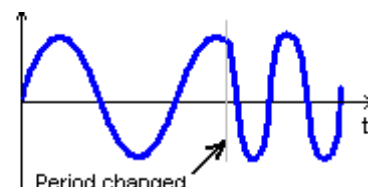
Sinusoidal signal starts after **Delay** time. **Phase** is sine phase in degrees at the moment when signal starts. If transient is paused, sine period changed, and then transient is continued, the phase of the signal remains continuous, providing smooth sine signal of variable frequency. If **Decay** is not zero, the sine signal is exponentially damped with time constant =  $1/\text{Decay}$ .



Phase = 0



Phase = 90



Period changed

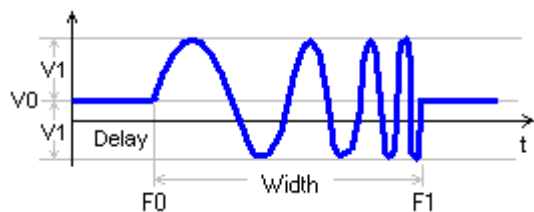
| Model        | Parameter    | Units | Description                 |
|--------------|--------------|-------|-----------------------------|
| <b>Sweep</b> | <b>V1</b>    | V     | Voltage amplitude.          |
|              | <b>V0</b>    | V     | Voltage baseline.           |
|              | <b>Width</b> | s     | Width of the signal.        |
|              | <b>F0</b>    | Hz    | Start frequency.            |
|              | <b>F1</b>    | Hz    | End frequency.              |
|              | <b>Type</b>  |       | Signal type: Linear/Exp.    |
|              | <b>Delay</b> | s     | Delay before signal starts. |

Sinusoidal signal with variable frequency starts after **Delay** time. Signal frequency changes during **Width** interval from **F0** to **F1** linearly or exponentially, depending on specified **Type**.

If **F0 = F1**, then one period of frequency  $1/\text{Width}$  will be generated.

If lowest frequency is set to zero and **Type** = Exp, then lowest frequency  $0.01/\text{Width}$  will be used.

If needed, the highest frequency will be increased to provide integer number of signal periods, so that signal phase at the beginning and at the end of **Width** interval is exactly zero.



| Model           | Parameter | Units | Description |
|-----------------|-----------|-------|-------------|
| <b>Function</b> | <b>f</b>  | V     | Function    |

Arbitrary function. **f** defines voltage as a function of the following variables:

- t** - current time
- V(name)** - voltage on the component **name**
- I(name)** - current through the component **name**
- P(name)** – power on the component **name**
- S(name)** – state of the component **name**

where **name** is the name of the component in the schematic. If **f** is blank, voltage is zero.

Example:

```
f = sin(t) * (1+cos(t*.01))
f = V(R1) * I(R1)
```

Please note that **V**, **I**, **P**, and **S** variables are taken at previous calculation step. This may affect stability of the schematic with closed loop.

| Model       | Parameter    | Units | Description               |
|-------------|--------------|-------|---------------------------|
| <b>List</b> | <b>List</b>  |       | Comma-separated string.   |
|             | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|             | <b>Delay</b> | s     | Delay.                    |

Piecewise linear signal is defined by **List** parameter in the **csv** (comma-separated values) format, as follows:

```
t0,V0,t1,V1,...,tn,Vn
```

where all **t** and **V** can be numerical values or expressions.

If  $t < t_0$ , signal is **V0**.

If  $t_0 < t < t_1$ , signal value is linearly interpolated between **V0** and **V1**, etc.

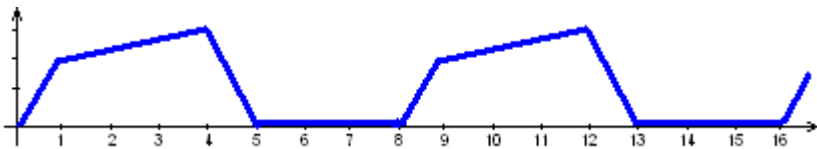
If  $t > t_n$ , and **Cycle** parameter is set to **No**, the signal value is **Vn**. Otherwise the signal defined in  $t_0 \dots t_n$  interval is repeated continuously.

Signal start is delayed by **Delay** time.

Example:

```
List = 0,0,1,2,4,3,5,0,8,0
```

If **Cycle = Yes**, **Delay = 0**, the following voltage will be generated:



See *List model* chapter of NL5 Manual for more details.

| Model       | Parameter    | Units | Description               |
|-------------|--------------|-------|---------------------------|
| <b>File</b> | <b>File</b>  |       | File name.                |
|             | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|             | <b>Delay</b> | s     | Delay.                    |

Piecewise linear signal is defined in the text file. If **File** parameter does not have a full path, NL5 will search for the file in the directory where schematic file is located, then in the Library directories, and then in the Library directories relative to schematic file directory (see NL5 Manual, *Schematic Properties*).

Signal is defined in the **csv** (comma-separated values) format, as follows:

```
<if first line does not start with a number, it is ignored >
t0,V0
t1,V1
.....
tn,Vn
```

where all t and V can be numerical values or expressions.

If  $t < t_0$ , signal is V0.

If  $t_0 < t < t_1$ , signal value is linearly interpolated between V0 and V1, etc.

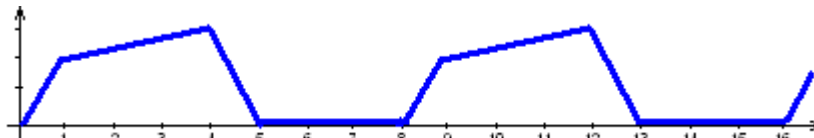
If  $t > t_n$ , and **Cycle** parameter is set to **No**, the signal value is Vn. otherwise the signal defined in  $t_0 \dots t_n$  interval is repeated continuously.

Signal start is delayed by **Delay** time.

Example. File content:

```
0,0
1,2
4,3
5,0
8,0
```

If **Cycle = Yes**, **Delay = 0**, the following voltage will be generated:



| Model        | Parameter    | Units | Description               |
|--------------|--------------|-------|---------------------------|
| <b>Trace</b> | <b>Trace</b> |       | Trace name.               |
|              | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|              | <b>Delay</b> | s     | Delay.                    |

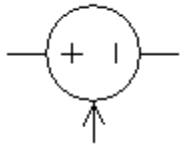
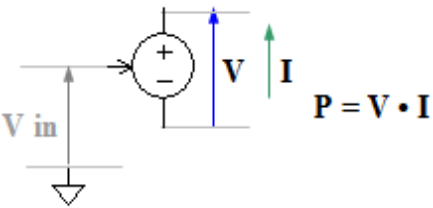
Signal is defined by an existing trace of **Data** type with “static” data: entered manually, duplicated, imported, pasted from the clipboard, etc. Signal value at required simulation point is obtained by linear interpolation between existing data points of the trace.

**Trace** is a name of the existing trace shown in the Trace list.

If **Cycle** parameter is set to **Yes**, the signal is repeated continuously.

Signal start is delayed by **Delay** time.

## V – Logic controlled voltage source

| Symbol                                                                            | Models                                                                                                                                                                                                                      | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <div>V</div> <div>One-shot</div> <div>Step</div> <div>Single</div> <div>Pulse</div> <div>Clock</div> <div>Sin</div> <div>Sweep</div> <div>Function</div> <div>List</div> <div>File</div> <div>Trace</div> <div>SubCir</div> |  |

| Model                 | Parameter | Units | Description |
|-----------------------|-----------|-------|-------------|
| V                     | V         | V     | Voltage.    |
| Constant voltage = V. |           |       |             |

| Model                                                                                                                                                                                                                                                                                                                                                          | Parameter | Units | Description        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|--------------------|
| One-shot                                                                                                                                                                                                                                                                                                                                                       | V1        | V     | Pulse On voltage.  |
|                                                                                                                                                                                                                                                                                                                                                                | V0        | V     | Pulse Off voltage. |
|                                                                                                                                                                                                                                                                                                                                                                | Width     | s     | Pulse width.       |
| <p>One-shot pulse generator. When increasing input voltage <math>V_{in}</math> crosses logical threshold, voltage pulse of <b>Width</b> duration is generated. <b>V0</b> is pulse Off level, <b>V1</b> is pulse On level.</p> <p>If increasing <math>V_{in}</math> crosses logical threshold value while pulse is being generated, the pulse is restarted.</p> |           |       |                    |

| Model                                                                                                                                                                                                                                                                                                                                                                                           | Parameter | Units | Description                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|----------------------------|
| Step                                                                                                                                                                                                                                                                                                                                                                                            | V1        | V     | Step On voltage.           |
|                                                                                                                                                                                                                                                                                                                                                                                                 | V0        | V     | Step Off voltage.          |
|                                                                                                                                                                                                                                                                                                                                                                                                 | Slope     |       | Slope type: Linear/Cos/Exp |
|                                                                                                                                                                                                                                                                                                                                                                                                 | Rise      | s     | Step rise length.          |
|                                                                                                                                                                                                                                                                                                                                                                                                 | Delay     | s     | Delay before step starts.  |
| <p>When control signal <math>V_{in}</math> is below logical threshold, output is in Off state. When increasing control signal <math>V_{in}</math> crosses logical threshold, a signal similar to <b>Step</b> model of <b>Voltage source</b> component is generated. When decreasing control signal <math>V_{in}</math> drops below logical threshold, output goes to Off state immediately.</p> |           |       |                            |

| Model         | Parameter    | Units | Description                |
|---------------|--------------|-------|----------------------------|
| <b>Single</b> | <b>V1</b>    | V     | Pulse On voltage.          |
|               | <b>V0</b>    | V     | Pulse Off voltage.         |
|               | <b>Width</b> | s     | Pulse width.               |
|               | <b>Slope</b> |       | Slope type: Linear/Cos/Exp |
|               | <b>Rise</b>  | s     | Pulse rise length.         |
|               | <b>Fall</b>  | s     | Pulse fall length.         |
|               | <b>Delay</b> | s     | Delay before pulse starts. |

When control signal *Vin* is below logical threshold, output is in Off state. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Single** model of **Voltage source** component is generated. When decreasing control signal *Vin* drops below logical threshold, output goes to Off state immediately.

| Model        | Parameter     | Units | Description                      |
|--------------|---------------|-------|----------------------------------|
| <b>Pulse</b> | <b>V1</b>     | V     | Pulse On voltage.                |
|              | <b>V0</b>     | V     | Pulse Off voltage.               |
|              | <b>Period</b> | s     | Period.                          |
|              | <b>Width</b>  | s     | Pulse width.                     |
|              | <b>Slope</b>  |       | Slope type: Linear/Cos/Exp       |
|              | <b>Rise</b>   | s     | Pulse rise length.               |
|              | <b>Fall</b>   | s     | Pulse fall length.               |
|              | <b>Delay</b>  | s     | Delay before first pulse starts. |

When control signal *Vin* is below logical threshold, output is in Off state. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Pulse** model of **Voltage source** component is generated. When decreasing control signal *Vin* drops below logical threshold, output goes to Off state immediately.

| Model        | Parameter     | Units | Description                       |
|--------------|---------------|-------|-----------------------------------|
| <b>Clock</b> | <b>V1</b>     | V     | Pulse On voltage.                 |
|              | <b>V0</b>     | V     | Pulse Off voltage.                |
|              | <b>Period</b> | s     | Period.                           |
|              | <b>Step</b>   | s     | Simulation step of rise and fall. |
|              | <b>Delay</b>  | s     | Delay before first pulse starts.  |

When control signal *Vin* is below logical threshold, output is in Off state. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Clock** model of **Voltage source** component is generated. When decreasing control signal *Vin* drops below logical threshold, output goes to Off state immediately.

| Model      | Parameter     | Units | Description                      |
|------------|---------------|-------|----------------------------------|
| <b>Sin</b> | <b>V1</b>     | V     | Voltage amplitude.               |
|            | <b>V0</b>     | V     | Voltage baseline.                |
|            | <b>Period</b> | s     | Period.                          |
|            | <b>Phase</b>  | deg   | Phase.                           |
|            | <b>Decay</b>  | 1/s   | Decay constant                   |
|            | <b>Delay</b>  | s     | Delay before sine signal starts. |

When control signal *Vin* is below logical threshold, output voltage is **V0**. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Sin** model of **Voltage source** component is generated. When decreasing control signal *Vin* drops below logical threshold, output goes to **V0** immediately.

| Model        | Parameter    | Units | Description                       |
|--------------|--------------|-------|-----------------------------------|
| <b>Sweep</b> | <b>V1</b>    | V     | Voltage amplitude.                |
|              | <b>V0</b>    | V     | Voltage baseline.                 |
|              | <b>Width</b> | s     | Width of the signal.              |
|              | <b>F0</b>    | Hz    | Start frequency.                  |
|              | <b>F1</b>    | Hz    | End frequency.                    |
|              | <b>Type</b>  |       | Signal type: Linear/Exp.          |
|              | <b>Delay</b> | s     | Delay before first signal starts. |

When control signal *Vin* is below logical threshold, output voltage is **V0**. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Sweep** model of **Voltage source** component is generated. When decreasing control signal *Vin* drops below logical threshold, output goes to **V0** immediately.

| Model           | Parameter | Units | Description |
|-----------------|-----------|-------|-------------|
| <b>Function</b> | <b>f</b>  | A     | Function    |

When control signal *Vin* is below logical threshold, output is zero. When increasing control signal *Vin* crosses logical threshold, a signal similar to **Function** model of **Voltage source** component is generated. If the function is using current time variable *t*, this moment will be considered as *t*=0. When decreasing control signal *Vin* drops below logical threshold, output goes to zero immediately.

| Model       | Parameter    | Units | Description               |
|-------------|--------------|-------|---------------------------|
| <b>List</b> | <b>List</b>  |       | Comma-separated string.   |
|             | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|             | <b>Delay</b> | s     | Delay.                    |

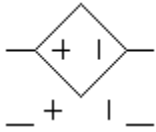
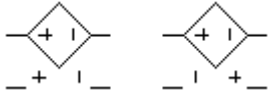
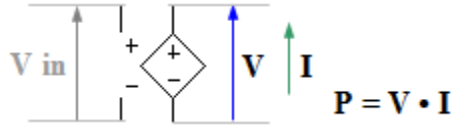
When control signal *Vin* is below logical threshold, output is equal to **V0** value of **List** signal. When increasing control signal *Vin* crosses logical threshold, a signal similar to **List** model of **Voltage source** component is generated. This moment is also considered as *t*=0 for the **List** signal. When decreasing control signal *Vin* drops below logical threshold, output goes to **V0** immediately.

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Parameter    | Units | Description               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>File</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>File</b>  |       | File name.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <i>V<sub>in</sub></i> is below logical threshold, output is equal to <b>V<sub>0</sub></b> value specified in the <b>File</b>. When increasing control signal <i>V<sub>in</sub></i> crosses logical threshold, a signal similar to <b>File</b> model of <b>Voltage source</b> component is generated. This moment is also considered as <math>t=0</math> for the <b>File</b> signal. When decreasing control signal <i>V<sub>in</sub></i> drops below logical threshold, output goes to <b>V<sub>0</sub></b> immediately.</p> |              |       |                           |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Parameter    | Units | Description               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>Trace</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Trace</b> |       | Trace name.               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <i>V<sub>in</sub></i> is below logical threshold, output is zero. When increasing control signal <i>V<sub>in</sub></i> crosses logical threshold, a signal similar to <b>Trace</b> model of <b>Voltage source</b> component is generated. This moment is also considered as <math>t=0</math> for the <b>Trace</b> signal. When decreasing control signal <i>V<sub>in</sub></i> drops below logical threshold, output goes to zero immediately.</p> |              |       |                           |



V – Voltage controlled voltage source

| Symbol                                                                                     | Models                                                                              | Signals                          |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------|
|           | Linear<br>V<br>Function<br>PWL                                                      | VCO<br>One-shot<br>PWM<br>SubCir |
| Views<br> |  |                                  |

| Model                                                        | Parameter | Units | Description |
|--------------------------------------------------------------|-----------|-------|-------------|
| Linear                                                       | K         | V/V   | Gain        |
| Linear voltage controlled voltage source: $V = K * V_{in}$ . |           |       |             |

| Model                    | Parameter | Units | Description |
|--------------------------|-----------|-------|-------------|
| V                        | V         | V     | Voltage.    |
| Constant voltage = $V$ . |           |       |             |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Parameter | Units | Description                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | f         | V     | Output as function of the input.   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | IC        | V     | Initial condition: output voltage. |
| <p>Arbitrary function <b>f</b> defines output voltage as a function of the following variables:</p> <ul style="list-style-type: none"><li><b>x</b> – input voltage <math>V_{in}</math></li><li><b>t</b> - current time</li><li><b>V(name)</b> - voltage on the component <b>name</b></li><li><b>I(name)</b> - current through the component <b>name</b></li><li><b>P(name)</b> – power on the component <b>name</b></li><li><b>S(name)</b> – state of the component <b>name</b></li></ul> <p>where <b>name</b> is the name of the component in the schematic. If <b>f</b> is blank, output is zero.</p> <p>Example:</p> <pre>f = x*x<br/>f = x * sin(t)<br/>f = P(r1) + P(r2)</pre> <p>When calculating DC operating point, and in AC analysis, output is set to specified output voltage <b>IC</b>. Please note that variable <b>x</b> (input voltage <math>V_{in}</math>) and variables <b>V</b>, <b>I</b>, <b>P</b>, and <b>S</b> are taken at previous calculation step. This may affect stability of the schematic with closed loop.</p> |           |       |                                    |

| Model      | Parameter  | Units | Description                    |
|------------|------------|-------|--------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, K(Vin) |

Piecewise linear voltage controlled voltage source. **pwl** string defines gain as a function of input voltage K(Vin). The transfer function of the source V(Vin) is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that K(Vin) is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

| Model      | Parameter    | Units | Description                                |
|------------|--------------|-------|--------------------------------------------|
| <b>VCO</b> | <b>V1</b>    | V     | Voltage amplitude.                         |
|            | <b>V0</b>    | V     | Voltage baseline or Off level.             |
|            | <b>dFdV</b>  | Hz/V  | Gain.                                      |
|            | <b>Type</b>  |       | Signal type: Sin/Square/Triangle/Sawtooth. |
|            | <b>Phase</b> | deg   | Phase.                                     |

Voltage controlled oscillator. Output voltage is a signal with frequency equal to:

$$f(\text{Hz}) = \text{dFdV} * V_{in}.$$

For **Sine** signal, **V0** is baseline, and **V1** is amplitude. For **Square**, **Triangle**, and **Sawtooth** signals, **V0** is Off level, **V1** is On level. **Phase** is additional phase of the signal, in degrees.

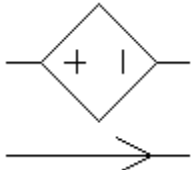
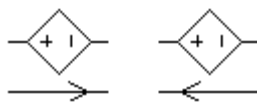
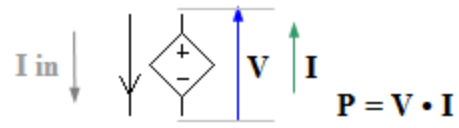
| Model           | Parameter        | Units | Description        |
|-----------------|------------------|-------|--------------------|
| <b>One-shot</b> | <b>V1</b>        | V     | Pulse On voltage.  |
|                 | <b>V0</b>        | V     | Pulse Off voltage. |
|                 | <b>Width</b>     | s     | Pulse width.       |
|                 | <b>Threshold</b> | V     | Voltage threshold. |

One-shot pulse generator. When increasing input voltage  $V_{in}$  crosses **Threshold** value, voltage pulse of **Width** duration is generated. **V0** is pulse Off level, **V1** is pulse On level.

If increasing  $V_{in}$  crosses **Threshold** value while pulse is being generated, the pulse is restarted.

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Parameter    | Units | Description                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|-------------------------------------------|
| <b>PWM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>V1</b>    | V     | Pulse On voltage.                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>V0</b>    | V     | Pulse Off voltage.                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>F</b>     | Hz    | Frequency.                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Vmax</b>  | V     | Input voltage corresponding to 100% duty. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Phase</b> | deg   | Phase.                                    |
| <p>Voltage controlled Pulse-Width Modulator. Output voltage is a pulse signal of frequency <b>F</b> shifted by <b>Phase</b>. Input voltage <i>Vin</i> is sampled at the beginning of each cycle of the signal, and width of the output pulse during this cycle is calculated according to the equation:</p> $\text{width} = 1/F * (V_{in} / V_{max})$ <p>or</p> $\text{duty} = 100\% * (V_{in} / V_{max});$ <p>If the width is equal or less than zero, a short On pulse with the width equal to the minimum calculation step at that moment will be generated. If the width is equal or greater than period of frequency <b>F</b>, a short Off pulse at the end of the period will be generated. As a result, the frequency of the output signal is always <b>F</b>.</p> |              |       |                                           |

## V – Current controlled voltage source

| Symbol                                                                                     | Models                         | Signals                                                                            |
|--------------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------|
|           | Linear<br>V<br>Function<br>PWL | CCO<br>One-shot<br>PWM<br>SubCir                                                   |
| Views<br> |                                |  |

| Model                                                        | Parameter | Units | Description |
|--------------------------------------------------------------|-----------|-------|-------------|
| <b>Linear</b>                                                | <b>K</b>  | V/V   | Gain        |
| Linear current controlled voltage source. $V = K * I_{in}$ . |           |       |             |

| Model                         | Parameter | Units | Description |
|-------------------------------|-----------|-------|-------------|
| <b>V</b>                      | <b>V</b>  | V     | Voltage.    |
| Constant voltage = <b>V</b> . |           |       |             |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Parameter | Units | Description                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------|
| <b>Function</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>f</b>  | V     | Output as function of the input.   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>IC</b> | V     | Initial condition: output current. |
| <p>Arbitrary function <b>f</b> defines output current as a function of the following variables:</p> <ul style="list-style-type: none"> <li><b>x</b> – input current <i>I<sub>in</sub></i></li> <li><b>t</b> - current time</li> <li><b>V(name)</b> - voltage on the component <b>name</b></li> <li><b>I(name)</b> - current through the component <b>name</b></li> <li><b>P(name)</b> – power on the component <b>name</b></li> <li><b>S(name)</b> – state of the component <b>name</b></li> </ul> <p>where <b>name</b> is the name of the component in the schematic. If <b>f</b> is blank, output is zero.</p> <p>Example:</p> <pre>f = x*x f = x * sin(t) f = P(r1) + P(r2)</pre> <p>When calculating DC operating point, and in AC analysis, output is set to specified output current <b>IC</b>. Please note that variable <b>x</b> (input current <i>I<sub>in</sub></i>) and variables <b>V</b>, <b>I</b>, <b>P</b>, and <b>S</b> are taken at previous calculation step. This may affect stability of the schematic with closed loop.</p> |           |       |                                    |

| Model      | Parameter  | Units | Description                             |
|------------|------------|-------|-----------------------------------------|
| <b>PWL</b> | <b>pwl</b> |       | Comma-separated string, K( <i>lin</i> ) |

Piecewise linear current controlled voltage source. **pwl** string defines gain as a function of input current K(*lin*). The transfer function of the source V(*lin*) is piecewise linear function, and it always goes through the origin (0,0). See *PWL model* chapter of NL5 Manual for details.

Please note that K(*lin*) is **piecewise constant** function, although the model and parameter are still called **pwl** for historical reasons.

| Model      | Parameter    | Units | Description                                |
|------------|--------------|-------|--------------------------------------------|
| <b>CCO</b> | <b>V1</b>    | V     | Voltage amplitude.                         |
|            | <b>V0</b>    | V     | Voltage baseline or Off level.             |
|            | <b>dFdl</b>  | Hz/A  | Gain.                                      |
|            | <b>Type</b>  |       | Signal type: Sin/Square/Triangle/Sawtooth. |
|            | <b>Phase</b> | deg   | Phase.                                     |

Current controlled oscillator. Output voltage is a signal with frequency equal to:

$$f(\text{Hz}) = \text{dFdl} * \text{lin}.$$

For **Sine** signal, **V0** is baseline, and **V1** is amplitude. For **Square**, **Triangle**, and **Sawtooth** signals, **V0** is Off level, **V1** is On level. **Phase** is additional phase of the signal, in degrees.

| Model           | Parameter        | Units | Description        |
|-----------------|------------------|-------|--------------------|
| <b>One-shot</b> | <b>V1</b>        | V     | Pulse On voltage.  |
|                 | <b>V0</b>        | V     | Pulse Off voltage. |
|                 | <b>Width</b>     | s     | Pulse width.       |
|                 | <b>Threshold</b> | A     | Current threshold. |

One-shot pulse generator. When increasing input current *lin* crosses **Threshold** value, voltage pulse of **Width** duration is generated. **V0** is pulse Off level, **V1** is pulse On level.

If increasing *lin* crosses **Threshold** value while pulse is being generated, the pulse is restarted.

| Model      | Parameter              | Units | Description                               |
|------------|------------------------|-------|-------------------------------------------|
| <b>PWM</b> | <b>V1</b>              | V     | Pulse On voltage.                         |
|            | <b>V0</b>              | V     | Pulse Off voltage.                        |
|            | <b>F</b>               | Hz    | Frequency.                                |
|            | <b>I<sub>max</sub></b> | A     | Input current corresponding to 100% duty. |
|            | <b>Phase</b>           | deg   | Phase.                                    |

Current controlled Pulse-Width Modulator. Output voltage is a pulse signal of frequency **F** shifted by **Phase**. Input current *I<sub>in</sub>* is sampled at the beginning of each cycle of the signal, and width of the output pulse during this cycle is calculated according to the equation:

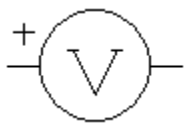
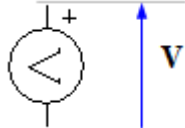
$$\text{width} = 1/F * (I_{in} / I_{max})$$

or

$$\text{duty} = 100\% * (I_{in} / I_{max});$$

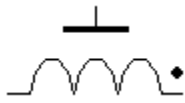
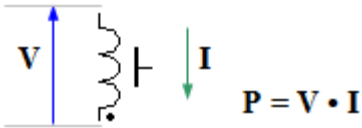
If the width is equal or less than zero, a short On pulse with the width equal to the minimum calculation step at that moment will be generated. If the width is equal or greater than period of frequency **F**, a short Off pulse at the end of the period will be generated. As a result, the frequency of the output signal is always **F**.

## V – Voltmeter

| Symbol                                                                            | Models    | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------|
|  | Voltmeter |  |

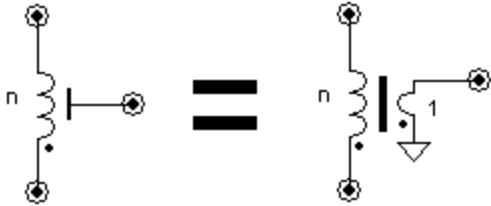
| Model         | Parameter      | Units | Description |
|---------------|----------------|-------|-------------|
| Voltmeter     | No parameters. |       |             |
| Open circuit. |                |       |             |

W – Winding

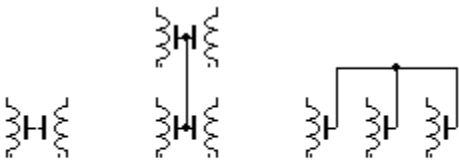
| Symbol                                                                            | Models  | Signals                                                                             |
|-----------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------|
|  | Winding |  |

| Model   | Parameter | Units | Description      |
|---------|-----------|-------|------------------|
| Winding | n         | turns | Number of turns. |

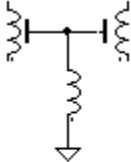
The Winding is actually an ideal transformer, with 1-turn second winding, one end of which is grounded, and another end is shown as a “core” pin of the winding:



To make an ideal transformer, connect cores of two or more windings by wire. Core magnetizing can be modeled by setting linear or non-linear inductor from core to ground:



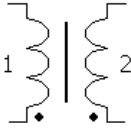
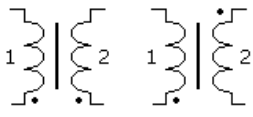
Ideal transformers



Transformer with magnetizing inductor

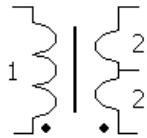
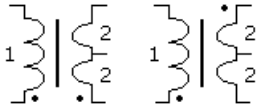


W – Transformer

| Symbol                                                                            |                                                                                   | Models                | Signals |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------|---------|
|  |                                                                                   | Transformer<br>SubCir |         |
| Views                                                                             |  |                       |         |

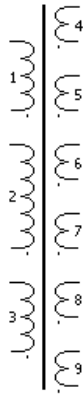
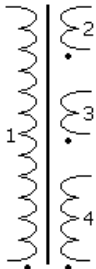
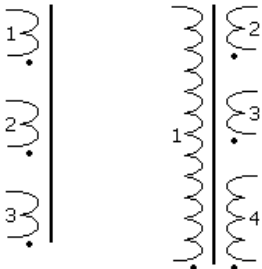
| Model                                                        | Parameter | Units | Description                            |
|--------------------------------------------------------------|-----------|-------|----------------------------------------|
| Transformer                                                  | n1        | turns | Number of turns in the first winding.  |
|                                                              | n2        | turns | Number of turns in the second winding. |
| Ideal transformer with 2 windings. Coupling coefficient = 1. |           |       |                                        |

W – Differential transformer

| Symbol                                                                                     | Models                | Signals |
|--------------------------------------------------------------------------------------------|-----------------------|---------|
|           | Transformer<br>SubCir |         |
| Views<br> |                       |         |

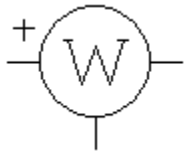
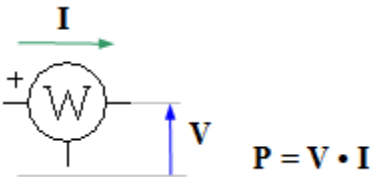
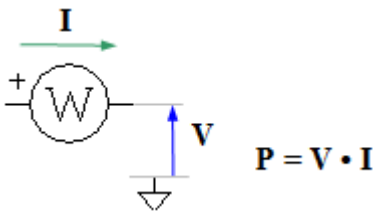
| Model                                                                                                                                                                                           | Parameter | Units | Description                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-----------------------------------------------|
| Transformer                                                                                                                                                                                     | n1        | turns | Number of turns in the first winding.         |
|                                                                                                                                                                                                 | n2        | turns | Number of turns in second and third windings. |
| Ideal differential transformer with 3 windings. Coupling coefficient = 1. Second and third windings have the same number of turns <b>n2</b> , and connected to form a differential transformer. |           |       |                                               |

W – Custom transformer

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Models                | Signals |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Transformer<br>SubCir |         |
| <p>This is a customized component. A component can be edited in the <b>Edit Component</b> dialog box. See <i>Customized component</i> chapter of NL5 Manual for instructions on editing a component.</p> <p>This component may have:</p> <ul style="list-style-type: none"><li>- height from 2 to 32,</li><li>- up to 32 windings (total),</li><li>- arbitrary length of a winding.</li></ul> <p>Examples of Custom transformer component:</p> <div></div> |                       |         |

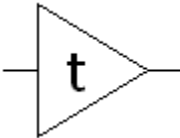
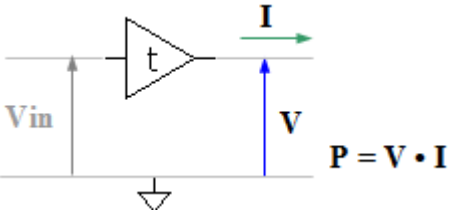
| Model                                                        | Parameter | Units | Description                                     |
|--------------------------------------------------------------|-----------|-------|-------------------------------------------------|
| Transformer                                                  | n1        | turns | Number of turns in the first winding.           |
|                                                              | ...       | ...   | ...                                             |
|                                                              | nN        | turns | Number of turns in the N <sup>th</sup> winding. |
| Ideal transformer with N windings. Coupling coefficient = 1. |           |       |                                                 |

W – Wattmeter

| Symbol                                                                            | Models    | Signals                                                                                             |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------|
|  | Wattmeter |  $P = V \cdot I$ |
|  | Wattmeter |  $P = V \cdot I$ |

| Model                                                                                                                                   | Parameter      | Units | Description |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------|-------|-------------|
| Wattmeter                                                                                                                               | No parameters. |       |             |
| Short circuit between current ports, open circuit between voltage ports. Can be used to measure power in grounded or non-grounded load. |                |       |             |

X – Delay

| Symbol                                                                            | Models          | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------|
|  | Delay<br>SubCir |  |

| Model | Parameter | Units | Description                        |
|-------|-----------|-------|------------------------------------|
| Delay | t0        | s     | Delay.                             |
|       | IC        | V     | Initial condition: output voltage. |

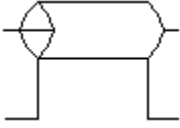
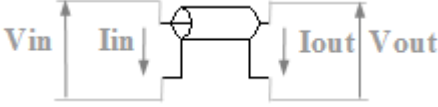
Output voltage is equal to input voltage, delayed by delay time **t0**:

$$V(t) = Vin(t - t0), \text{ where } t \text{ is current time.}$$

When calculating DC operating point, output is set to specified output voltage **IC**, or to input voltage, if **IC** is blank. Then output voltage is not changing until delay time **t0**.

The model allocates memory for storing delayed data only when needed and frees it immediately when possible.

X – Transmission line

| Symbol                                                                            | Models     | Signals                                                                            |
|-----------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------|
|  | Line Lossy |  |

| Model | Parameter | Units | Description                 |
|-------|-----------|-------|-----------------------------|
| Line  | t0        | s     | Delay.                      |
|       | z0        | Ohm   | Characteristic impedance.   |
|       | VIC       | V     | Initial condition: voltage. |
|       | IIC       | A     | Initial condition: current. |

Lossless transmission line. The voltage and current in the line are represented as a superposition of forward and reflected waves, with V/I ratio in each wave equal to line characteristic impedance **z0**. V and I values of each wave are calculated based on boundary (input and output) conditions. The line functionality can also be described by the following equations:

$$Vin(t) = z0 * (lin(t) - Iout(t - t0))$$
$$Vout(t) = z0 * (Iout(t) - lin(t - t0))$$

where t is current time.

Input and output are galvanically isolated: no current is flowing between input and output, and any voltage difference between input and output may exist.

When calculating DC operating point, initial forward and reflected voltage and current are calculated based on the following conditions:

if **VIC** and **IIC** are blank . . . . . :  $Vin = Vout$ ,  $lin = -Iout$ .  
if **VIC** is specified and **IIC** is blank . . :  $Vin = Vout = VIC$ .  
if **VIC** is blank and **IIC** is specified . . :  $lin = IIC$ ,  $Iout = -IIC$ .  
if **VIC** and **IIC** are specified . . . . . :  $Vin = Vout = VIC$ ,  $lin = IIC$ ,  $Iout = -IIC$ .

The model allocates memory for storing forward and reflected wave data only when needed, and frees it immediately when possible.

If real line characteristics are given in line capacitance and inductance per length, the following equations can be used to derive **t0** and **z0** parameters:

$$t0 = \sqrt{L * C} * D$$
$$z0 = \sqrt{L / C}$$

where:

C – line capacitance per length, F/m  
L – line inductance per length, H/m  
D – line length, m

| Model        | Parameter  | Units    | Description                                |
|--------------|------------|----------|--------------------------------------------|
| <b>Lossy</b> | <b>t0</b>  | s        | Delay.                                     |
|              | <b>z0</b>  | Ohm      | Characteristic impedance.                  |
|              | <b>R</b>   | Ohm/ns   | Series resistance per ns.                  |
|              | <b>fr</b>  | MHz      | Skin losses cutoff (3 dB) frequency.       |
|              | <b>G</b>   | 1/Ohm/ns | Shunt conductance per ns.                  |
|              | <b>fG</b>  | MHz      | Dielectric losses cutoff (3 dB) frequency. |
|              | <b>VIC</b> | V        | Initial condition: voltage.                |
|              | <b>IIC</b> | A        | Initial condition: current.                |

Lossy transmission line. Lossy line modeling is similar to lossless transmission line, with addition of losses due to series resistance, skin effect, shunt conductance, and dielectric losses.

Constant series resistance is defined by **r** parameter. Skin losses are modeled by a number of RL chains, providing series impedance increase as a square root of frequency. The number of chains is automatically optimized based on calculation step value; however, the maximum impedance increase due to skin effect is limited to 40 dB (100 times). **fr** parameter defines a frequency where effective series impedance is approximately 3 dB higher than **r**. Skin losses are calculated only if **r** > 0, and **fr** is not infinite.

Constant shunt conductance is defined by **G** parameter. Dielectric losses are modeled by a shunt capacitance, providing shunt admittance increase proportional to frequency. **fG** parameter defines a frequency where effective shunt admittance is approximately 3 dB higher than **G**. Dielectric losses are calculated only if **G** > 0, and **fG** is not infinite.

Input and output are galvanically isolated: no current is flowing between input and output, and any voltage difference between input and output may exist.

When calculating DC operating point initial forward and reflected voltage and current are calculated based on the following conditions:

if **VIC** and **IIC** are blank . . . . . :  $V_{in} = V_{out}$ ,  $I_{in} = -I_{out}$ .  
 if **VIC** is specified and **IIC** is blank . . :  $V_{in} = V_{out} = \mathbf{VIC}$ .  
 if **VIC** is blank and **IIC** is specified . . :  $I_{in} = \mathbf{IIC}$ ,  $I_{out} = -\mathbf{IIC}$ .  
 if **VIC** and **IIC** are specified . . . . . :  $V_{in} = V_{out} = \mathbf{VIC}$ ,  $I_{in} = \mathbf{IIC}$ ,  $I_{out} = -\mathbf{IIC}$ .

The model allocates all the required memory immediately at transient start. The amount of memory is proportional to line delay and inverse proportional to calculation step.

If real line characteristics are given in line capacitance and inductance per length, the following equations can be used to derive **t0** and **z0** parameters:

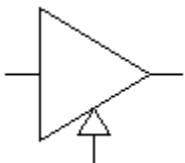
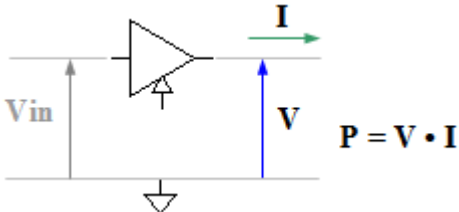
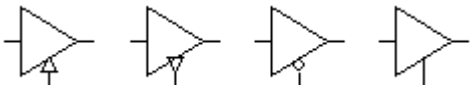
$$\mathbf{t0} = \sqrt{L * C} * D$$

$$\mathbf{z0} = \sqrt{L / C}$$

where:

C – line capacitance per length, F/m  
 L – line inductance per length, H/m  
 D – line length, m

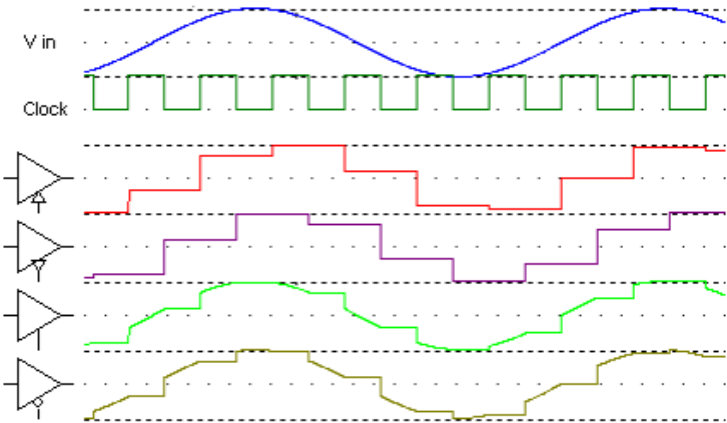
X – Sample/hold

| Symbol                                                                            | Models                                                                            | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | SH<br>SubCir                                                                      |  |
| Views                                                                             |  |                                                                                     |

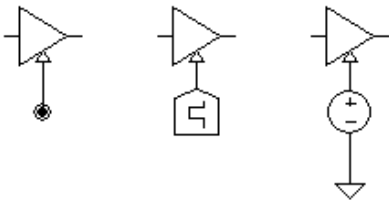
| Model | Parameter | Units | Description                        |
|-------|-----------|-------|------------------------------------|
| SH    | IC        | V     | Initial condition: output voltage. |

Depending on view, the model is functioning as a sample/hold, or as a track/hold. In sample/hold mode, input voltage is sampled at rising (or falling) edge of a logical clock signal. In track/hold mode, output voltage tracks input voltage while clock signal is above (or below, for inverted control pin) the logical threshold, and holds it while clock signal is below (or above, for inverted control pin) the logical threshold.

A waveforms example for different modes:

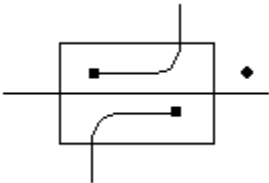
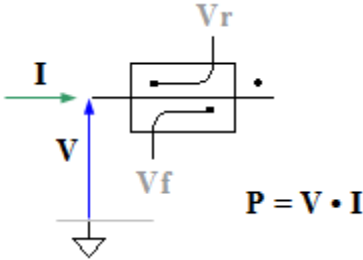


When calculating DC operating point output is set to specified output voltage **IC**.  
When calculating Linearized AC, and the clock pin is connected to the Label, Voltage source, or Logic generator with Pulse or Clock model, the component operates as a linear buffer with a delay equal to the Period of the source:





X – Directional coupler

| Symbol                                                                            | Models  | Signals                                                                             |
|-----------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------|
|  | Coupler |  |

| Model   | Parameter | Units | Description              |
|---------|-----------|-------|--------------------------|
| Coupler | z0        | Ohm   | Characteristic impedance |
|         | CF        | dB    | Coupling factor          |

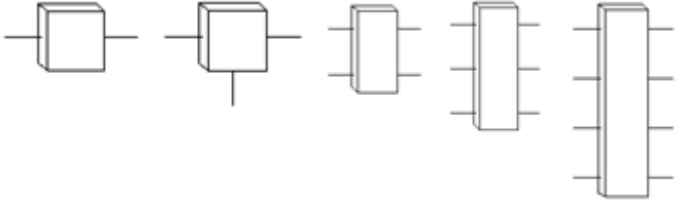
Directional coupler is a short circuit (no insertion loss) with two output ports: forward Vf, and reflected Vr. Output ports are voltage sources with zero output impedance and coupling factor **CF**. The output voltages are calculated as follows:

$$V_f = K * (V + I * z_0) / 2$$
$$V_r = K * (V - I * z_0) / 2$$

where  $K = 10^{-CF/20}$ .

All voltages are referenced to ground.

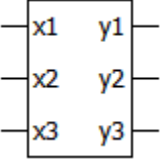
**X – Block-2...Block-8**

| Symbol                                                                                            | Models |
|---------------------------------------------------------------------------------------------------|--------|
|                  | SubCir |
| Blocks with predefined size and number of pins, to be used as subcircuits ( <b>SubCir</b> model). |        |

X – Custom block

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                  | Models | Signals |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
|                                                                                                                                                                                                                                                                                                                                                                                                                         | SubCir |         |
| <p>This is a customized component. A component can be edited in the <b>Edit Component</b> dialog box. See <i>Customized component</i> chapter of NL5 Manual for instructions on editing a component.</p> <p>This component may have:</p> <ul style="list-style-type: none"><li>- arbitrary size up to 32(width) X 32(height),</li><li>- up to 32 pins on each side</li></ul> <p>Examples of Custom block component:</p> |        |         |

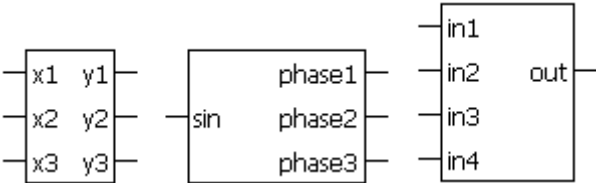
X – NL5 circuit

| Symbol                                                                            | Models | Signals |
|-----------------------------------------------------------------------------------|--------|---------|
|  | SubCir |         |

This is a customized component. A component can be edited in the **Edit Component** dialog box. See *Customized component* chapter of NL5 Manual for instructions on editing a component.

- This component may have:
- arbitrary size up to 32(width) X 256(height),
  - up to 256 inputs on the left side,
  - up to 256 outputs on the right side,

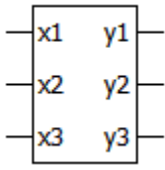
Examples of NL5 circuit component:



| Model  | Parameter | Units | Description                          |
|--------|-----------|-------|--------------------------------------|
| SubCir | File      |       | File name of subcircuit schematic.   |
|        | Cmd       |       | Subcircuit start-up command string   |
|        | IC        |       | Subcircuit Initial conditions string |

This model is similar to a standard **SubCir** model, with one difference: instead of parameters, subcircuit names are entered in the **Edit Component** dialog as a component pin names. See *Subcircuit model* chapter of NL5 Manual for details on subcircuit operation.

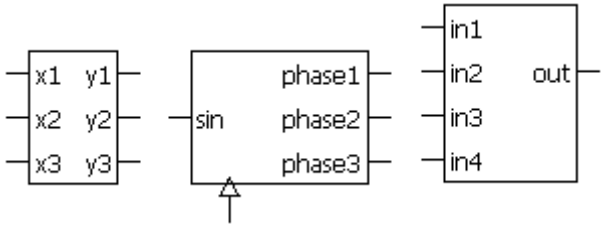
X – C-code

| Symbol                                                                            | Models    | Signals |
|-----------------------------------------------------------------------------------|-----------|---------|
|  | C<br>File |         |

This is a customized component. A component can be edited in the **Edit Component** dialog box. See *Customized component* chapter of NL5 Manual for instructions on editing a component.

- This component may have:
- arbitrary size up to 32(width) X 256(height),
  - up to 256 inputs on the left side,
  - up to 256 outputs on the right side,
  - one or no clock pins on the bottom side.
  - custom or default input and output names.

Examples of C-code component:



| Model    | Parameter   | Units | Description         |
|----------|-------------|-------|---------------------|
| <b>C</b> | <b>Code</b> |       | C-code.             |
|          | <b>IC</b>   |       | Initial conditions. |

C-code block. The model contains code written in simplified C language.

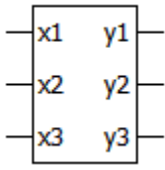
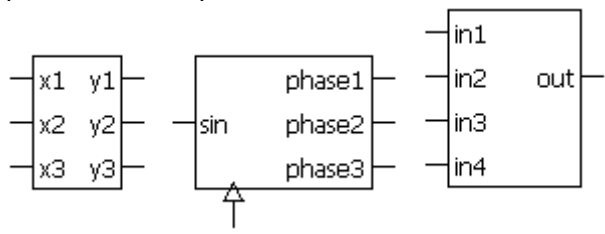
**Code** contains global variables declaration, initialization code, and main code.

**IC** may contain assignments of initial values to global variables. If not empty, **IC** will be executed after initialization code.

See *C-code component* chapter of NL5 Manual for details of the model functionality and instructions on creating the code.

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Parameter   | Units | Description         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------|---------------------|
| <b>File</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>File</b> |       | C-code file name    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>IC</b>   |       | Initial conditions. |
| <p>The model executes C-code from the text file. If <b>File</b> parameter does not have a full path, NL5 will search for the file in the directory where schematic file is located, then in the Library directories, and then in the Library directories relative to schematic file directory (see NL5 Manual, <i>Schematic Properties</i>).</p> <p><b>IC</b> may contain assignments of initial values to global variables. If not empty, <b>IC</b> will be executed after initialization code.</p> <p>See <i>C-code component</i> chapter of NL5 Manual for details of the model functionality and instructions on creating the code.</p> |             |       |                     |

X – DLL

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Models | Signals |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DLL    |         |
| <p>This is a customized component. A component can be edited in the <b>Edit Component</b> dialog box. See <i>Customized component</i> chapter of NL5 Manual for instructions on editing a component.</p> <p>This component may have:</p> <ul style="list-style-type: none"><li>- arbitrary size up to 32(width) X 256(height),</li><li>- up to 256 inputs on the left side,</li><li>- up to 256 outputs on the right side,</li><li>- one or no clock pins on the bottom side.</li><li>- custom or default input and output names.</li></ul> <p>Examples of DLL component:</p>  |        |         |

| Model | Parameter    | Units | Description                   |
|-------|--------------|-------|-------------------------------|
| DLL   | <b>DLL</b>   |       | DLL file name                 |
|       | <b>Init</b>  |       | Initialization function name. |
|       | <b>Main</b>  |       | Main function name.           |
|       | <b>Pause</b> |       | Pause function name.          |
|       | <b>Exit</b>  |       | Exit function name.           |
|       | <b>IC</b>    |       | Initial conditions.           |

DLL block. Component's code is written in C, compiled, and placed in the **64-bit** DLL file. DLL functions will be called by NL5 during transient simulation.

**DLL** parameter is a DLL file name. If **DLL** parameter does not have a full path, NL5 will search for the file in the directory where schematic file is located, then in the Library directories, and then in the Library directories relative to schematic file directory (see NL5 Manual, *Schematic Properties*).

**Init** - initialization function. Called once at the beginning of transient simulation at  $t=0$ . Leave it blank if not used.

**Main** - main function. Called on every simulation step, or in rising edge of the clock, if clock pin exists.

**Pause** - called when transient simulation is paused. Leave it blank if not used.

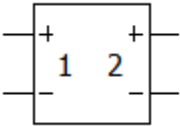
**Exit** - called when DLL is being closed, and DLL component destroyed. Leave it blank if not used.

**IC** may contain assignments of initial values to outputs and component variables. If not empty, **IC** will be executed after initialization code.

See *DLL component* chapter of NL5 Manual for details of the model functionality and instructions on creating code and DLL.



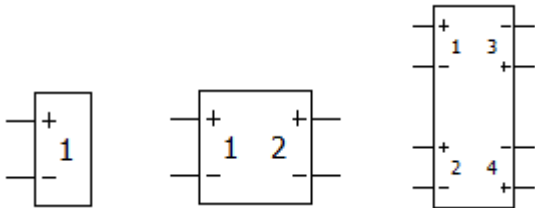
X – S-matrix

| Symbol                                                                            | Models         | Signals |
|-----------------------------------------------------------------------------------|----------------|---------|
|  | File<br>SubCir |         |

This is a customized component. A component can be edited in the **Edit Component** dialog box. See *Customized component* chapter of NL5 Manual for instructions on editing a component.

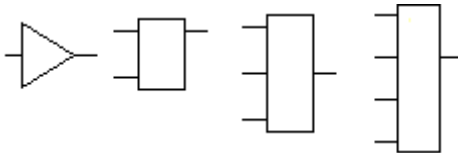
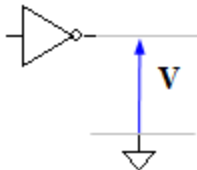
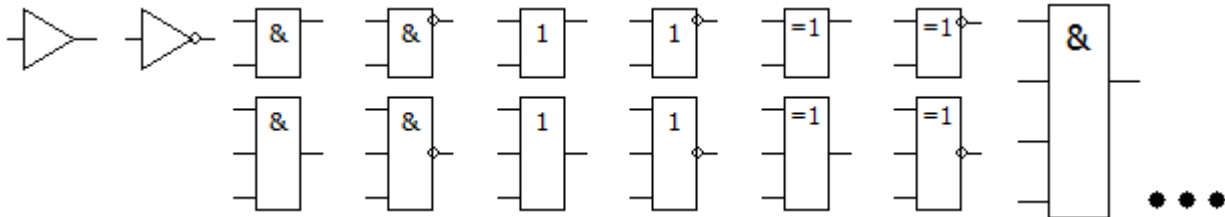
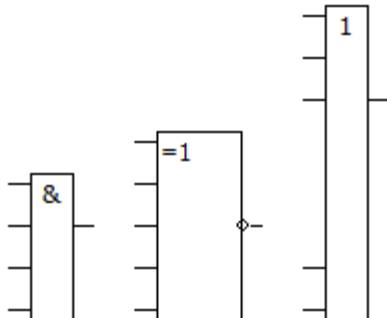
- This component may have:
- arbitrary size up to 32(width) X 32(height),
  - up to 18 pins (9 ports),
  - selected position of +/- port inputs.

Examples of S-matrix component:



| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Parameter   | Units | Description          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------|----------------------|
| <b>File</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>File</b> |       | Touchstone file name |
| <p>The model is used to perform AC simulation (Linearized schematic method only) with data from <b>touchstone</b> file. It supports up to 9 ports (touchstone files s1p...s9p, ts), data types S, Y, Z, H, G.</p> <p>If <b>File</b> parameter does not have a full path, NL5 will search for the file in the directory where schematic file is located, then in the Library directories, and then in the Library directories relative to schematic file directory (see NL5 Manual, <i>Schematic Properties</i>).</p> <p>At transient and DC operation point calculation, each port represents a constant resistance equal to specified port resistance.</p> |             |       |                      |

Y – Gates

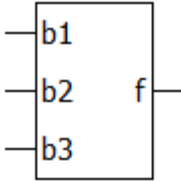
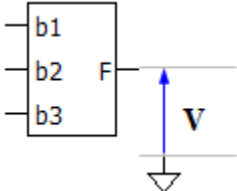
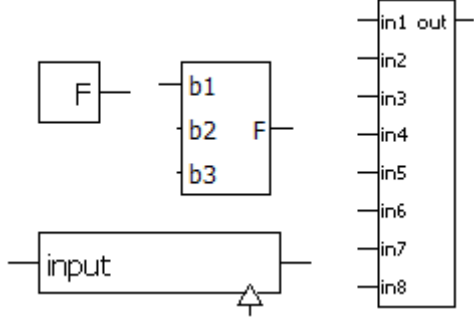
| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                    |  |  | Models                   |  | Signals                                                                             |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--|--|--------------------------|--|-------------------------------------------------------------------------------------|--|--|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                    |  |  | Logic<br>Delay<br>Delay2 |  |  |  |  |
| Views                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |  |                          |  |                                                                                     |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                    |  |  |                          |  |                                                                                     |  |  |
| <p>For all gates, select <b>view</b> for logical function (AND, OR, XOR), and output pin inversion.</p> <p>XOR (=1) function is <b>odd function</b> (modulo-2 sum): the output is high if odd number of inputs are high.</p> <p>When calculating DC operating point, output is set to specified <b>IC</b> level.</p> <p><b>Custom gate</b> component can be edited in the <b>Edit Component</b> dialog box. See <i>Customized component</i> chapter of NL5 Manual for instructions on editing a component.</p> <p>This component may have:</p> <ul style="list-style-type: none"><li>- arbitrary size up to 32(width) X 32(height),</li><li>- up to 32 inputs on the left side,</li><li>- one output on the right side.</li></ul> <p>Examples of Custom gate component:</p>  |                                                                                    |  |  |                          |  |                                                                                     |  |  |

| Model                                             | Parameter | Units | Description                  |
|---------------------------------------------------|-----------|-------|------------------------------|
| <b>Logic</b>                                      | <b>IC</b> |       | Initial condition: Low/High. |
| Output signal is delayed by one calculation step. |           |       |                              |

| Model                                                                                                                                                   | Parameter    | Units | Description                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|------------------------------|
| <b>Delay</b>                                                                                                                                            | <b>Delay</b> | s     | Output delay.                |
|                                                                                                                                                         | <b>IC</b>    |       | Initial condition: Low/High. |
| Output signal is delayed by specified <b>Delay</b> time.<br>Short signals (shorter than <b>Delay</b> ) may not pass through and will not affect output. |              |       |                              |

| Model                                                                                                                                                                                                                                                      | Parameter   | Units | Description                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------|-------------------------------|
| <b>Delay2</b>                                                                                                                                                                                                                                              | <b>Up</b>   | s     | Output delay of rising edge.  |
|                                                                                                                                                                                                                                                            | <b>Down</b> | s     | Output delay of falling edge. |
|                                                                                                                                                                                                                                                            | <b>IC</b>   |       | Initial condition: Low/High.  |
| Output signal is delayed by <b>Up</b> time for rising edge of output signal, and <b>Down</b> time for falling edge of output signal. Short signals (comparable or shorter than <b>Up</b> or <b>Down</b> ) may not pass through and will not affect output. |             |       |                               |

Y – Logical function

| Symbol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Models   | Signals                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Function |  |
| <p>This is a customized component. A component can be edited in the <b>Edit Component</b> dialog box. See <i>Customized component</i> chapter of NL5 Manual for instructions on editing a component.</p> <p>This component may have:</p> <ul style="list-style-type: none"><li>- arbitrary size up to 32(width) X 8(height),</li><li>- up to 8 inputs on the left side,</li><li>- one output on the right side,</li><li>- one or no clock pins on the bottom side.</li><li>- custom input and output names.</li></ul> <p>Examples of Logical function component:</p>  |          |                                                                                     |

| Model           | Parameter | Units | Description                               |
|-----------------|-----------|-------|-------------------------------------------|
| <b>Function</b> | <b>f</b>  | V     | Output as function of the logical inputs. |
|                 | <b>IC</b> | V     | Initial condition: Low/High.              |

Arbitrary logical function **f** defines output logical state as a function of the following variables:

**pin\_name** – logical value of the input voltage on the input pin **pin\_name**.  
**t** - current time

First, input voltages are converted to logical values, then the function is calculated. A logical result of the function is converted to the output voltage, which may have only Low or High logical level.

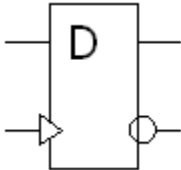
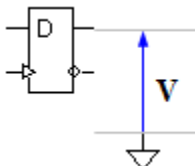
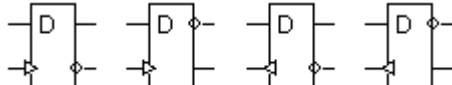
Example:

```
f = (b1 && b2) || b2
f = selector ? in1 : in2
```

If **clock** pin does not exist, the model operates in **continuous** mode: the function is calculated and applied to the output on every calculation step. If **clock** pin exists, the model operates in **synchronized** mode: the function is calculated and applied to the output only on rising (or falling) edge of logical clock signal. As a result, this mode may provide faster simulation than **continuous** mode.

When calculating DC operating point, and in AC analysis, output is set to specified output voltage **IC**. Output voltage is always delayed by one calculation step.

Y – D flip-flop

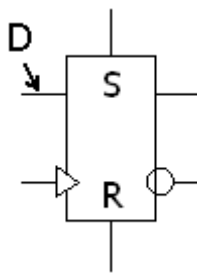
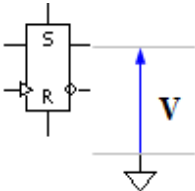
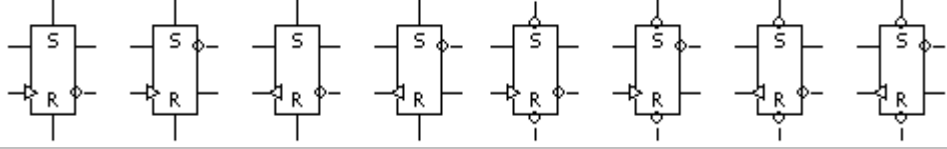
| Symbol                                                                                           |                                                                                   | Models                   | Traces                                                                              |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------|
|                 |                                                                                   | Logic<br>Delay<br>Delay2 |  |
| Views                                                                                            |  |                          |                                                                                     |
| For all models, when calculating DC operating point, output is set to specified <b>IC</b> level. |                                                                                   |                          |                                                                                     |

| Model                                             | Parameter | Units | Description                  |
|---------------------------------------------------|-----------|-------|------------------------------|
| <b>Logic</b>                                      | <b>IC</b> |       | Initial condition: Low/High. |
| Output signal is delayed by one calculation step. |           |       |                              |

| Model                                                                                                                                                   | Parameter    | Units | Description                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|------------------------------|
| <b>Delay</b>                                                                                                                                            | <b>Delay</b> | s     | Output delay.                |
|                                                                                                                                                         | <b>IC</b>    |       | Initial condition: Low/High. |
| Output signal is delayed by specified <b>Delay</b> time.<br>Short signals (shorter than <b>Delay</b> ) may not pass through and will not affect output. |              |       |                              |

| Model                                                                                                                                                                                                                                                      | Parameter   | Units | Description                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------|-------------------------------|
| <b>Delay2</b>                                                                                                                                                                                                                                              | <b>Up</b>   | s     | Output delay of rising edge.  |
|                                                                                                                                                                                                                                                            | <b>Down</b> | s     | Output delay of falling edge. |
|                                                                                                                                                                                                                                                            | <b>IC</b>   |       | Initial condition: Low/High.  |
| Output signal is delayed by <b>Up</b> time for rising edge of output signal, and <b>Down</b> time for falling edge of output signal. Short signals (comparable or shorter than <b>Up</b> or <b>Down</b> ) may not pass through and will not affect output. |             |       |                               |

Y – SR trigger

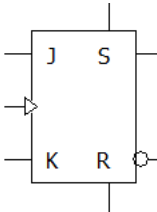
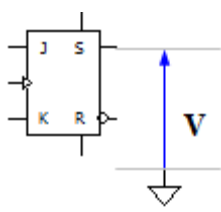
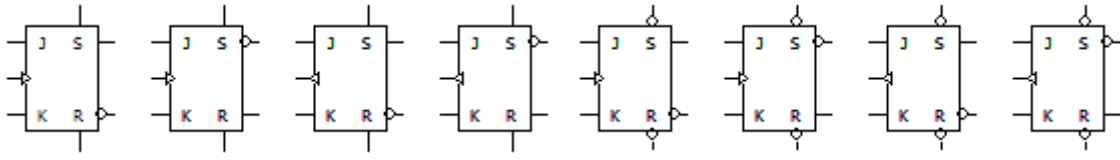
| Symbol                                                                                                                                                                                                                                                                                                         | Models                                                                             | Traces                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                               | Logic<br>Delay<br>Delay2                                                           |  |
| Views                                                                                                                                                                                                                                                                                                          |  |                                                                                     |
| <p>For all models, For all models, when calculating DC operating point, output is set to specified <b>IC</b> level.</p> <p><b>Dominance</b> defines trigger output states in case both Set and Reset inputs are active.</p> <p>Please note that 'D' input is not shown on the symbol due to limited space.</p> |                                                                                    |                                                                                     |

| Model                                             | Parameter | Units | Description                  |
|---------------------------------------------------|-----------|-------|------------------------------|
| Logic                                             | Dominance |       | Dominance: None/Set/Reset    |
|                                                   | IC        |       | Initial condition: Low/High. |
| Output signal is delayed by one calculation step. |           |       |                              |

| Model                                                                                                                                                   | Parameter | Units | Description                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------|
| Delay                                                                                                                                                   | Delay     | s     | Output delay.                |
|                                                                                                                                                         | Dominance |       | Dominance: None/Set/Reset    |
|                                                                                                                                                         | IC        |       | Initial condition: Low/High. |
| Output signal is delayed by specified <b>Delay</b> time.<br>Short signals (shorter than <b>Delay</b> ) may not pass through and will not affect output. |           |       |                              |

| Model                                                                                                                                                                                                                                                      | Parameter | Units | Description                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-------------------------------|
| Delay2                                                                                                                                                                                                                                                     | Up        | s     | Output delay of rising edge.  |
|                                                                                                                                                                                                                                                            | Down      | s     | Output delay of falling edge. |
|                                                                                                                                                                                                                                                            | Dominance |       | Dominance: None/Set/Reset     |
|                                                                                                                                                                                                                                                            | IC        |       | Initial condition: Low/High.  |
| Output signal is delayed by <b>Up</b> time for rising edge of output signal, and <b>Down</b> time for falling edge of output signal. Short signals (comparable or shorter than <b>Up</b> or <b>Down</b> ) may not pass through and will not affect output. |           |       |                               |

## Y – JK trigger

| Symbol                                                                                                                                                                                                                   |                                                                                    | Models                   | Traces                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------|
|                                                                                                                                         |                                                                                    | Logic<br>Delay<br>Delay2 |  |
| Views                                                                                                                                                                                                                    |  |                          |                                                                                     |
| <p>For all models, For all models, when calculating DC operating point, output is set to specified <b>IC</b> level.<br/><b>Dominance</b> defines trigger output states in case both Set and Reset inputs are active.</p> |                                                                                    |                          |                                                                                     |

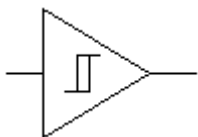
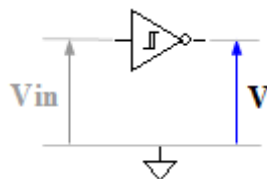
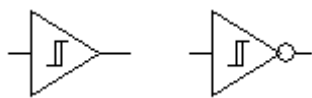
| Model                                             | Parameter        | Units | Description                  |
|---------------------------------------------------|------------------|-------|------------------------------|
| <b>Logic</b>                                      | <b>Dominance</b> |       | Dominance: None/Set/Reset    |
|                                                   | <b>IC</b>        |       | Initial condition: Low/High. |
| Output signal is delayed by one calculation step. |                  |       |                              |

| Model                                                                                                                                                           | Parameter        | Units | Description                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------|------------------------------|
| <b>Delay</b>                                                                                                                                                    | <b>Delay</b>     | s     | Output delay.                |
|                                                                                                                                                                 | <b>Dominance</b> |       | Dominance: None/Set/Reset    |
|                                                                                                                                                                 | <b>IC</b>        |       | Initial condition: Low/High. |
| <p>Output signal is delayed by specified <b>Delay</b> time.<br/> Short signals (shorter than <b>Delay</b>) may not pass through and will not affect output.</p> |                  |       |                              |

| Model                                                                                                                                                                                                                                                            | Parameter        | Units | Description                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------|-------------------------------|
| <b>Delay2</b>                                                                                                                                                                                                                                                    | <b>Up</b>        | s     | Output delay of rising edge.  |
|                                                                                                                                                                                                                                                                  | <b>Down</b>      | s     | Output delay of falling edge. |
|                                                                                                                                                                                                                                                                  | <b>Dominance</b> |       | Dominance: None/Set/Reset     |
|                                                                                                                                                                                                                                                                  | <b>IC</b>        |       | Initial condition: Low/High.  |
| <p>Output signal is delayed by <b>Up</b> time for rising edge of output signal, and <b>Down</b> time for falling edge of output signal. Short signals (comparable or shorter than <b>Up</b> or <b>Down</b>) may not pass through and will not affect output.</p> |                  |       |                               |



## Y – Schmitt trigger

| Symbol                                                                            | Models                                                                            | Traces                                                                              |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Logic<br>Delay<br>Delay2                                                          |  |
| Views                                                                             |  |                                                                                     |

For all models, when calculating DC operating point, output is set to specified **IC** level.

Output is set to Low or High level as follows (non-inverted output, **Threshold** is a logical level threshold):

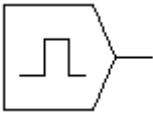
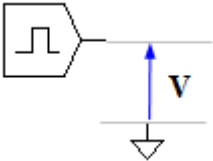
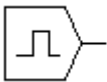
$V_{in} > \text{Threshold} + \text{Hysteresis}/2 \dots : V = \text{High}$   
 $V_{in} < \text{Threshold} - \text{Hysteresis}/2 \dots : V = \text{Low}$   
Otherwise ..... :  $V = \text{previous state}$

| Model                                             | Parameter         | Units | Description                  |
|---------------------------------------------------|-------------------|-------|------------------------------|
| <b>Logic</b>                                      | <b>Hysteresis</b> | V     | Hysteresis.                  |
|                                                   | <b>IC</b>         |       | Initial condition: Low/High. |
| Output signal is delayed by one calculation step. |                   |       |                              |

| Model                                                                                                                                                   | Parameter         | Units | Description                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------|------------------------------|
| <b>Delay</b>                                                                                                                                            | <b>Hysteresis</b> | V     | Hysteresis.                  |
|                                                                                                                                                         | <b>Delay</b>      | s     | Output delay                 |
|                                                                                                                                                         | <b>IC</b>         |       | Initial condition: Low/High. |
| Output signal is delayed by specified <b>Delay</b> time.<br>Short signals (shorter than <b>Delay</b> ) may not pass through and will not affect output. |                   |       |                              |

| Model                                                                                                                                                                                                                                                      | Parameter         | Units | Description                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------|-------------------------------|
| <b>Delay2</b>                                                                                                                                                                                                                                              | <b>Hysteresis</b> | V     | Hysteresis.                   |
|                                                                                                                                                                                                                                                            | <b>Up</b>         | s     | Output delay of rising edge.  |
|                                                                                                                                                                                                                                                            | <b>Down</b>       | s     | Output delay of falling edge. |
|                                                                                                                                                                                                                                                            | <b>IC</b>         |       | Initial condition: Low/High.  |
| Output signal is delayed by <b>Up</b> time for rising edge of output signal, and <b>Down</b> time for falling edge of output signal. Short signals (comparable or shorter than <b>Up</b> or <b>Down</b> ) may not pass through and will not affect output. |                   |       |                               |

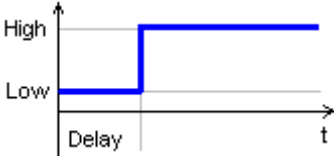
Y – Logic generator

| Symbol                                                                            | Models                                                                                                                                                              |                                | Traces                                                                              |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------|
|  | Low<br>High<br>Step<br>Single                                                                                                                                       | Pulse<br>Clock<br>List<br>File |  |
| Views                                                                             |   |                                |                                                                                     |

| Model                                                               | Parameter      | Units | Description |
|---------------------------------------------------------------------|----------------|-------|-------------|
| Low                                                                 | No parameters. |       |             |
| Output is always <b>Low</b> , regardless of output inversion state. |                |       |             |

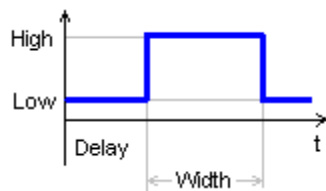
| Model                                                               | Parameter      | Units | Description |
|---------------------------------------------------------------------|----------------|-------|-------------|
| High                                                                | No parameters. |       |             |
| Output is always <b>High</b> , regardless of output inversion state |                |       |             |

| Model                                                                                                                | Parameter | Units | Description                |
|----------------------------------------------------------------------------------------------------------------------|-----------|-------|----------------------------|
| Step                                                                                                                 | Delay     | s     | Delay before active state. |
| Output goes to active state after <b>Delay</b> time. The following signal will be generated for non-inverted output: |           |       |                            |



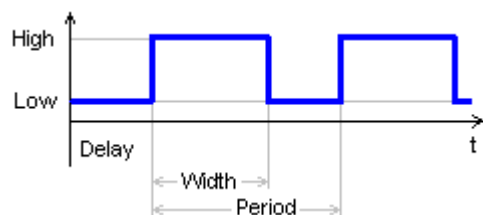
| Model         | Parameter    | Units | Description                      |
|---------------|--------------|-------|----------------------------------|
| <b>Single</b> | <b>Width</b> | s     | Pulse width.                     |
|               | <b>Delay</b> | s     | Delay before first pulse starts. |

The pulse starts after **Delay** time. The following signal will be generated for non-inverted output:



| Model        | Parameter     | Units | Description                      |
|--------------|---------------|-------|----------------------------------|
| <b>Pulse</b> | <b>Period</b> | s     | Period.                          |
|              | <b>Width</b>  | s     | Pulse width.                     |
|              | <b>Delay</b>  | s     | Delay before first pulse starts. |

Pulses start after **Delay** time. The following signal will be generated for non-inverted output:

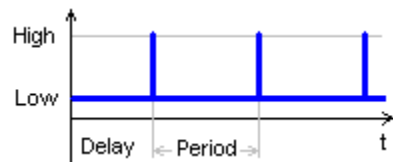


| Model | Parameter | Units | Description                       |
|-------|-----------|-------|-----------------------------------|
| Clock | Period    | s     | Period.                           |
|       | Step      | s     | Simulation step of rise and fall. |
|       | Delay     | s     | Delay before first pulse starts.  |

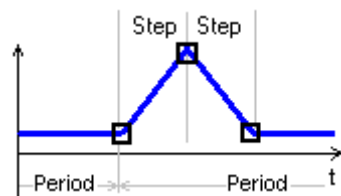
Pulses start after **Delay** time. Output goes to active state for one simulation step only.

**Clock** model is recommended to produce a constant frequency clock signal for C-code, DLL, logical components, etc. Unlike **Pulse** model, it won't force unnecessary step reduction at the end of the pulse, which may help to accelerate simulation.

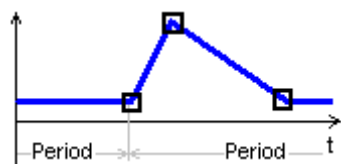
The following signal will be generated for non-inverted output:



If **Step** parameter is not zero **and** is less than current schematic simulation step, the step is adjusted to provide rise and fall of clock pulse to be equal to **Step**:



Otherwise, the clock pulse is created using current schematic simulation step, which depends on many factors, and cannot be easily predicted:



| Model | Parameter | Units | Description               |
|-------|-----------|-------|---------------------------|
| List  | List      |       | Comma-separated string.   |
|       | Cycle     |       | Cycling (repeat): No/Yes. |
|       | Delay     | s     | Delay.                    |

Output sequence is defined in the **List** parameter in the **csv** (comma-separated values) format, as follows:

$t_0, s_0, t_1, s_1, \dots, t_n, s_n$

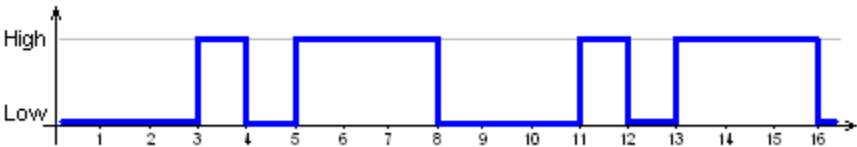
$s_0 \dots s_n$  defines output logical level: positive number corresponds to High, zero or negative number to Low.  
If  $t < t_0$ , output level is  $s_0$ .  
At  $t_0$  output level is  $s_0$ , at  $t_1$  output level is  $s_1$ , and so on.  
If  $t > t_n$ , and **Cycle** parameter is set to **No**, output level remains at  $s_n$ . Otherwise the sequence is repeated continuously.

Sequence start is delayed by **Delay** time.

Example:

`List = 0,0,3,1,4,0,5,1,8,0`

If **Cycle = Yes**, **Delay = 0**, the following logical output will be generated:



See *List model* chapter of NL5 Manual for more details.

| Model | Parameter | Units | Description               |
|-------|-----------|-------|---------------------------|
| File  | File      |       | File name.                |
|       | Cycle     |       | Cycling (repeat): No/Yes. |
|       | Delay     | s     | Delay.                    |

Output sequence defined in the text file. If **File** parameter does not have a full path, NL5 will search for the file in the directory where schematic file is located, then in the Library directories, and then in the Library directories relative to schematic file directory (see NL5 Manual, *Schematic Properties*).

Logical output sequence is defined in the **csv** (comma-separated values) format, as follows:

```
<if first line does not start with a number, it is ignored>
t0,s0
t1,s1
.....
tn,sn
```

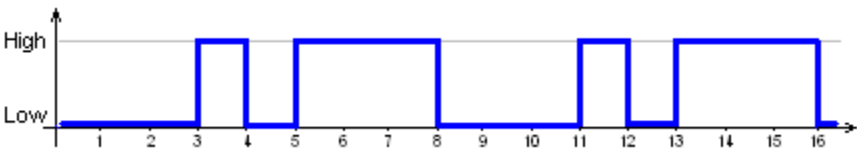
s0...sn defines output logical level: positive number corresponds to High, zero or negative number to Low.  
If  $t < t_0$ , output level is s0.  
At  $t_0$  output level is s0, at  $t_1$  output level is s1, and so on.  
If  $t > t_n$ , and **Cycle** parameter is set to **No**, output level remains at sn. Otherwise the sequence is repeated continuously.

Sequence start is delayed by **Delay** time.

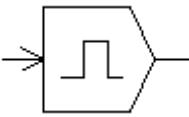
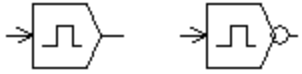
Example:

```
0,0
3,1
4,0
5,1
8,0
```

If **Cycle = Yes**, **Delay = 0**, the following sequence will be generated:



Y – Logic controlled logic generator

| Symbol                                                                            | Models                                                                            | Signals                                  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------|
|  | Gate<br>Low<br>High<br>One-shot<br>Step                                           | Single<br>Pulse<br>Clock<br>List<br>File |
| Views                                                                             |  |                                          |

| Model                                                                                  | Parameter | Units | Description                  |
|----------------------------------------------------------------------------------------|-----------|-------|------------------------------|
| Gate                                                                                   | IC        |       | Initial condition: Low/High. |
| Component operates similar to <b>Logic</b> model of <b>Gate</b> component.             |           |       |                              |
| When calculating DC operating point, output is set to the state defined in <b>IC</b> . |           |       |                              |

| Model                                                               | Parameter      | Units | Description |
|---------------------------------------------------------------------|----------------|-------|-------------|
| Low                                                                 | No parameters. |       |             |
| Output is always <b>Low</b> , regardless of output inversion state. |                |       |             |

| Model                                                               | Parameter      | Units | Description |
|---------------------------------------------------------------------|----------------|-------|-------------|
| High                                                                | No parameters. |       |             |
| Output is always <b>High</b> , regardless of output inversion state |                |       |             |

| Model                                                                                                                                            | Parameter | Units | Description  |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|--------------|
| One-shot                                                                                                                                         | Width     | s     | Pulse width. |
| One-shot pulse generator. When increasing input voltage $V_{in}$ crosses logical threshold, logical pulse of <b>Width</b> duration is generated. |           |       |              |
| If increasing $V_{in}$ crosses logical threshold value while pulse is being generated, the pulse is restarted.                                   |           |       |              |

| Model                                                                                                                                                                                                                                                                                                                                                                 | Parameter    | Units | Description                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|----------------------------|
| <b>Step</b>                                                                                                                                                                                                                                                                                                                                                           | <b>Delay</b> | s     | Delay before active state. |
| <p>When control signal <i>Vin</i> is below logical threshold, output is in Off state. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>Step</b> model of <b>Logic generator</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to Off state immediately.</p> |              |       |                            |

| Model                                                                                                                                                                                                                                                                                                                                                                   | Parameter    | Units | Description                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|----------------------------------|
| <b>Single</b>                                                                                                                                                                                                                                                                                                                                                           | <b>Width</b> | s     | Pulse width.                     |
|                                                                                                                                                                                                                                                                                                                                                                         | <b>Delay</b> | s     | Delay before first pulse starts. |
| <p>When control signal <i>Vin</i> is below logical threshold, output is in Off state. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>Single</b> model of <b>Logic generator</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to Off state immediately.</p> |              |       |                                  |

| Model                                                                                                                                                                                                                                                                                                                                                                  | Parameter     | Units | Description                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|----------------------------------|
| <b>Pulse</b>                                                                                                                                                                                                                                                                                                                                                           | <b>Period</b> | s     | Period.                          |
|                                                                                                                                                                                                                                                                                                                                                                        | <b>Width</b>  | s     | Pulse width.                     |
|                                                                                                                                                                                                                                                                                                                                                                        | <b>Delay</b>  | s     | Delay before first pulse starts. |
| <p>When control signal <i>Vin</i> is below logical threshold, output is in Off state. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>Pulse</b> model of <b>Logic generator</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to Off state immediately.</p> |               |       |                                  |

| Model                                                                                                                                                                                                                                                                                                                                                                  | Parameter     | Units | Description                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|-----------------------------------|
| <b>Clock</b>                                                                                                                                                                                                                                                                                                                                                           | <b>Period</b> | s     | Period.                           |
|                                                                                                                                                                                                                                                                                                                                                                        | <b>Step</b>   | s     | Simulation step of rise and fall. |
|                                                                                                                                                                                                                                                                                                                                                                        | <b>Delay</b>  | s     | Delay before first pulse starts.  |
| <p>When control signal <i>Vin</i> is below logical threshold, output is in Off state. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>Clock</b> model of <b>Logic generator</b> component is generated. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to Off state immediately.</p> |               |       |                                   |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Parameter    | Units | Description               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>List</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>List</b>  |       | Comma-separated string.   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <i>Vin</i> is below logical threshold, output is equal to <b>s0</b> value of <b>List</b> signal. When increasing control signal <i>Vin</i> crosses logical threshold, a signal similar to <b>List</b> model of <b>Logic generator</b> component is generated. This moment is also considered as <b>t=0</b> for the <b>List</b> signal. When decreasing control signal <i>Vin</i> drops below logical threshold, output goes to <b>s0</b> immediately.</p> |              |       |                           |



| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Parameter    | Units | Description               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|---------------------------|
| <b>File</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>File</b>  |       | File name.                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Cycle</b> |       | Cycling (repeat): No/Yes. |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Delay</b> | s     | Delay.                    |
| <p>When control signal <math>V_{in}</math> is below logical threshold, output is equal to <b>s0</b> value specified in the <b>File</b>. When increasing control signal <math>V_{in}</math> crosses logical threshold, a signal similar to <b>File</b> model of <b>Logic generator</b> component is generated. This moment is also considered as <math>t=0</math> for the <b>File</b> signal. When decreasing control signal <math>V_{in}</math> drops below logical threshold, output goes to <b>s0</b> immediately.</p> |              |       |                           |

Y – Voltage controlled logic generator

| Symbol | Models                          | Signals |
|--------|---------------------------------|---------|
|        | Gate<br>Low<br>High<br>One-shot |         |
| Views  |                                 |         |

| Model | Parameter  | Units | Description                  |
|-------|------------|-------|------------------------------|
| Gate  | Threshold  | V     | Voltage threshold.           |
|       | Hysteresis | V     | Hysteresis.                  |
|       | IC         |       | Initial condition: Low/High. |

Gate with hysteresis. Output is set to Low or High level as follows (non-inverted output):

$V_{in} > \text{Threshold} + \text{Hysteresis}/2 \dots : V = \text{High}$   
 $V_{in} < \text{Threshold} - \text{Hysteresis}/2 \dots : V = \text{Low}$   
Otherwise  $\dots : V = \text{previous state}$

When calculating DC operating point, output is set to the state defined in **IC**.

| Model | Parameter      | Units | Description |
|-------|----------------|-------|-------------|
| Low   | No parameters. |       |             |

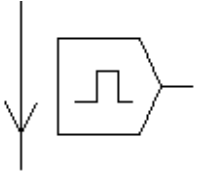
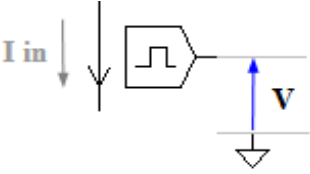
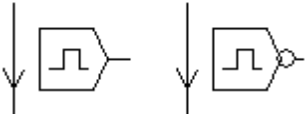
Output is always **Low**, regardless of output inversion state.

| Model | Parameter      | Units | Description |
|-------|----------------|-------|-------------|
| High  | No parameters. |       |             |

Output is always **High**, regardless of output inversion state

| Model                                                                                                                                                                                                                                                                                             | Parameter        | Units | Description        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------|--------------------|
| <b>One-shot</b>                                                                                                                                                                                                                                                                                   | <b>Threshold</b> | V     | Voltage threshold. |
|                                                                                                                                                                                                                                                                                                   | <b>Width</b>     | s     | Pulse width.       |
| <p>One-shot pulse generator. When increasing input voltage <math>V_{in}</math> crosses <b>Threshold</b>, logical pulse of <b>Width</b> duration is generated.</p> <p>If increasing <math>V_{in}</math> crosses <b>Threshold</b> value while pulse is being generated, the pulse is restarted.</p> |                  |       |                    |

## Y – Current controlled logic generator

| Symbol                                                                            | Models                                                                            | Signals                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Gate<br>Low<br>High<br>One-shot                                                   |  |
| Views                                                                             |  |                                                                                     |

| Model       | Parameter         | Units | Description                  |
|-------------|-------------------|-------|------------------------------|
| <b>Gate</b> | <b>Threshold</b>  | A     | Current threshold.           |
|             | <b>Hysteresis</b> | A     | Hysteresis.                  |
|             | <b>IC</b>         |       | Initial condition: Low/High. |

Gate with hysteresis. Output is set to Low or High level as follows (non-inverted output):

$I_{in} > \text{Threshold} + \text{Hysteresis}/2 \dots : V = \text{High}$   
 $I_{in} < \text{Threshold} - \text{Hysteresis}/2 \dots : V = \text{Low}$   
 Otherwise ..... :  $V = \text{previous state}$

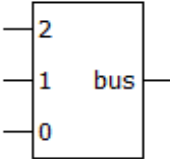
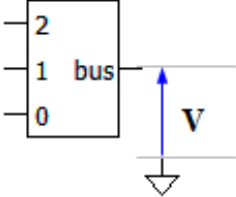
When calculating DC operating point, output is set to the state defined in **IC**.

| Model                                                               | Parameter      | Units | Description |
|---------------------------------------------------------------------|----------------|-------|-------------|
| Low                                                                 | No parameters. |       |             |
| Output is always <b>Low</b> , regardless of output inversion state. |                |       |             |

| Model                                                               | Parameter      | Units | Description |
|---------------------------------------------------------------------|----------------|-------|-------------|
| High                                                                | No parameters. |       |             |
| Output is always <b>High</b> , regardless of output inversion state |                |       |             |

| Model                                                                                                                                                                                                                                                                                             | Parameter        | Units | Description        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------|--------------------|
| <b>One-shot</b>                                                                                                                                                                                                                                                                                   | <b>Threshold</b> | A     | Current threshold. |
|                                                                                                                                                                                                                                                                                                   | <b>Width</b>     | s     | Pulse width.       |
| <p>One-shot pulse generator. When increasing input current <math>I_{in}</math> crosses <b>Threshold</b>, logical pulse of <b>Width</b> duration is generated.</p> <p>If increasing <math>I_{in}</math> crosses <b>Threshold</b> value while pulse is being generated, the pulse is restarted.</p> |                  |       |                    |

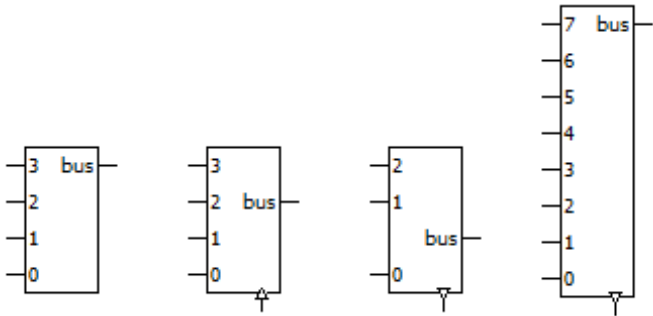
Y – Bus

| Symbol                                                                            | Models | Traces                                                                              |
|-----------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------|
|  | Bus    |  |

This is a customized component. A component can be edited in the **Edit Component** dialog box. See *Customized component* chapter of NL5 Manual for instructions on editing a component.

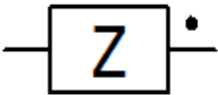
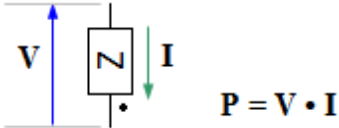
- This component may have:
- arbitrary size up to 32(width) X 32(height),
  - up to 32 inputs on the left side,
  - one output on the right side,
  - one or no clock pins on the bottom side.

Examples of **Bus** component:



| Model                                                                                                                                                                                                                                                                                                                                                                                                                                          | Parameter     | Units | Description                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|------------------------------------|
| <b>Bus</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Format</b> |       | Bus format: Signed/Unsigned        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>IC</b>     | V     | Initial condition: output voltage. |
| <p>Digital-to-bus converter. Converts logical inputs to a bus signal, considering logical inputs as bits of binary number. Input <b>0</b> is <b>LSB</b>. If <b>Format</b> is set to Signed, an input number is considered to be in two's complement format.</p> <p>When calculating DC operating point, and in AC analysis, output is set to specified output voltage <b>IC</b>. Output voltage is always delayed by one calculation step.</p> |               |       |                                    |

Z – Impedance

| Symbol                                                                            | Models                                               |                                          | Signals                                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|
|  | Function<br>Poly<br>Poly1<br>Poly2<br>Poly3<br>Poly4 | Poly5<br>Roots<br>Table<br>File<br>Trace |  |

| Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Parameter | Units | Description |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-------------|
| Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | f         | Ohm   | Impedance.  |
| <p>Arbitrary function <b>f</b> defines impedance in <b>s</b> domain. The following variables can represent frequency in the function:</p> <p><b>f</b> – current AC frequency, Hz<br/><b>w</b> – angular AC frequency, <b>w = 2πf</b> .<br/><b>s</b> or <b>p</b> – Laplace parameter, <b>s = p = j*2πf</b>.</p> <p>Example:</p> <p>f = 1/(1+s)<br/>f = exp(-R1*C1*s)</p> <p>Only operators and functions that support complex numbers can be used in this function. If <b>f</b> is blank, it is assumed to be zero. If division by zero occurs during function calculation, this frequency is ignored, and data point is not added to the trace.</p> <p>At transient and DC operation point calculation for AC (if enabled), the impedance is equal to <b>f(0)</b>.</p> |           |       |             |

| Model       | Parameter   | Units | Description        |
|-------------|-------------|-------|--------------------|
| <b>Poly</b> | <b>Poly</b> |       | Polynomial string. |
|             | <b>IC</b>   |       | Initial condition. |

Impedance is a ratio of polynomials of Laplace parameter  $s$ :

$$f(s) = (b_0 + b_1*s + b_2*s^2 + \dots) / (a_0 + a_1*s + a_2*s^2 + \dots)$$

The model supports transient as well.

**Poly** parameter is a comma-separated string:

$$n, b_0, \dots, b_n, a_0, \dots, a_n$$

where **n** is maximum order of numerator and denominator ( $n = 0 \dots 5$ ), **bi** are numerator coefficients, and **ai** are denominator coefficients ( $i = 0 \dots n$ ). Please note that all **n** coefficients should be provided, even if they are zeroes. **Poly** parameter can be entered in the table in the **Poly window** of NL5 GUI.

Initial condition **IC** is a **csv** (comma separated values) string that keeps internal state of the component when transient is paused and **Save IC** command is performed. Modifying **IC** parameter manually is not recommended, except clearing it to blank (no **IC**).

If division by zero occurs during polynomial calculation, this frequency is ignored, and data point is not added to the trace.

At DC operation point calculation, **f(0)** is used.  
See *Poly model* chapter of NL5 Manual for more details.

| Model        | Parameter | Units | Description                            |
|--------------|-----------|-------|----------------------------------------|
| <b>Poly1</b> | <b>b0</b> |       | Numerator polynomial coefficients 0.   |
| <b>Poly2</b> | ...       |       | ...                                    |
| <b>Poly3</b> | <b>a0</b> |       | Denominator polynomial coefficients 0. |
| <b>Poly4</b> | ...       |       | ...                                    |
| <b>Poly5</b> | <b>IC</b> |       | Initial condition.                     |

Impedance is a ratio of polynomials of Laplace parameter  $s$ :

$$f(s) = (b_0 + b_1*s + b_2*s^2 + \dots) / (a_0 + a_1*s + a_2*s^2 + \dots)$$

If division by zero occurs during polynomial calculation, this frequency is ignored, and data point is not added to the trace.

These models support transient as well.

Initial condition **IC** is a **csv** (comma separated values) string that keeps internal state of the component when transient is paused and **Save IC** command is performed. Modifying **IC** parameter manually is not recommended, except clearing it to blank (no **IC**).

At DC operation point calculation, **f(0)** is used.



| Model        | Parameter    | Units | Description               |
|--------------|--------------|-------|---------------------------|
| <b>Roots</b> | <b>K</b>     |       | Gain.                     |
|              | <b>Roots</b> |       | Roots (zeroes and poles). |
|              | <b>IC</b>    | V     | Initial condition.        |

Impedance is defines by zeroes and poles:

$$f(s) = K * (s-z1)*(s-z2).../(s-p1)/(s-p2)...$$

where **K** is gain, z1...zn are zeroes, p1...pN are poles.

Roots are defined by **Roots** parameter in the **csv** (comma-separated values) format, as follows:

Nz,Rez1,Imz1,...,Np,Rep1,Imp1,...

where:

Nz - number of zeroes

Rezi – real part of zi

Imzi – imaginary part of zi

Np - number of poles,

Repi – real part of pi

Impi – imaginary part of pi

There could be any number of zeroes and poles, however the resulting numerator and denominator polynomials order should not exceed 5. See *Roots model* chapter of NL5 Manual for details on entering/editing roots.

If division by zero occurs during calculation, this frequency is ignored, and data point is not added to the trace.

The model supports transient as well.

Initial condition **IC** is a **csv** (comma separated values) string that keeps internal state of the component when transient is paused and **Save IC** command is performed. Modifying **IC** parameter manually is not recommended, except clearing it to blank (no **IC**).

At DC operation point calculation, **f(0)** is used.

| Model        | Parameter    | Units | Description                               |
|--------------|--------------|-------|-------------------------------------------|
| <b>Table</b> | <b>Table</b> |       | Comma-separated string.                   |
|              | <b>R</b>     | Ohm   | Resistance used for transient simulation. |

Look-up table. Impedance is defined by **Table** parameter in the **csv** (comma separated values) format, as follows:

```
f1,mag1,phase1,f2,mag2,phase2,...,fN,magN,phaseN
```

where mag and phase are impedance magnitude and phase at specified frequency. Output value between specified frequencies is interpolated using linear or logarithmic interpolation, depending on simulation scale. Below first and above last specified frequency, simulation result is not provided. Frequencies should be given in ascending order.

At transient simulation, constant resistance **R** is used.

See *Table model* chapter of NL5 Manual for more details.

| Model       | Parameter   | Units | Description                               |
|-------------|-------------|-------|-------------------------------------------|
| <b>File</b> | <b>File</b> |       | File name.                                |
|             | <b>R</b>    | Ohm   | Resistance used for transient simulation. |

Impedance is defined in the text file in the following format.

```
f1,mag1,phase1
f2,mag2,phase2
...
```

At transient simulation, constant resistance **R** is used

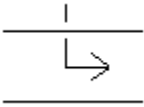
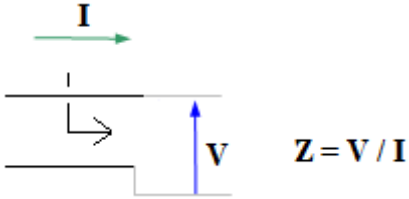
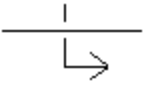
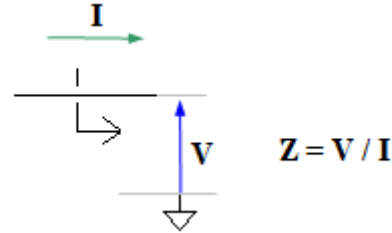
| Model        | Parameter    | Units | Description                               |
|--------------|--------------|-------|-------------------------------------------|
| <b>Trace</b> | <b>Trace</b> |       | Trace name.                               |
|              | <b>R</b>     | Ohm   | Resistance used for transient simulation. |

Impedance is defined by an existing AC trace of **Data** type with “static” data: entered manually, duplicated, imported, pasted from the clipboard, etc.

**Trace** is a name of the existing trace shown in the Trace list.

At transient simulation, constant resistance **R** is used.

**Z – Z-meter**

| Symbol                                                                            | Models  | Signals                                                                             |
|-----------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------|
|  | Z-meter |  |
|  | Z-meter |  |

| Model                                                                                                                       | Parameter      | Units | Description |
|-----------------------------------------------------------------------------------------------------------------------------|----------------|-------|-------------|
| Z-meter                                                                                                                     | No parameters. |       |             |
| Impedance meter (floating or grounded load) for AC, <i>Linearize schematic</i> method only.<br>Short circuit for transient. |                |       |             |

# Operators

Operators are listed in descending precedence order (1 - most, 14 - least).

The table is based on [http://en.cppreference.com/w/cpp/language/operator\\_precedence](http://en.cppreference.com/w/cpp/language/operator_precedence)

| Precedence | Operator                                                                           | Description                                                                                                                                                                                                                                                                                              |
|------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | ()<br>[]<br>x++<br>x--<br>++x<br>--x                                               | Function call<br>Array subscripting<br>Postfix increment: x=x+1 after use<br>Postfix decrement: x=x-1 after use<br>Prefix increment: x=x+1 before use<br>Prefix decrement: x=x-1 before use                                                                                                              |
| 2          | +<br>-<br>!<br>~<br>(bool)<br>(int)<br>(int64)<br>(float)<br>(double)<br>(complex) | Unary plus<br>Unary minus<br>Logical NOT<br>Bitwise NOT<br>Type cast to <b>bool</b><br>Type cast to <b>int</b><br>Type cast to <b>int64</b><br>Type cast to <b>float</b><br>Type cast to <b>double</b><br>Type cast to <b>complex</b>                                                                    |
| 3          | *<br>/<br>%                                                                        | Multiplication<br>Division<br>Remainder                                                                                                                                                                                                                                                                  |
| 4          | +<br>-                                                                             | Addition<br>Subtraction                                                                                                                                                                                                                                                                                  |
| 5          | <<<br>>>                                                                           | Bitwise left shift<br>Bitwise right shift                                                                                                                                                                                                                                                                |
| 6          | <<br><=<br>><br>>=                                                                 | Relation operator "less than"<br>Relation operator "less than or equal to"<br>Relation operator "greater than"<br>Relation operator "greater than or equal to"                                                                                                                                           |
| 7          | ==<br>!=                                                                           | Relation operator "equal to"<br>Relation operator "not equal to"                                                                                                                                                                                                                                         |
| 8          | &                                                                                  | Bitwise AND                                                                                                                                                                                                                                                                                              |
| 9          | ^                                                                                  | Bitwise XOR (exclusive OR)                                                                                                                                                                                                                                                                               |
| 10         |                                                                                    | Bitwise OR                                                                                                                                                                                                                                                                                               |
| 11         | &&                                                                                 | Logical AND                                                                                                                                                                                                                                                                                              |
| 12         |                                                                                    | Logical OR                                                                                                                                                                                                                                                                                               |
| 13         | ?:                                                                                 | Ternary conditional operator                                                                                                                                                                                                                                                                             |
| 14         | =<br>+=<br>-=<br>*=<br>/=<br>%=<br><<=<br>>>=<br>&=<br>^=<br> =                    | Assignment<br>Assignment by sum<br>Assignment by difference<br>Assignment by product<br>Assignment by quotient<br>Assignment by remainder<br>Assignment by bitwise left shift<br>Assignment by bitwise right shift<br>Assignment by bitwise AND<br>Assignment by bitwise XOR<br>Assignment by bitwise OR |

# Functions

## List of C functions:

|         |            |          |        |             |              |
|---------|------------|----------|--------|-------------|--------------|
| bool    | fabs       | sqrt     | sin    | random,rand | min          |
| int     | fmod       | sqr      | cos    | gauss       | max          |
| int64   | fmin       | sq       | tan,tg | islow       | mean,average |
| float   | fmax       | pow      | asin   | ishigh      | sum          |
| double  | abs,mag    | pwr      | acos   | lim,limit   | re           |
| complex | round      | exp      | atan   | sign        | im           |
|         | round(x,y) | exp2     | atan2  | par         |              |
|         | floor      | log,ln   | phase  |             |              |
|         | ceil       | log2,lb  |        |             |              |
|         |            | log10,lg |        |             |              |
|         |            | log(x,y) |        |             |              |
|         |            | db       |        |             |              |

**bool C-keyword**

|                    |                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------|
| <b>Description</b> | Declares boolean variable or array.                                                        |
| <b>Examples</b>    | <pre>bool b; bool var = false; bool array[10]; bool array[] = { true, false, true };</pre> |

**(bool) type-casting operator**

|                    |                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Converts a number of arbitrary type to boolean.<br>(bool)x works exactly the same as function bool(x). |
| <b>Examples</b>    | b = (bool)x;                                                                                           |

**bool**

|                    |                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre><b>bool</b> bool(<b>bool</b>) <b>bool</b> bool(<b>int</b>) <b>bool</b> bool(<b>int64</b>) <b>bool</b> bool(<b>double</b>) <b>bool</b> bool(<b>complex</b>)</pre> |
| <b>Description</b> | Returns false if argument is equal to zero, returns true if argument is non-zero.<br>bool(x) works exactly the same as type-casting operator (bool)x.                 |
| <b>Examples</b>    | <pre>bool(0) = false bool(1.5) = true bool(1.0+2.0j) = true</pre>                                                                                                     |

**int C-keyword**

|                    |                                                                              |
|--------------------|------------------------------------------------------------------------------|
| <b>Description</b> | Declares integer variable or array.                                          |
| <b>Examples</b>    | <pre>int i; int var = 10; int array[10]; int array[] = { 0, 1, 2, 3 };</pre> |

**(int) type-casting operator**

|                    |                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Converts a number of arbitrary type to integer.<br>(int)x works exactly the same as function int(x). |
| <b>Examples</b>    | i = (int)x;                                                                                          |

**int**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre><b>int</b> int(<b>bool</b>) <b>int</b> int(<b>int</b>) <b>int</b> int(<b>int64</b>) <b>int</b> int(<b>double</b>) <b>int</b> int(<b>complex</b>)</pre>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Description</b> | <p>Returns argument value converted to <code>int</code> type.</p> <p><code>int(bool x)</code> returns 0 if <code>x=false</code>, and returns 1 if <code>x=true</code>.</p> <p><code>int(double x)</code> converts <code>double</code> to <code>int</code> by truncating (discarding the fractional part).</p> <p><code>int(complex x)</code> converts <code>double</code> real part of a complex number to <code>int</code> by truncating (discarding the fractional part).</p> <p><code>int(x)</code> works exactly the same as type-casting operator <code>(int)x</code>.</p> |
| <b>Examples</b>    | <pre>int(true) = 1 int(1.6) = 1 int(-1.6) = -1 int(1.1+2.2j) = 1</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



## int64 C-keyword

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| <b>Description</b> | Declares 64-bit integer variable or array.                                           |
| <b>Examples</b>    | <pre>int64 i; int64 var = 10; int64 array[10]; int64 array[] = { 0, 1, 2, 3 };</pre> |

## (int64) type-casting operator

|                    |                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Converts a number of arbitrary type to 64-bit integer.<br>(int64)x works exactly the same as function int64(x). |
| <b>Examples</b>    | <pre>i = (int64)x;</pre>                                                                                        |

## int64

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre><b>int64</b> int64(<b>bool</b>) <b>int64</b> int64(<b>int</b>) <b>int64</b> int64(<b>int64</b>) <b>int64</b> int64(<b>double</b>) <b>int64</b> int64(<b>complex</b>)</pre>                                                                                                                                                                                                                                            |
| <b>Description</b> | Returns argument value converted to int64 type.<br><br>int64(bool x) returns 0i64 if x=false, and returns 1i64 if x=true.<br>int64(double x) converts double to int64 by truncating (discarding the fractional part).<br>int64(complex x) converts double real part of a complex number to int64 by truncating (discarding the fractional part).<br><br>int64(x) works exactly the same as type-casting operator (int64)x. |
| <b>Examples</b>    | <pre>int64(true) = 1i64 int64(1.6) = 1i64 int64(-1.6) = -1i64 int64(1.1+2.2j) = 1i64</pre>                                                                                                                                                                                                                                                                                                                                 |

**double** *C-keyword*

|                    |                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------|
| <b>Description</b> | Declares double variable or array                                                              |
| <b>Examples</b>    | <pre>double x; double var = 12.34; double array[10]; double array[] = { 1.2, 3.4, 5.6 };</pre> |

**(double)** *type-casting operator*

|                    |                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | <p>Converts a number of arbitrary type to double.</p> <p><code>(double)x</code> works exactly the same as function <code>double(x)</code>.</p> |
| <b>Examples</b>    | <code>x = (double)1/2;</code>                                                                                                                  |

**double**

|                    |                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre><b>double</b> double(<b>bool</b>) <b>double</b> double(<b>int</b>) <b>double</b> double(<b>int64</b>) <b>double</b> double(<b>double</b>) <b>double</b> double(<b>complex</b>)</pre>                                                                                                                                                                                            |
| <b>Description</b> | <p>Returns argument value converted to <code>double</code> type.</p> <p><code>double(bool x)</code> returns 0.0 if <code>x=false</code>, and returns 1.0 if <code>x=true</code>.</p> <p><code>double(complex x)</code> returns real part of a complex number <code>x</code>.</p> <p><code>double()</code> works exactly the same as type-casting operator <code>(double)</code>.</p> |
| <b>Examples</b>    | <pre>double(true) = 1.0 double(1) = 1.0 double(1.1+2.2j) = 1.1</pre>                                                                                                                                                                                                                                                                                                                 |

**float** keyword, type-casting operator, and function can be used in NL5, however it is identical to **double**.

**complex** *C-keyword*

|                    |                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Declares complex variable or array.                                                                            |
| <b>Examples</b>    | <pre>complex c; complex var = 1.2+3.4j; complex array[10]; complex array[] = { 1.0, 1.0j, -1.0, -1.0j };</pre> |

**(complex)** *type-casting operator*

|                    |                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | <p>Converts a number of arbitrary type to complex.</p> <p><code>(complex)x</code> works exactly the same as function <code>complex(x)</code>.</p> |
| <b>Examples</b>    | <pre>x = sqrt((complex)(-1));</pre>                                                                                                               |

**complex**

|                    |                                                                                                                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre><b>complex</b> complex(<b>bool</b>) <b>complex</b> complex(<b>int</b>) <b>complex</b> complex(<b>int64</b>) <b>complex</b> complex(<b>double</b>) <b>complex</b> complex(<b>complex</b>)</pre>                                                                                            |
| <b>Description</b> | <p>Returns argument value converted to <code>complex</code> type.</p> <p><code>complex(bool x)</code> returns 0.0 if <code>x=false</code>, and returns 1.0 if <code>x=true</code>.</p> <p><code>complex(x)</code> works exactly the same as type-casting operator <code>(complex)x</code>.</p> |
| <b>Examples</b>    | <pre>complex(true) = 1.0+0j complex(2) = 2.0+0j</pre>                                                                                                                                                                                                                                          |

## fabs

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double fabs(double)</code><br><code>double fabs(complex)</code> |
| <b>Description</b> | Absolute value.<br>Magnitude of complex number for complex argument.  |
| <b>Examples</b>    | <code>fabs(-3.5) = 3.5</code><br><code>fabs(3.5) = 3.5</code>         |

## fmod

|                    |                                              |
|--------------------|----------------------------------------------|
| <b>Prototype</b>   | <code>double fmod(double x, double y)</code> |
| <b>Description</b> | Remainder of the division $x / y$ .          |
| <b>Examples</b>    | <code>fmod(10.2, 3.0) = 1.2</code>           |

## fmin

|                    |                                              |
|--------------------|----------------------------------------------|
| <b>Prototype</b>   | <code>double fmin(double x, double y)</code> |
| <b>Description</b> | Smaller of two numbers.                      |
| <b>Examples</b>    | <code>fmin(10.2, 3.0) = 3.0</code>           |

## fmax

|                    |                                              |
|--------------------|----------------------------------------------|
| <b>Prototype</b>   | <code>double fmax(double x, double y)</code> |
| <b>Description</b> | Larger of two numbers.                       |
| <b>Examples</b>    | <code>fmin(10.2, 3.0) = 10.2</code>          |

## abs, mag

|                    |                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre> <b>int</b> abs(<b>int</b>) <b>int64</b> abs(<b>int64</b>) <b>double</b> abs(<b>double</b>) <b>double</b> abs(<b>complex</b>)  mag() can be used instead of abs(). </pre> |
| <b>Description</b> | <p>Absolute value (magnitude).</p> <p>For complex argument: <math>\text{abs} = \sqrt{re^2 + im^2}</math>.</p>                                                                  |
| <b>Examples</b>    | <pre> abs(-23) = 23 abs(1.0) = 1.0 abs(3.0+4.0j) = 5.0 </pre>                                                                                                                  |

## round

|                    |                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre> <b>double</b> round(<b>double</b>) <b>double</b> round(<b>double</b> x, <b>double</b> y) </pre>                           |
| <b>Description</b> | <p>round(x) rounds x to the nearest integer.</p> <p>round(x,y) rounds x to the nearest multiple of y. Returns x if y&lt;=0.</p> |
| <b>Examples</b>    | <pre> round(1.5) = 2.0 round(-1.5) = -1.0 round(3.1415,0.1) = 3.1 </pre>                                                        |

## floor

|                    |                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre> <b>double</b> floor(<b>double</b>) </pre>                                                                |
| <b>Description</b> | <p>Rounds down: finds the largest integer not greater than the argument and returns it as a <b>double</b>.</p> |
| <b>Examples</b>    | <pre> floor(1.6) = 1.0 floor(-1.6) = -2.0 </pre>                                                               |

## ceil

|                    |                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre> <b>double</b> ceil(<b>double</b>) </pre>                                                             |
| <b>Description</b> | <p>Rounds up: finds the smallest integer not less than the argument and returns it as a <b>double</b>.</p> |
| <b>Examples</b>    | <pre> ceil(1.6) = 2.0 ceil(-1.6) = -1.0 </pre>                                                             |

**sqrt**

|                    |                                                                                 |
|--------------------|---------------------------------------------------------------------------------|
| <b>Prototype</b>   | <i>double</i> sqrt( <i>double</i> )<br><i>complex</i> sqrt( <i>complex</i> )    |
| <b>Description</b> | Square root. If argument is <i>double</i> , negative argument will cause error. |
| <b>Examples</b>    | sqrt(4.0) = 2<br>sqrt(-4.0) : math error<br>sqrt(2j) = 1+1j                     |

**sqr**

|                    |                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <i>double</i> sqr( <i>double</i> )                                                                                                                                                                                    |
| <b>Description</b> | “Signed” square root. <i>sqr</i> ( <i>x</i> ) returns: <ul style="list-style-type: none"> <li>• <math>\sqrt{x}</math> if <math>x \geq 0</math></li> <li>• <math>-\sqrt{-x}</math> if <math>x &lt; 0</math></li> </ul> |
| <b>Examples</b>    | sqr(4) = 2<br>sqr(-4) = -2                                                                                                                                                                                            |

**sq**

|                    |                                                                          |
|--------------------|--------------------------------------------------------------------------|
| <b>Prototype</b>   | <i>complex</i> sq( <i>complex</i> )<br><i>double</i> sq( <i>double</i> ) |
| <b>Description</b> | <i>sq</i> ( <i>x</i> ) calculates $x*x$ : square of the argument.        |
| <b>Examples</b>    | sq(2) = 4<br>sq(1+1j) = 0+2j                                             |

**pow**

|                    |                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <i>double</i> pow( <i>double</i> <i>x</i> , <i>double</i> <i>y</i> )<br><i>complex</i> pow( <i>complex</i> <i>x</i> , <i>double</i> <i>y</i> ) |
| <b>Description</b> | Calculates $x^y$ : <i>x</i> to the power of <i>y</i> .<br>If double argument <i>x</i> is negative, math error may occur.                       |
| <b>Examples</b>    | pow(10.0,2.0) = 100.0<br>pow(1j,3) = 0-1j<br>pow(-4.0,0.5) : math error<br>pow(-4.0+0j,0.5) = 0+2j                                             |

**pwr**

|                    |                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <i>double</i> pwr( <i>double</i> x, <i>double</i> y)                                                                                                                                                      |
| <b>Description</b> | <p>"Signed" power function. pwr(x,y) returns:</p> <ul style="list-style-type: none"> <li>• <math>x^y</math> if <math>x \geq 0</math>,</li> <li>• <math>-(-x)^y</math> if <math>x &lt; 0</math></li> </ul> |
| <b>Examples</b>    | <pre>pwr(10.0,2.0) = 100.0 pwr(-10.0,2.0) = -100.0</pre>                                                                                                                                                  |

**exp**

|                    |                                                                            |
|--------------------|----------------------------------------------------------------------------|
| <b>Prototype</b>   | <i>double</i> exp( <i>double</i> )<br><i>complex</i> exp( <i>complex</i> ) |
| <b>Description</b> | Calculates the exponential $e$ to the $x$ .                                |
| <b>Examples</b>    | <pre>exp(3.0) = 20.0855 exp(PI*0.5j) = 0+1j</pre>                          |

**exp2**

|                    |                                            |
|--------------------|--------------------------------------------|
| <b>Prototype</b>   | <i>double</i> exp2( <i>double</i> )        |
| <b>Description</b> | Calculates $2^y$ : 2 to the power of $y$ . |
| <b>Examples</b>    | <pre>exp2(4.0) = 16.0</pre>                |

**log, ln**

|                    |                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <i>double</i> log( <i>double</i> )<br><i>complex</i> log( <i>complex</i> )<br><br>ln() can be used instead of log(). |
| <b>Description</b> | Calculates the natural logarithm.                                                                                    |
| <b>Examples</b>    | <pre>log(100) = 4.60517 log(-1.0) : math error log(-1.0+0j) = 0+3.14159j</pre>                                       |

## log2, lb

|                    |                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double log2(double)</code><br><code>complex log2(complex)</code><br><br><code>lb()</code> can be used instead of <code>log2()</code> . |
| <b>Description</b> | Calculates logarithm to base two.                                                                                                            |
| <b>Examples</b>    | <code>log2(128) = 7</code><br><code>log2(-8.0) : math error</code><br><code>log2(-8.0+0j) = 3+4.53236j</code>                                |

## log10, lg

|                    |                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double log10(double)</code><br><code>complex log10(complex)</code><br><br><code>lg()</code> can be used instead of <code>log10()</code> . |
| <b>Description</b> | Calculates logarithm to base ten.                                                                                                               |
| <b>Examples</b>    | <code>log10(100.0) = 2</code><br><code>log10(-100.0) : math error</code><br><code>log10(-100.0+0j) = 2+1.36437j</code>                          |

## log(x,y)

|                    |                                                                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>complex log(complex x, double y)</code><br><code>double log(double x, double y)</code>                                                                                                         |
| <b>Description</b> | Calculates logarithm $x$ to base $y$ .                                                                                                                                                               |
| <b>Examples</b>    | <code>log(128,2) = 7</code><br><code>log(PI,PI) = 1.0</code><br><code>log(-10.0,10.0) : math error</code><br><code>log(-10.0+0j,10.0) = 1+1.36437j</code><br><code>log(1j,10.0) = 0+682.1e-3j</code> |

## db

|                    |                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double db(double)</code><br><code>double db(double x, double y)</code>                                                                                                                                     |
| <b>Description</b> | <code>db(x)</code> calculates value of $x$ in decibel, as: $20 \cdot \log_{10}(\text{abs}(x))$<br><code>db(x,y)</code> calculates value of the ratio $x/y$ in decibel, as: $20 \cdot \log_{10}(\text{abs}(x/y))$ |
| <b>Examples</b>    | <code>db(100)=40</code><br><code>db(0.1,20.0) = -46.0205999133</code>                                                                                                                                            |



## sin, cos, tan, tg

|                    |                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double sin(double)</code><br><code>double cos(double)</code><br><code>double tan(double)</code><br><br>tg() can be used instead of tan(). |
| <b>Description</b> | Calculates sine, cosine, and tangent.                                                                                                           |
| <b>Examples</b>    | <code>sin(1.570796327) = 1.0</code><br><code>cos(1.570796327) = 0.0</code><br><code>tan(0.78539816339) = 1.0</code>                             |

## asin, acos, atan

|                    |                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double asin(double)</code><br><code>double acos(double)</code><br><code>double atan(double)</code>                                                                                                        |
| <b>Description</b> | Calculates arcsine, arccosine, arctangent.<br>asin returns angle in the range $-\pi/2 \dots \pi/2$<br>acos returns angle in the range $0 \dots \pi$ .<br>atan returns angle in the range $-\pi/2 \dots \pi/2$ . |
| <b>Examples</b>    | <code>asin(1.0) = 1.57079632679</code><br><code>acos(1.0) = 0</code><br><code>atan(1.0) = 0.785398163397</code>                                                                                                 |

## atan2

|                    |                                                                                |
|--------------------|--------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double atan2(double x, double y)</code>                                  |
| <b>Description</b> | Calculates arctangent of $x/y$ . Returns angle in the range $-\pi \dots \pi$ . |
| <b>Examples</b>    | <code>atan2(1.0, 1.0) = 0.785398163397</code>                                  |

## phase

|                    |                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double phase(complex)</code><br><code>double phase(double x, double y)</code>                                                                                                                               |
| <b>Description</b> | phase( <code>complex</code> ) calculates phase of complex number.<br>phase( <code>double x, double y</code> ) calculates arctangent of $x/y$ .<br>Returns phase of complex number in the range $-\pi \dots \pi$ . |
| <b>Examples</b>    | <code>phase(1+1j) = 0.785398 (Pi/4)</code>                                                                                                                                                                        |

## random, rand

|                    |                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double random(double)</code><br><br><code>rand()</code> can be used instead of <code>random()</code> |
| <b>Description</b> | Returns random number with uniform distribution in the range 0...x.                                        |
| <b>Examples</b>    | <code>random(3.0) = 1.2937463</code>                                                                       |

## gauss

|                    |                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double gauss(double m, double d)</code>                                                                     |
| <b>Description</b> | Returns normally distributed random number with mean value <code>m</code> and standard deviation <code>d</code> . |
| <b>Examples</b>    | <code>gauss(0,2) = -.8678275</code>                                                                               |

## islow, ishigh

|                    |                                                                                                                                                                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>bool islow(double)</code><br><code>bool ishigh(double)</code>                                                                                                                                                                                                                    |
| <b>Description</b> | Compares argument with logical threshold.<br><code>islow(x)</code> returns true if <code>x</code> is less than circuit logical threshold, otherwise false.<br><code>ishigh(x)</code> returns true if <code>x</code> is greater or equal to circuit logical threshold, otherwise false. |
| <b>Examples</b>    | <code>islow(1.0) = true</code><br><code>ishigh(1.0) = false</code>                                                                                                                                                                                                                     |

## lim, limit

|                    |                                                                                                                                                                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double lim(double x, double min, double max)</code><br><br><code>limit()</code> can be used instead of <code>lim()</code> .                                                                                                                                             |
| <b>Description</b> | Limiting function. <code>lim(x, min, max)</code> returns: <ul style="list-style-type: none"> <li><code>x</code>, if <code>min &lt;= x &lt;= max</code></li> <li><code>min</code>, if <code>x &lt; min</code></li> <li><code>max</code>, if <code>x &gt; max</code></li> </ul> |
| <b>Examples</b>    | <code>lim(0,-1,2) = 0</code><br><code>lim(-2,-1,2) = -1</code><br><code>lim(10,-1,2) = 2</code>                                                                                                                                                                               |

## sign

|                    |                                                                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>int sign(double)</code>                                                                                                                                                                     |
| <b>Description</b> | Indicates whether a numeric value is positive, negative, or zero.<br>sign(x) returns: <ul style="list-style-type: none"> <li>• 0 if x=0</li> <li>• 1 if x&gt;0</li> <li>• -1 if x&lt;0</li> </ul> |
| <b>Examples</b>    | <pre>sign(1.234) = 1 sign(0) = 0 sign(-5) = -1</pre>                                                                                                                                              |

## par

|                    |                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>complex par(complex,...)</code><br><code>double par(double,...)</code>                   |
| <b>Description</b> | Parallel connection of real or complex impedances. Number of arguments is from 1 to unlimited. |
| <b>Examples</b>    | <pre>par(1.0,1.0) = 0.5 par(1.0,2.0,3.0,4.0) = par(par(1.0,2.0),par(3.0,4.0)) = 0.48</pre>     |

## min

|                    |                                                                                                                                                                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code>double min(double,...)</code><br><code>double min(double[])</code><br><code>int64 min(int64,...)</code><br><code>int64 min(int64[])</code><br><code>int min(int,...)</code><br><code>int min(int[])</code><br><code>bool min(bool,...)</code><br><code>bool min(bool[])</code> |
| <b>Description</b> | min(x,...) returns smaller value of arguments. Number of arguments is not limited.<br>If x is an array x[N], min(x) returns smaller value of all array elements.                                                                                                                     |
| <b>Examples</b>    | <pre>min(1.0,2.0,3.0) = 1.0 min(1,2,3) = 1 min(false, true, true) = false  double x[] = { -1.0, 2.0, -3.0, 4.0 }; min(x) = -3.0;</pre>                                                                                                                                               |

## max

|                    |                                                                                                                                                                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre> <b>double</b> max(<b>double</b>,...) <b>double</b> max(<b>double</b>[]) <b>int64</b> max(<b>int64</b>,...) <b>int64</b> max(<b>int64</b>[]) <b>int</b> max(<b>int</b>,...) <b>int</b> max(<b>int</b>[]) <b>bool</b> max(<b>bool</b>,...) <b>bool</b> max(<b>bool</b>[]) </pre> |
| <b>Description</b> | <p>max(x, ...) returns larger value of arguments. Number of arguments is not limited.</p> <p>If x is an array x[N], max(x) returns larger value of all array elements.</p>                                                                                                           |
| <b>Examples</b>    | <pre> max(1.0,2.0,3.0) = 3.0 max(1,2,3) = 3 max(false, true, true) = true  double x[] = { -1.0, 2.0, -3.0, 4.0 }; max(x) = 4.0; </pre>                                                                                                                                               |

## mean, average

|                    |                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <pre> <b>complex</b> mean(<b>complex</b>,...) <b>complex</b> mean(<b>complex</b>[]) <b>double</b> mean(<b>double</b>,...) <b>double</b> mean(<b>double</b>[])  average() can be used instead of mean(). </pre> |
| <b>Description</b> | <p>mean(x, ...) returns mean (average) value of arguments. Number of arguments is not limited.</p> <p>If x is an array x[N], sum(x) returns mean (average) value of all array elements.</p>                    |
| <b>Examples</b>    | <pre> mean(1.0,2.0,3.0) = 2.0 mean(1.0+1.0j,2.0+2.0j) = 1.5+1.5j double x[] = { 1.0, 2.0, 3.0, 4.0 }; mean(x) = 2.5; </pre>                                                                                    |

**sum**

|                    |                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code><i>complex</i> sum(<i>complex</i>,...)</code><br><code><i>complex</i> sum(<i>complex</i>[])</code><br><code><i>double</i> sum(<i>double</i>,...)</code><br><code><i>double</i> sum(<i>double</i>[])</code> |
| <b>Description</b> | <p><code>sum(x,...)</code> returns sum of arguments. Number of arguments is not limited.</p> <p>If <code>x</code> is an array <code>x[N]</code>, <code>sum(x)</code> returns sum of all array elements.</p>      |
| <b>Examples</b>    | <pre>sum(1.0,2.0,3.0) = 6.0 sum(1.0+1.0j,2.0+2.0j) = 3.0+3.0j double x[] = { 1.0, 2.0, 3.0, 4.0 }; sum(x) = 10.0;</pre>                                                                                          |

**re, im**

|                    |                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------|
| <b>Prototype</b>   | <code><i>double</i> re(<i>complex</i>)</code><br><code><i>double</i> im(<i>complex</i>)</code> |
| <b>Description</b> | Real and imaginary part of complex number.                                                     |
| <b>Examples</b>    | <pre>re(1.2+3.4j) = 1.2 im(1.2+3.4j) = 3.4</pre>                                               |

# Script commands

*In alphabetical order.*

## List of script commands:

|         |         |          |                  |
|---------|---------|----------|------------------|
| ac      | exit    | return   | stop             |
| clear   | export  | rununtil | store            |
| close   | import  | save     | storetext        |
| cmd     | logdata | saveic   | traces           |
| cont    | open    | show     | <i>tracename</i> |
| cursors | pause   | silent   | tran             |
| display | ready   | sleep    |                  |

**ac**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>ac; ac from; ac from, to; ac from, to, points; ac from, to, points, scale;</pre>                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Description</b> | <p>Set AC analysis parameters and perform AC analysis.</p> <p><i>from</i> : start frequency<br/> <i>to</i> : stop frequency<br/> <i>points</i> : number of points<br/> <i>scale</i> = log or lin : logarithmic or linear frequency scale.</p> <p>If called from the script, command will not return until AC analysis is completed. If called from console or HTTP link, returns immediately. Use <code>ready</code> command to check for analysis completion.</p> |
| <b>Examples</b>    | <pre>ac; ac 1M; ac 1M, 100M; ac 1M, 100M, 500; ac 1M, 100M, 500, lin;</pre>                                                                                                                                                                                                                                                                                                                                                                                        |

**clear**

|                    |                     |
|--------------------|---------------------|
| <b>Usage</b>       | <code>clear;</code> |
| <b>Description</b> | Clear storage.      |

**close**

|                    |                        |
|--------------------|------------------------|
| <b>Usage</b>       | <code>close;</code>    |
| <b>Description</b> | Close active document. |

**cmd**

|                    |                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <code>cmd <i>command_line</i>;</code>                                                            |
| <b>Description</b> | Execute Windows command line <i>command_line</i> .                                               |
| <b>Examples</b>    | <pre>cmd nl5.exe rc.nl5; cmd "C:\Arduino\arduino.exe" --upload "C:\Arduino\demo\demo.ino";</pre> |

## cont

|                    |                                                                                                                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>cont; cont screen; cont screen, step;</pre>                                                                                                                                                                                                                                                                      |
| <b>Description</b> | <p>Continue transient.</p> <p><i>screen</i> : screen size<br/> <i>step</i> : calculation step</p> <p>If called from the script, command will not return until transient is completed. If called from console or HTTP link, returns immediately. Use <code>ready</code> command to check for transient completion.</p> |
| <b>Examples</b>    | <pre>cont; cont 1m; cont 1m, 10n;</pre>                                                                                                                                                                                                                                                                               |

## cursors

|                    |                                                                                                                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>cursors left, right; cursors on; cursors off;</pre>                                                                                                                                                                                                                                                    |
| <b>Description</b> | <p><code>cursors left, right</code> : set cursors (transient or AC) to specified positions and show cursors.</p> <p><i>left</i> : position of the left cursor   <i>step</i> : position of the right cursor</p> <p><code>cursors on</code> : show cursors.<br/> <code>cursors off</code> : hide cursors.</p> |
| <b>Examples</b>    | <pre>cursors 1.5, 2.5; cursors off;</pre>                                                                                                                                                                                                                                                                   |

## display

|                    |                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>display on; display off;</pre>                                                                                            |
| <b>Description</b> | <p><code>display on</code> : show transient and AC windows.<br/> <code>display off</code> : hide transient and AC windows.</p> |

## exit

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>exit;</pre>                                                                     |
| <b>Description</b> | <p>Close all documents and exit NL5. Cannot be called from console command line.</p> |



**export (transient)**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre> export; export filename; export filename, from; export filename, from, to; export filename, from, to, step; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Description</b> | <p>Export transient traces into csv file.</p> <p><i>filename</i> : name of the file to export traces<br/> <i>from</i> : start of the data interval<br/> <i>to</i> : end of the data interval<br/> <i>step</i> : time step</p> <p>If <i>filename</i> is omitted, name of the file to export is the same as script file name, with “csv” extension. If file path is not specified, export in the script file directory. Extension “csv” can be omitted.</p> <p>Number of points cannot exceed <b>Max number of points</b> value defined in the <b>Preferences</b> dialog box, <b>Transient</b> page.</p> <p>If <i>step</i> is omitted, 101 points will be exported.</p> <p>Only traces currently shown on the graph will be exported.</p> |
| <b>Examples</b>    | <pre> export; export rc_traces; export rc_traces, 0, 100; export rc_traces, 0, 100, 0.1; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**export (AC)**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre> export; export filename; export filename, from; export filename, from, to; export filename, from, to, points; export filename, from, to, points, scale; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Description</b> | <p>Export AC traces into csv file.</p> <p><i>filename</i> : name of the file to export traces.<br/> <i>from</i> : start frequency.<br/> <i>to</i> : end frequency.<br/> <i>points</i> : number of points.<br/> <i>scale</i> = log or lin : logarithmic or linear frequency scale.</p> <p>If <i>filename</i> is omitted, name of the file to export is the same as script file name, with “csv” extension. If file path is not specified, export in the script file directory. Extension “csv” can be omitted.</p> <p>Only traces currently shown on the graph will be exported.</p> |
| <b>Examples</b>    | <pre> export; export ac_traces; export ac_traces, 1m, 1k; export ac_traces, 1m, 1k, 100; export ac_traces, 1m, 1k, 100, lin; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**import** (*transient*)

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>import filename; import filename, cf, cn, rf, rn, hr, tc, ts;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description</b> | <p>Import transient traces from text file or scope data file.</p> <p><i>filename</i> : name of the file to import traces (with extension).</p> <p><i>filename</i> should contain extension to specify type of data file. The following extensions/types are supported:</p> <ul style="list-style-type: none"> <li>"txt" and "csv" – text file, comma-separated.</li> <li>"wfm" – Tektronix waveform format.</li> <li>"isf" – Tektronix interval format.</li> <li>"bin" – Keysight Technologies (Agilent) binary format.</li> <li>"trc" – LeCroy binary format.</li> </ul> <p>More parameters can be used for import from text (comma-separated) files only.</p> <ul style="list-style-type: none"> <li><i>cf</i> : first data column to import (1 is first column of the file).</li> <li><i>cn</i> : number of data columns to import. If <i>cn</i> = -1, import all available columns after first.</li> <li><i>rf</i> : first data row to import (1 is first row of the file).</li> <li><i>rn</i> : number of data rows to import. If <i>rn</i> = -1, import all available rows after first.</li> <li><i>hr</i> : header row. If <i>hr</i> = 0, add header row: "<i>trace(s), trace1, trace2, ...</i>"</li> <li><i>tc</i> : time column. If <i>tc</i> = 0, add time column with time step <i>ts</i>.</li> <li><i>ts</i> : time step of added time column floating point (set any value if not used).</li> </ul> |
| <b>Examples</b>    | <pre>import scope_traces.isf; import rc_traces.csv, 2, -1, 2, -1, 1, 1, 0;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## logdata

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>logdata filename, expr1,...; logdata +, filename, expr1,...; logdata;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description</b> | <p><code>logdata</code> with parameters is the first data logging command.</p> <p><i>filename</i> : name of the file to export traces<br/> <code>+</code> : flag to append the data into existing file<br/> <i>exprN</i> : expression to be logged</p> <p>If a file <i>filename</i> does not exist, creates a new log file and writes a header.<br/> If a file <i>filename</i> already exists, and a first parameter is <code>+</code>, a new data will be appended to existing data, otherwise old data will be overwritten.<br/> Extension “csv” in the file name can be omitted. If file path is not specified, creates log file in the script file directory.</p> <p><code>logdata</code> without parameters evaluates expressions <i>exprN</i> specified in the first <code>logdata</code> command and writes results into the log file as comma-separated string.</p> |
| <b>Examples</b>    | <pre>logdata rclog, r1, v(r1), v(c1).rms; logdata +, rcapp, r1, v(r1), v(c1).rms; logdata;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## open

|                    |                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>open filename;</pre>                                                                                                                 |
| <b>Description</b> | Open schematic file <i>filename</i> . Extension “nl5” can be omitted. If file path is not specified, search in the script file directory. |
| <b>Examples</b>    | <pre>open "c:\Project files\nl5\rc.nl5"; open rc;</pre>                                                                                   |

## pause

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>pause;</pre>                                                                    |
| <b>Description</b> | Pause transient. Command can be called from console command line and HTTP link only. |

## ready

|                    |                                                                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>ready;</pre>                                                                                                                                                                                     |
| <b>Description</b> | <p>Check if transient or AC analysis is completed. Returns “0” if analysis is still running, returns “1” if completed.</p> <p>Command can be called from console command line and HTTP link only.</p> |

**return**

|                    |                           |
|--------------------|---------------------------|
| <b>Usage</b>       | <code>return;</code>      |
| <b>Description</b> | Stop executing the script |
| <b>Examples</b>    | <code>return;</code>      |

**rununtil**

|                    |                                                                                                                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <code>rununtil;</code><br><code>rununtil <i>expr</i>;</code>                                                                                                                                                                          |
| <b>Description</b> | Set up “run until” transient mode.<br>If parameter <i>expr</i> is omitted, turn off “run until” mode and clear “run until” expression.<br>Otherwise turn on “run until” mode and use parameter <i>expr</i> as “run until” expression. |
| <b>Examples</b>    | <code>rununtil;</code><br><code>rununtil V(C1)&lt;0;</code>                                                                                                                                                                           |

**save**

|                    |                                                                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <code>save;</code><br><code>save <i>filename</i>;</code>                                                                                                                                                               |
| <b>Description</b> | Save schematic into a file <i>filename</i> . Extension “nl5” can be omitted.<br>If file path is not specified, save in the script file directory.<br>If parameter <i>filename</i> is omitted, save into the same file. |
| <b>Examples</b>    | <code>save;</code><br><code>save rcnew;</code>                                                                                                                                                                         |

**saveic**

|                    |                               |
|--------------------|-------------------------------|
| <b>Usage</b>       | <code>saveic;</code>          |
| <b>Description</b> | Save Initial Conditions (IC). |

## show

|                    |                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <code>show window;</code>                                                                                                                                                                                                                                                                                                                                                        |
| <b>Description</b> | <p>Show or activate window specified by parameter <i>window</i>. The following <i>window</i> values can be used:</p> <pre> tran : transient; ac    : AC; dc    : DC sweep; xy    : XY diagram; ed    : Eye diagram; ah    : Amplitude histogram; fft   : FFT; th    : Transient histogram; pow   : Power; smith : Smith chart; ny    : Nyquist plot; ach   : AC histogram.</pre> |
| <b>Examples</b>    | <code>show tran;</code>                                                                                                                                                                                                                                                                                                                                                          |

## silent

|                    |                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre> silent on; silent off;</pre>                                                                |
| <b>Description</b> | <pre> silent on : do not show script execution log. silent off : show script execution log.</pre> |

## sleep

|                    |                                            |
|--------------------|--------------------------------------------|
| <b>Usage</b>       | <code>sleep time;</code>                   |
| <b>Description</b> | Pause script execution for <i>time</i> ms. |
| <b>Examples</b>    | <code>sleep 1000;</code>                   |

## stop

|                    |                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <code>stop;</code>                                                                                                                          |
| <b>Description</b> | Stop transient. This command can be used to free memory allocated for transient analysis. Transient cannot be continued after this command. |

## store

|                    |                                                                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <code>store;</code><br><code>store <i>expr</i>;</code>                                                                                                                                                |
| <b>Description</b> | Move run into storage. The parameter <i>expr</i> is evaluated as an expression, and the result is used as a storage name. If parameter <i>expr</i> is omitted, a default storage name "RunN" is used. |
| <b>Examples</b>    | <code>store;</code><br><code>store R1*C1;</code>                                                                                                                                                      |

## storetext

|                    |                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <code>storetext;</code><br><code>storetext <i>text</i>;</code>                                                                                  |
| <b>Description</b> | Move run into storage with parameter <i>text</i> as a storage name. If parameter <i>text</i> is omitted, a default storage name "RunN" is used. |
| <b>Examples</b>    | <code>storetext;</code><br><code>storetext This is a first run;</code>                                                                          |

## traces

|                    |                                                                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <code>traces <i>stateN</i>,...;</code>                                                                                                                                                                                                                             |
| <b>Description</b> | Hide or show traces on the graph. The parameter <i>stateN</i> specifies show/hide status of the trace number N (traces are listed in the same order as in the Transient/Data or AC/Data window).<br><br><i>stateN</i> = 0 – hide trace;<br>otherwise – show trace. |
| <b>Examples</b>    | <code>traces 0,1,1,0,0,1;</code>                                                                                                                                                                                                                                   |

***tracename*** (*transient*)

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>tracename; tracename from; tracename from, to; tracename from, to, step;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Description</b> | <p>Request transient trace data as a comma-separated string.</p> <p><i>from</i> : start of the data interval.<br/> <i>to</i> : end of the data interval.<br/> <i>step</i> : step.</p> <p><i>tracename</i>; - returns 101 points of entire <i>tracename</i> interval.<br/> <i>tracename from</i>; - returns only one trace value at <i>t=from</i>.<br/> <i>tracename from, to</i>; - returns 101 points in specified interval.<br/> <i>tracename from, to, step</i>; - returns data points in specified interval with specified step.</p> <p>Trace <i>tracename</i> should be specified in the Transient Data, however it does not need to be displayed on the graph or in the table.</p> <p>Please note that length of the returned string may be limited, and the limit may be different for different applications. If you got a time-out on this command, please reduce the number of data points requested by one command.</p> <p>This command can be called from HTTP link only.</p> |
| <b>Examples</b>    | <pre>V(R1) ; V(R1) 1.23; V(R1) 0, 100; V(R1) 0, 10, 0.1;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

***tracename* (AC)**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre> tracename; tracename from; tracename from, to; tracename from, to, points; tracename from, to, points, scale; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description</b> | <p>Request AC trace data as a comma-separated string.</p> <p><i>from</i> : start frequency.<br/> <i>to</i> : end frequency.<br/> <i>points</i> : number of points.<br/> <i>scale</i> = log or lin : logarithmic or linear frequency scale.</p> <p><i>tracename</i>; - returns all calculated data points of <i>tracename</i> trace.<br/> <i>tracename from</i>; - returns only one trace value at <math>f=from</math>.<br/> <i>tracename from, to</i>; - returns all calculated data points in the specified interval.<br/> <i>tracename from, to, points</i>; - returns specified number of points in the specified interval.<br/> <i>tracename from, to, points, scale</i>; - returns data with specified scale type.</p> <p>Trace <i>tracename</i> should be specified in the AC Data, however it does not need to be displayed on the graph or in the table.</p> <p>Please note that length of the returned string may be limited, and the limit may be different for different applications. If you got a time-out on this command, please reduce the number of data points requested by one command.</p> <p>This command can be called from HTTP link only.</p> |
| <b>Examples</b>    | <pre> V(R1) ; V(R1) 12.34; V(R1) 1, 100; V(R1) 1, 10, 100; V(R1) 1, 10, 100, lin; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



## tran

|                    |                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usage</b>       | <pre>tran;<br/>tran start;<br/>tran start, screen;<br/>tran start, screen, step;</pre>                                                                                                                                                                                                                                                                                                       |
| <b>Description</b> | <p>Set transient parameters and start transient.</p> <p><i>start</i> : start of transient display<br/><i>screen</i> : screen size<br/><i>step</i> : calculation step</p> <p>If called from the script, command will not return until transient is completed. If called from console or HTTP link, returns immediately. Use <code>ready</code> command to check for transient completion.</p> |
| <b>Examples</b>    | <pre>tran;<br/>tran 0;<br/>tran 0, 10m;<br/>tran 0, 10m, 1u;</pre>                                                                                                                                                                                                                                                                                                                           |

# Script examples

**Set component parameters.** Component parameters have been calculated in external application (for instance, Excel), or entered manually and saved into the text file in the ***name=value*** format:

```
R1 = 5.1;
C1 = 12e-9;
V3.Period = 0.01;
```

Run the script to apply new parameters to components.

**Sweep parameter.** Component parameter is changing in specified range, transient analysis performed for each parameter, results placed into storage:

```
for( R1=1; R1<=10; R1+=1 )
{
    tran;
    store R1;
}
```

**Sweep parameter from the list.** Component parameter is assigned value from the list, AC analysis performed for each parameter, results placed into storage:

```
for( R1 =1m, 2m, 10m, 50, 100m )
{
    ac;
    store R1;
}
```

**Sweep variable.** Local variable is changing in some range, component parameters modified, transient analysis performed, results placed into storage:

```
double freq;
for( freq=1; freq<=10; freq*=1.1 )
{
    V2.Period = 1 / freq;
    R2 = 1 / freq;
    tran;
    store freq;
}
```

**Wait for condition.** Transient is running until peak-to-peak value of the trace is less than specified threshold. When done, Initial Conditions are saved.

```
double threshold = 1e-6;
tran;
while( V(C1).pp > threshold ) cont;
saveic;
```

**Perform analysis for specified file, save data, exit application.** Schematic file is loaded into NL5, component parameters changed, transient analysis performed, traces exported into “csv” file, NL5 closed. This script can be executed from command line.

```
open lcr.nl5;
R1=100;
C1=1n5;
tran;
export data.csv;
exit;
```

**Perform analysis for specified file, log data, exit application.** Schematic file is loaded into NL5, component parameter swept, transient analysis performed, traces data logged into text file, NL5 closed. This script can be executed from command line.

```
open lcr.nl5;
logdata lcrdata.csv, r1, V(R1).mean, V(R1).rms;
for( R1=100; R1<=1000; R1+=100 )
{
    tran;
    logdata;
}
exit;
```

# HTTP functions

## List of HTTP functions.

### General

NL5\_GetInfo  
 NL5\_New  
 NL5\_Open  
 NL5\_Save  
 NL5\_Close  
 NL5\_Show  
 NL5\_GetActive  
 NL5\_GetParam  
 NL5\_SetParam  
 NL5\_DisableCmp  
 NL5\_EnableCmp  
 NL5\_SetCmpState

### Transient

NL5\_GetTracesSize  
 NL5\_AddTrace  
 NL5\_DeleteTrace  
 NL5\_GetTrace  
 NL5\_GetTraceValue  
 NL5\_SetTrace  
 NL5\_ShowTran  
 NL5\_SetScreen  
 NL5\_Cursors  
 NL5\_Start  
 NL5\_Pause  
 NL5\_Continue  
 NL5\_Stop  
 NL5\_SaveIC  
 NL5\_SaveAsWithIC  
 NL5\_IsRunning  
 NL5\_GetDataSize  
 NL5\_GetDataAt  
 NL5\_GetData  
 NL5\_AddData  
 NL5\_Export  
 NL5\_GetStorageSize  
 NL5\_Store  
 NL5\_DeleteStorage

### AC

NL5\_GetACTracesSize  
 NL5\_AddACTrace  
 NL5\_DeleteACTrace  
 NL5\_GetACTrace  
 NL5\_GetACTraceValue  
 NL5\_SetACTrace  
 NL5\_ShowAC  
 NL5\_SetACScreen  
 NL5\_ACCursors  
 NL5\_StartAC  
 NL5\_StopAC  
 NL5\_GetACDataSize  
 NL5\_GetACDataAt  
 NL5\_GetACStorageSize  
 NL5\_StoreAC  
 NL5\_DeleteACStorage

## NL5\_GetInfo

Get NL5 version info

**Request:** NL5\_GetInfo

**Response:** 0,Version Ver.Rev.Core.GUI, Date MM/DD/YYYY

## NL5\_New

Create new document

**Request:** NL5\_New

**Response:** *ndoc*

*ndoc* : document handle

## NL5\_Open

Open document *name*

**Request:** NL5\_Open, *name*

**Response:** *ndoc*

*ndoc* : document handle

## NL5\_Save

Save active document

**Request:** NL5\_Save

NL5\_Save, *name*

**Response:** 0

*name* : file name. If not provided, save active document with existing file name.

## NL5\_Close

Close active document

**Request:** NL5\_Close

**Response:** 0

## NL5\_Show

Show schematic of active document, or activate document and show schematic.

**Request:** NL5\_Show

NL5\_Show, *ndoc*

**Response:** *ndoc*

*ndoc* : document handle

If *ndoc* not provided, show schematic of active document.

If *ndoc* provided, activate document with document handle *ndoc* and show schematic.

## NL5\_GetActive

Get document handle of active document

**Request:** NL5\_GetActive

**Response:** *ndoc*

*ndoc* : document handle

## NL5\_GetParam

Get value of component parameter.

**Request:** NL5\_GetValue, *param*

**Response:** 0, *value*  
0, *value*, *text*

*param* : *cmp.model* – return model name of component *cmp*  
*cmp.param* – return value of *cmp.param*  
*cmp* - return value of *cmp* first parameter

If parameter is a number, return number: 0,1.23.

If parameter has a formula, return number and text of the formula: 0,2.46,=R1.R\*2.

If parameter is a text, return text: 0,filename.nl5.

If parameter is a text from the list, return index of the text and text: 0,1,On.

## NL5\_SetParam

Set value of component parameters.

**Request:** NL5\_SetParam, *expr1*, *expr2*, ...

**Response:** 0

*expr* : *name=value*  
*name* : *cmp.model* – set model of component *cmp*  
*cmp.param* – set value of *cmp.param*  
*cmp* - set value of *cmp* first parameter  
*value* : number or text

Any number of *expr* can be provided.

Text with commas should be placed into double quotes: "1,2,3,4", or HEX formatted.

Text or number with forbidden characters should be HEX formatted.

Set formula for parameter as "*=formula*": =R1.R\*2.

To clear the parameter text, give empty *value*: IC=.

If parameter is a text from the list, it can be set as an index of the text or as a text: D1.IC=1, D1.IC=On.

## NL5\_DisableCmp

Disable one or more components.

**Request:** NL5\_DisableCmp, *name1*, *name2*, ...

**Response:** 0

*name* : component name: R1

Any number of components can be disabled by one function.

If Label component is being disabled, all Labels with that name will be disabled.

## NL5\_EnableCmp

Enable one or more components.

**Request:** NL5\_EnableCmp, *name1*, *name2*, ...

**Response:** 0

*name* : component name: R1.

Any number of components can be enabled by one function.

If Label component is being enabled, all Labels with that name will be enabled.

## NL5\_SetCmpState

Set short/open state of the component

**Request:** NL5\_EnableCmp, *name*, *state*

**Response:** 0

*name* : component name: R1.

*state* : short: set component to short state  
open: set component to open state  
*empty*: set component to normal state

## NL5\_GetTracesSize

Get number of transient traces.

**Request:** NL5\_GetTracesSize

**Response:** *number*

*number*: number of traces



## NL5\_AddTrace

Add new transient trace.

**Request:** NL5\_AddTrace, *type*, *expr*

**Response:** *index*, *handle*, *name*

*type* : V, I, P, State, Variable, Function, Data (*text*)

*expr* :  
for V, I, P, State: component name  
for Variable: variable name  
for Function: function expression  
for Data: trace name (optional)

*index* : trace index ( $\geq 0$ )

*handle* : trace handle ( $< 0$ )

*name* : trace name

## NL5\_DeleteTrace

Delete all or one transient trace.

**Request:** NL5\_DeleteTrace, *trace*

**Response:** 0

*trace* :  
all (*text*) – delete all traces  
*number*  $\geq 0$  - trace index  
*number*  $< 0$  - trace handle  
*text* - trace name

## NL5\_GetTrace

Get transient trace index, handle, and name.

**Request:** NL5\_GetTrace, *trace*

**Response:** *index*, *handle*, *name*

*trace* :  
*number*  $\geq 0$  - trace index  
*number*  $< 0$  - trace handle  
*text* - trace name

*index* : trace index ( $\geq 0$ )

*handle* : trace handle ( $< 0$ )

*name* : trace name

## NL5\_GetTraceValue

Get value *name* of transient trace, or trace value at specified time.

**Request:** NL5\_GetTraceValue, *trace*, *name* [, storage=*index*]

**Response:** 0, *value*

*trace* :        *number* >= 0 - trace index  
                  *number* < 0 - trace handle  
                  *text* - trace name

*name* :        *number*: get trace value at time=*number*  
                  last, left, right, delta, min, max, pp, mean, rms, sd, freq, period, integral:  
                  name of requested trace value (text, same as trace values displayed in the table).

storage=*index* : (optional) storage index. If not provided, return value for the last Run.

*value* :        trace value

## NL5\_SetTrace

Set properties of all or selected transient traces.

**Request:** NL5\_SetTrace, *trace1*,..., *tracen*, *expr1*, *expr2*, ...

**Response:** 0

*trace* :        all (*text*) – set all traces  
                  *number* >= 0 - trace index  
                  *number* < 0 - trace handle  
                  *text* - trace name

*expr* :        *property\_name* = *value* . All expressions are optional.

show=on, off – display trace on the screen

name=*text* – set trace name to *text*. To erase name, provide “name=” without text.

type=analog, digital, bus

analog=1,2,3,4 – set trace analog section

scale=*number*

mid=*number*

shift=*number* - time shift

color=*text* or *number*

Trace color can be defined using the following predefined texts:

aqua, black, blue, gray, green, lime, navy, olive, purple, red, white, yellow

or hexadecimal number 0xBBGGRR, where BB, GG, RR are hexadecimal values of blue, green, and red parts of the color. For example, 0xff0000 = blue, etc.

width=*number* – trace width

line=on, off - show line

dp=on, off - show data points

## NL5\_ShowTran

Show transient window.

**Request:** NL5\_ShowTran  
NL5\_ShowTran, *ndoc*

**Response:** *ndoc*

*ndoc* : document handle. Activate document and show transient window.  
If not provided, show transient window of active document.

## NL5\_SetScreen

Set properties of transient screen.

**Request:** NL5\_SetScreen, *expr1*, *expr2*, ...

**Response:** 0

*expr* : *name=value*. All expressions are optional.

fit=hor, vert, screen : fit horizontal, fit vertical, fit the screen

analog=add : add analog section

analog=remove: remove active analog section

analog=*n* : activate analog section *n* = 1...4

maximize=on, off : maximize active analog section

separate=on, off : separate traces in active analog section

start=*number* : horizontal start, s

screen= *number* : horizontal screen, s

## NL5\_Cursors

Show/hide transient cursors, set cursors to specified positions, set Table mode to Cursors. Return current cursors position.

**Request:** NL5\_Cursors, *expr1*, *expr2*, ...

**Response:** 0, *left*, *right*

*expr* : *name* or *name=value*. All expressions are optional.

on : show cursors

off : hide cursors

left=*number* : set left cursor position

right= *number* : set right cursor position

*left* : left cursor position.

*right* : right cursor position.

## NL5\_Start

Start transient simulation.

**Request:** NL5\_Start, *extr1*, *expr2*, ...

**Response:** 0

*expr* : *name=value*. All expressions are optional.

start= *number* : left edge of the screen simulation

screen= *number* : transient screen size

step= *number* : simulation step

## NL5\_Pause

Pause transient simulation.

**Request:** NL5\_Pause

**Response:** 0

## NL5\_Continue

Continue transient simulation. Optional transient settings can be provided.

**Request:** NL5\_Continue, *extr1*, *expr2*, ...

**Response:** 0

*expr* : *name=value*. All expressions are optional.

screen= *number* : transient screen size

step= *number* : simulation step

## NL5\_Stop

Stop transient simulation.

**Request:** NL5\_Stop

**Response:** 0

## NL5\_SaveIC

Save Initial Conditions of transient simulation.

**Request:** NL5\_SaveIC

**Response:** 0

## NL5\_SaveAsWithIC

Save active document into a new file, then save current schematic state into components IC in the file.

**Request:** NL5\_SaveAsWithIC, *name*

**Response:** 0

*name* : file name

## NL5\_IsRunning

Check status of transient simulation, get last simulation time if paused or running.

**Request:** NL5\_IsRunning

**Response:** 0 - no active simulation  
 0,*time* - transient simulation is paused at *time*  
 1,*time* - transient simulation is running at *time*

## NL5\_GetDataSize

Get size of transient trace data (last Run or storage), and time of last data point.

**Request:** NL5\_GetDataSize, *trace* [,storage=*index*]

**Response:** *size*,*tlast*

*trace* :        *number* >= 0 - trace index  
                  *number* < 0 - trace handle  
                  *text* - trace name

storage=*index* : (optional) storage index. If not provided, return data size of the last Run

*size* :        number of data points  
*tlast* :        time of last data point

## NL5\_GetDataAt

Get transient trace data points as *time,value* comma-separated values.

**Request:** NL5\_GetDataAt, *trace*, *expr1*, *expr2*, ...

**Response:** *points*,*t1*,*v1*,*t2*,*v2*,...

*trace* :        *number* >= 0 - trace index  
                  *number* < 0 - trace handle  
                  *text* - trace name

*expr* :        *name=value*. All expressions are optional.

storage=*index* : storage index. If not provided, return data of the last Run  
 start=*number* : start data point >= 0. If not provided, start = 0  
 points=*number* : number of data points. If not provided, points=1

*points* :        number of data points returned (>=0)  
*t,v* :            comma-separated time and value

## NL5\_GetData

Get interpolated transient trace data points

**Request:** NL5\_GetData, *trace*, *expr1*, *expr2*, ...

**Response:** *points*, *t1*, *v1*, *t2*, *v2*, ...

*trace* : *number*  $\geq 0$  - trace index

*number*  $< 0$  - trace handle

*text* - trace name

*expr* : *name=value*. All expressions are optional.

*storage=index* : storage index. If not provided, return data of the last Run

*from=number* : start time. If not provided, from = 0

*to=number* : end time. If not provided, to = 0

*step=number* : time step. If not provided, step = 0

*points* : number of data points returned ( $\geq 0$ )

*t,v* : comma-separated time/value

## NL5\_AddData

Add data points to the transient trace of Data type.

**Request:** NL5\_AddData, *trace*, *t1*, *v1*, *t2*, *v2*, ...

**Response:** *size*

*trace* : *number*  $\geq 0$  - trace index

*number*  $< 0$  - trace handle

*text* - trace name

*t,v* : comma-separated time/value

*size* : new data size

## NL5\_Export

Export transient data of all or selected traces to a text file (comma-separated).

**Request:** NL5\_Export, *file, trace1,..., tracen, expr1, expr2, ...*

**Response:** *points* – number of points exported

*file* : file name

*trace* : all (*text*) – set all traces  
*number*  $\geq 0$  - trace index  
*number*  $< 0$  - trace handle  
*text* - trace name

*expr* : *name=value*. All expressions are optional.

*storage=index* : storage index. If not provided, return data of the last Run  
*from=number* : start time. If not provided, from = 0  
*to= number* : end time. If not provided, to = 0  
*step= number* : time step. If not provided, step = 0

*points*: number of points exported

## NL5\_GetStorageSize

Get size of the storage, last Run not included.

**Request:** NL5\_GetStorageSize

**Response:** *size*

*size*: storage size

## NL5\_Store

Copy Run data to storage

**Request:** NL5\_Store, *name*

**Response:** *index*

*name* : storage name. If not provided, default automatic name is used

*index*: index of added storage

## NL5\_DeleteStorage

Delete all or one storage.

**Request:** NL5\_DeleteStorage, *index*  
 NL5\_DeleteStorage

**Response:** *size*

*index* : index of storage to delete. If not provided, delete all storage.

*size*: storage size after deleting

## NL5\_GetACTracesSize

Get number of AC traces.

**Request:** NL5\_GetACTracesSize

**Response:** *number*

*number*: number of traces

## NL5\_AddACTrace

Add new AC trace.

**Request:** NL5\_AddTrace, *type*, *expr*

**Response:** *index*, *handle*, *name*

*type* : V, I, Function, Z, Y, Gamma, VSWR, Loop: trace type (text)

*expr* : for V, I: component name

for Function: function expression

for Z, Y: component name or not used (relative to AC source)

for Gamma, VSWR, Loop – not used (relative to AC source)

*index* : trace index ( $\geq 0$ )

*handle* : trace handle ( $< 0$ )

*name* : trace name

## NL5\_DeleteACTrace

Delete all or one AC trace.

**Request:** NL5\_DeleteACTrace, *trace*

**Response:** 0

*trace* : all (*text*) – delete all traces

*number*  $\geq 0$  - trace index

*number*  $< 0$  - trace handle

*text* - trace name

## NL5\_GetACTrace

Get AC trace index, handle, and name

**Request:** NL5\_GetACTrace, *trace*

**Response:** *index*, *handle*, *name*

*trace* : *number*  $\geq 0$  - trace index

*number*  $< 0$  - trace handle

*text* - trace name

*index* : trace index ( $\geq 0$ )

*handle* : trace handle ( $< 0$ )

*name* : trace name



## NL5\_GetACTraceValue

Get value *name* of AC trace.

Get value *name* of AC trace, or trace value at specified frequency.

**Request:** NL5\_GetACTraceValue, *trace*, *name* [, storage=*index*]

**Response:** 0, *value*

*trace* : *number* >= 0 - trace index

*number* < 0 - trace handle

*text* - trace name

*name* : *number*: get trace value at frequency=*number*

last, left, right, delta, min, max, pp, slope:

name of requested trace value (text, same as trace values displayed in the table).

storage=*index* : (optional) storage index. If not provided, return value for the last Run.

*value* : trace value

## NL5\_SetACTrace

Set properties of all or selected AC traces.

**Request:** NL5\_SetACTrace, *trace1*, ..., *tracen*, *expr1*, *expr2*, ...

NL5\_SetACTrace, all, *expr1*, *expr2*, ...

**Response:** 0

*trace* : all (*text*) – set all traces

*number* >= 0 - trace index

*number* < 0 - trace handle

*text* - trace name

*expr* : *property\_name* = *value*. All expressions are optional.

show=on, off – display trace on the screen

name=*text* – to erase name, provide “name=” without text

color=*text* or *number*

Trace color can be defined using the following predefined texts:

aqua, black, blue, gray, green, lime, navy, olive, purple, red, white, yellow

or hexadecimal number 0xBBGGRR, where BB, GG, RR are hexadecimal values of blue, green, and red parts of the color. For example, 0xff0000 = blue, etc.

width=*number* – trace width

phase\_width=*number*

line=on, off - show line

dp=on, off - show data points

## NL5\_ShowAC

Show AC window.

**Request:** NL5\_ShowAC  
NL5\_ShowAC, *ndoc*

**Response:** *ndoc*

*ndoc* : document handle. Activate document and show AC window.  
If not provided, show AC window of active document.

## NL5\_SetACScreen

Set properties of AC screen.

**Request:** NL5\_SetACScreen, *expr1*, *expr2*, ...  
**Response:** 0

*expr* : *name=value*. All expressions are optional.

fit=hor, vert, screen, phase : fit horizontal, fit vertical, fit the screen, fit phase vertical  
hor\_scale=lin, log : horizontal scale  
left= *number*: screen left, Hz  
right= *number*: screen right, Hz  
vert\_scale=lin, log : vertical scale  
db=off, on : show vertical scale in dB  
top= *number*: screen top  
bot= *number*: screen bottom  
phase=off, on, separate : phase display mode  
phase\_top= *number*: phase top  
phase\_bot= *number*: phase bottom

## NL5\_ACCursors

Show/hide AC cursors, set cursors to specified positions, set Table mode to Cursors, get current cursors position.

**Request:** NL5\_Cursors, *expr1*, *expr2*, ...  
**Response:** 0, *left*, *right*

*expr* : *name* or *name=value*. All expressions are optional.  
on : show cursors  
off : hide cursors  
left= *number*: set left cursor position  
right= *number*: set right cursor position

*left* : left cursor position.  
*right* : right cursor position.

## NL5\_StartAC

Start AC simulation.

**Request:** NL5\_StartAC, *extr1*, *extr2*, ...

**Response:** 0

*expr* : *name=value*. All expressions are optional.

*source=text* : AC source component name

*from= number*: start frequency

*to= number*: end frequency

*points= number*: number of points

## NL5\_StopAC

Stop AC simulation.

**Request:** NL5\_StopAC

**Response:** 0

## NL5\_GetACDataSize

Get size of AC trace data (last Run or storage), and frequency of last data point.

**Request:** NL5\_GetACDataSize, *trace* [,*storage=index*]

**Response:** *size,flast*

*trace* : *number* >= 0 - trace index

*number* < 0 - trace handle

*text* - trace name

*storage=index* : (optional) storage index. If not provided, return data size of the last Run

*size* : number of data points

*flast* : frequency of last data point

## NL5\_GetACDataAt

Get AC trace data points as *freq,mag,phase* comma-separated values.

**Request:** NL5\_GetACDataAt, *trace, expr1, expr2, ...*

**Response:** *points,f1,mag1,phase1,f2,mag2,phase2,,...*

*trace* : *number* >= 0 - trace index

*number* < 0 - trace handle

*text* - trace name

*expr* : *name=value*. All expressions are optional.

*storage=index* : storage index. If not provided, return data of the last Run

*start= number*: start data point >= 0. If not provided, start = 0

*points= number*: number of data points. If not provided, points=1

*points* number of data points returned (>=0)

*f,mag,phase* : comma-separated frequency, magnitude, phase

## NL5\_GetACStorageSize

Get size of AC storage, last Run not included.

**Request:** NL5\_GetACStorageSize

**Response:** *size*

*size*: storage size

## NL5\_StoreAC

Copy AC Run data to storage

**Request:** NL5\_StoreAC, *name*

**Response:** *index*

*name* : storage name. If not provided, default automatic name is used

*index*: index of added storage

## NL5\_DeleteStorage

Delete all or one AC storage.

**Request:** NL5\_DeleteACStorage, *index*

NL5\_DeleteACStorage

**Response:** *size*

*index* : index of AC storage to delete. If not provided, delete all AC storage.

*size*: storage size after deleting